MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
LARSEN & TOUBRO

Project: **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**

49PR POWER GENERATION : ENGINE HALL

DESIGN CALCULATION OF FOUNDATION

AS BUILT

MD-216-49PR-EG-CV-CAL-1600

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
D01	26.10.2023	AS BUILT	SHD	SGC	PRI	
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A01	29-04-2020	FIRST ISSUE	SHD	SGC	PRI	

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Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
MD-216-3000-EG-CV-RPT-1008	Geotechnical Report
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 304 :2018	Saudi Concrete Structures Code
ACI 318 :2014	Building Code Requirement of Structural Concrete

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DEFINITIONS AND ABBREVIATIONS

Employer	MA'ADEN GOLD AND BASE METALS or their representatives
EPC Contractor	Consortium of Outotec (OT) and Larsen & Toubro (L&T)
Project management Company (PMC)	BECHTEL appointed by EMPLOYER
Supplier	An entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Sub- supplier	A sub entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Shall	Is to be understood as mandatory in relation to the requirements of this Document
Should	Is to be understood as a strong recommendation to comply with the requirements of this Document
LRFD	Load and Resistance Factor Design
F.O.S	Factor of Safety

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1. INTENT

The document is prepared for Design Calculation of Column Foundation for Engine Hall Building of Power Plant.

2. SCOPE

The document is prepared for Design Calculation of Column Foundation for Engine Hall Building of Power Plant.
All the loads applied to the structure for analysis & design followed Civil -Structural Design Criteria and SBC 301:2007
Foundations mkd. F3, F4 are modelled in & analysed in Staad-Pro. . Support Reactions from Superstructure have been applied as loads to Foundations. As Seismic load is less then wind load so foundations are designed for wind load. Design of Reinforced Concrete Foundations followed Strength Design as per SBC 304:2018 & ACI 318:2008 (as supported by STAAD-Pro).
Net Safe Bearing Capacity is obtained from Geotechnical report BH28 . Approximate value of 50T/sq.m is considered

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3. Design of Engine Hall Building Column Foundation

3.1 Design of Foundation mkd. F1 (GRID F - grid Line-0,1,2,3,4,5,6)

Load Detail	FX (T)	FY (T)	FZ (T)
1 DL	-0.4	21.6	0.0
2 LLROOF	0.0	6.4	0.0
3 CL1(L+XS)	0.2	1.9	0.0
4 CL2(L-XS)	0.2	1.9	0.0
5 CL3(R+XS)	0.6	4.6	0.0
6 CL4(R-XS)	0.4	4.6	0.0
7 CL5(L+ZT)	0.2	1.9	0.0
8 CL6(L-ZT)	0.2	1.9	0.0
9 CL7(R+ZT)	0.5	4.6	0.0
10 CL8(R-ZT)	0.5	4.6	0.0
21 WL1(+X)S	0.6	-1.4	0.0
22 WL2(+X)P	1.7	-3.2	0.0
23 WL3(-X)S	-2.0	-2.9	0.0
24 WL4(-X)P	-0.9	-4.4	0.0
25 WL5(+Z)S	0.9	-2.6	0.0
26 WL6(+Z)P	1.9	-4.9	0.0
27 WL7(-Z)S	0.9	-2.6	0.0
28 WL8(-Z)P	1.9	-4.6	0.0

STRENGTH DESIGN	FOUNDATION IN F GRID		
Load Combinations	FX (T)	FY (T)	FZ (T)
1.4D	-0.56	30.20	0.00
1.4D+1.7(CL1)+0.5Lr	-0.2	36.6	0.0
1.4D+1.7(CL2)+0.5Lr	-0.2	36.6	0.0
1.4D+1.7(CL3)+0.5Lr	0.4	41.3	0.0
1.4D+1.7(CL4)+0.5Lr	0.0	41.3	0.0
1.4D+1.7(CL5)+0.5Lr	-0.2	36.6	0.0
1.4D+1.7(CL6)+0.5Lr	-0.2	36.6	0.0
1.4D+1.7(CL5)+0.5Lr	0.2	41.3	0.0
1.4D+1.7(CL6)+0.5Lr	0.2	41.3	0.0
1.2D+1.6Lr+1.0(CL1)	-0.3	38.0	0.0
1.2D+1.6Lr+1.0(CL2)	-0.3	38.0	0.0
1.2D+1.6Lr+1.0(CL3)	0.1	40.8	0.0
1.2D+1.6Lr+1.0(CL4)	-0.1	40.8	0.0
1.2D+1.6Lr+1.0(CL5)	-0.3	38.0	0.0
1.2D+1.6Lr+1.0(CL6)	-0.3	38.0	0.0
1.2D+1.6Lr+1.0(CL7)	0.0	40.8	0.0
1.2D+1.6Lr+1.0(CL8)	0.0	40.8	0.0
1.2D + 1.6WL1+1.0L+0.5Lr	0.4	26.8	0.0
1.2D + 1.6WL2+1.0L+0.5Lr	2.2	23.9	0.0
1.2D + 1.6WL3+1.0L+0.5Lr	-3.7	24.5	0.0
1.2D + 1.6WL4+1.0L+0.5Lr	-2.0	22.0	0.0
1.2D + 1.6WL5+1.0L+0.5Lr	0.9	24.9	0.0
1.2D + 1.6WL6+1.0L+0.5Lr	2.6	21.3	0.0
1.2D + 1.6WL7+1.0L+0.5Lr	0.9	24.9	0.0
1.2D + 1.6WL8+1.0L+0.5Lr	2.6	21.7	0.0
0.9D+1.6WL1	0.5	17.2	0.0
0.9D+1.6WL2	2.3	14.3	0.0
0.9D+1.6WL3	-3.6	14.8	0.0
0.9D+1.6WL4	-1.9	12.3	0.0
0.9D+1.6WL5	1.0	15.2	0.0
0.9D+1.6WL6	2.7	11.6	0.0
0.9D+1.6WL7	1.0	15.2	0.0
0.9D+1.6WL8	2.7	12.1	0.0

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DESIGN OF FOUNDATION MKD. F1

	ABOUT-X	ABOUT-Z
Size of column :	0.60m	0.60m
Size of fdn. raft :	2.00m	2.00m
Thickness of raft :	0.35 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-1.75 m	
U/S level Base Plate :	0.50 m	
Unit weight of conc :	2.5 T/m^3	
Unit weight of soil :	1.8 T/m^3	

SYMBOLS AND NOTATIONS

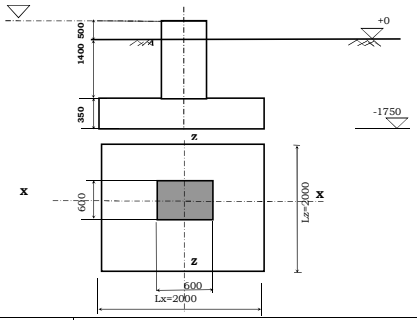
Dead Load	D
live load	LL
Roof live load	Lr
Wind Load	W
Crane Load	CL

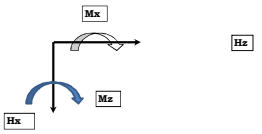
Calculation of Weight of Foundation :

Weight of raft & Col.= (2*2*0.35*2.5)+(0.6*0.6*1.9*2.5) = 5.21T

Weight of soil = ((2*2)-(0.6*0.6))*1.4*1.8 = 9.17T

U/S Base Pl Total = 14.38T





Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	75mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.12%
Mininum percentage of reinforcement at top=	0.06%

Effective depth of slab along X direction = 267mm = 0.267m

Effective depth of steel reinforcement along Z direction = 251mm = 0.251m

Clear projection along Z direction = 0.7m

Clear projection along X direction = 0.7m

WORKING STRESS DESIGN	Load at U/S Of Base Plt					Load at bottom of foundation				
Load Case	FX (T)	FY(T)	FZ(T)	Mx	Mz	V (T)	Hx (T)	MX (T-m)	Hz (T)	MZ (T-m)
DL+Lr	-0.40	27.97	0.00	0.0	0.00	42.35	-0.40	0.00	0.00	0.90
DL+CL1	-0.21	23.47	0.00	0.0	0.00	37.85	-0.21	0.00	0.00	0.48
DL+CL2	-0.21	23.47	0.00	0.0	0.00	37.85	-0.21	0.00	0.00	0.48
DL+CL3	0.17	26.21	0.00	0.0	0.00	40.59	0.17	0.00	0.00	0.39
DL+CL4	-0.05	26.22	0.00	0.0	0.00	40.60	-0.05	0.00	0.00	0.12
DL+CL5	-0.21	23.47	0.00	0.0	0.00	37.85	-0.21	0.00	0.00	0.48
DL+CL6	-0.21	23.47	0.00	0.0	0.00	37.85	-0.21	0.00	0.00	0.48
DL+CL5	0.06	26.21	0.00	0.0	0.00	40.59	0.06	0.00	0.00	0.14
DL+CL6	0.06	26.21	0.00	0.0	0.00	40.59	0.06	0.00	0.00	0.14
DL+0.75*CL1+0.75*Lr	-0.26	27.79	0.00	0.0	0.00	42.17	-0.26	0.00	0.00	0.58
DL+0.75*CL2+0.75*Lr	-0.26	27.79	0.00	0.0	0.00	42.17	-0.26	0.00	0.00	0.58
DL+0.75*CL3+0.75*Lr	0.03	29.85	0.00	0.0	0.00	44.23	0.03	0.00	0.00	0.07
DL+0.75*CL4+0.75*Lr	-0.14	29.86	0.00	0.0	0.00	44.24	-0.14	0.00	0.00	0.31
DL+0.75*CL5+0.75*Lr	-0.26	27.79	0.00	0.0	0.00	42.17	-0.26	0.00	0.00	0.58
DL+0.75*CL6+0.75*Lr	-0.26	27.79	0.00	0.0	0.00	42.17	-0.26	0.00	0.00	0.58
DL+0.75*CL7+0.75*Lr	-0.05	29.85	0.00	0.0	0.00	44.23	-0.05	0.00	0.00	0.12
DL+0.75*CL8+0.75*Lr	-0.05	29.85	0.00	0.0	0.00	44.23	-0.05	0.00	0.00	0.12
DL+0.75*Lr+0.75*WL1	0.02	25.31	0.00	0.0	0.00	39.69	0.02	0.00	0.00	0.05
DL+0.75*Lr+0.75*WL2	0.86	23.96	0.00	0.0	0.00	38.34	0.86	0.00	0.00	1.94
DL+0.75*Lr+0.75*WL3	-1.90	24.21	0.00	0.0	0.00	38.59	-1.90	0.00	0.00	4.28
DL+0.75*Lr+0.75*WL4	-1.10	23.04	0.00	0.0	0.00	37.42	-1.10	0.00	0.00	2.48
DL+0.75*Lr+0.75*WL5	0.26	24.39	0.00	0.0	0.00	38.77	0.26	0.00	0.00	0.57
DL+0.75*Lr+0.75*WL6	1.05	22.72	0.00	0.0	0.00	37.10	1.05	0.00	0.00	2.37
DL+0.75*Lr+0.75*WL7	0.26	24.39	0.00	0.0	0.00	38.77	0.26	0.00	0.00	0.57
DL+0.75*Lr+0.75*WL8	1.05	22.93	0.00	0.0	0.00	37.31	1.05	0.00	0.00	2.37
0.6*DL+WL1	0.32	11.53	0.00	0.0	0.00	25.91	0.32	0.00	0.00	0.72
0.6*DL+WL2	1.45	9.73	0.00	0.0	0.00	24.11	1.45	0.00	0.00	3.26
0.6*DL+WL3	-2.24	10.05	0.00	0.0	0.00	24.43	-2.24	0.00	0.00	5.04
0.6*DL+WL4	-1.18	8.50	0.00	0.0	0.00	22.88	-1.18	0.00	0.00	2.65
0.6*DL+WL5	0.63	10.30	0.00	0.0	0.00	24.68	0.63	0.00	0.00	1.43
0.6*DL+WL6	1.70	8.07	0.00	0.0	0.00	22.45	1.70	0.00	0.00	3.82
0.6*DL+WL7	0.63	10.30	0.00	0.0	0.00	24.68	0.63	0.00	0.00	1.43
0.6*DL+WL8	1.70	8.35	0.00	0.0	0.00	22.73	1.70	0.00	0.00	3.82

CHECK FOR OVERTURNING AND SLIDING

	Load at Foundation Base					Sliding Coeffecient, μ = 0.4			
	V (T)	Hx (T)	Mz (Tm)	Hz (T)	Mx (Tm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
Load Case									
DL+Lr	42.35	-0.40	0.90	0.00	0.00	46.91	NA	42.22	NA
DL+CL1	37.85	-0.21	0.48	0.00	0.00	79.16	NA	71.24	NA
DL+CL2	37.85	-0.21	0.48	0.00	0.00	79.16	NA	71.24	NA
DL+CL3	40.59	0.17	0.39	0.00	0.00	103.81	NA	93.43	NA
DL+CL4	40.60	-0.05	0.12	0.00	0.00	352.08	NA	316.87	NA
DL+CL5	37.85	-0.21	0.48	0.00	0.00	79.16	NA	71.24	NA
DL+CL6	37.85	-0.21	0.48	0.00	0.00	79.16	NA	71.24	NA
DL+CL5	40.59	0.06	0.14	0.00	0.00	294.55	NA	265.09	NA
DL+CL6	40.59	0.06	0.14	0.00	0.00	294.54	NA	265.09	NA
DL+0.75*CL1+0.75*Lr	42.17	-0.26	0.58	0.00	0.00	72.18	NA	64.96	NA
DL+0.75*CL2+0.75*Lr	42.17	-0.26	0.58	0.00	0.00	72.18	NA	64.96	NA
DL+0.75*CL3+0.75*Lr	44.23	0.03	0.07	0.00	0.00	655.22	NA	589.70	NA
DL+0.75*CL4+0.75*Lr	44.24	-0.14	0.31	0.00	0.00	141.70	NA	127.53	NA
DL+0.75*CL5+0.75*Lr	42.17	-0.26	0.58	0.00	0.00	72.18	NA	64.96	NA
DL+0.75*CL6+0.75*Lr	42.17	-0.26	0.58	0.00	0.00	72.18	NA	64.96	NA
DL+0.75*CL7+0.75*Lr	44.23	-0.05	0.12	0.00	0.00	361.55	NA	325.39	NA
DL+0.75*CL8+0.75*Lr	44.23	-0.05	0.12	0.00	0.00	361.54	NA	325.38	NA
DL+0.75*Lr+0.75*WL1	39.69	0.02	0.05	0.00	0.00	855.35	NA	769.81	NA
DL+0.75*Lr+0.75*WL2	38.34	0.86	1.94	0.00	0.00	19.72	NA	17.74	NA
DL+0.75*Lr+0.75*WL3	38.59	-1.90	4.28	0.00	0.00	9.02	NA	8.12	NA
DL+0.75*Lr+0.75*WL4	37.42	-1.10	2.48	0.00	0.00	15.06	NA	13.55	NA
DL+0.75*Lr+0.75*WL5	38.77	0.26	0.57	0.00	0.00	67.58	NA	60.82	NA
DL+0.75*Lr+0.75*WL6	37.10	1.05	2.37	0.00	0.00	15.67	NA	14.11	NA
DL+0.75*Lr+0.75*WL7	38.77	0.26	0.57	0.00	0.00	67.57	NA	60.82	NA
DL+0.75*Lr+0.75*WL8	37.31	1.05	2.37	0.00	0.00	15.76	NA	14.19	NA
0.6*DL+WL1	25.91	0.32	0.72	0.00	0.00	35.79	NA	32.21	NA
0.6*DL+WL2	24.11	1.45	3.26	0.00	0.00	7.41	NA	6.67	NA
0.6*DL+WL3	24.43	-2.24	5.04	0.00	0.00	4.85	NA	4.36	NA
0.6*DL+WL4	22.88	-1.18	2.65	0.00	0.00	8.63	NA	7.77	NA
0.6*DL+WL5	24.68	0.63	1.43	0.00	0.00	17.30	NA	15.57	NA
0.6*DL+WL6	22.45	1.70	3.82	0.00	0.00	5.88	NA	5.29	NA
0.6*DL+WL7	24.68	0.63	1.43	0.00	0.00	17.29	NA	15.57	NA
0.6*DL+WL8	22.73	1.70	3.82	0.00	0.00	5.95	NA	5.36	NA

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	C01	2
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

ULTIMATE STRESS DESIGN	Load at U/S Of Base Pit					Load at bottom of foundation		
Load Case	FX (T)	FY(T)	FZ(T)	M_x	M_z	V (T)	Mx (T-m)	Mz (T-m)
1.4D	-0.56	30.20	0.00	0.0	0.00	44.583	0.00	1.26
1.4D+1.7(CL1)+0.5Lr	-0.24	36.62	0.00	0.0	0.00	51.001	0.00	0.54
1.4D+1.7(CL2)+0.5Lr	-0.24	36.62	0.00	0.0	0.00	51.001	0.00	0.54
1.4D+1.7(CL3)+0.5Lr	0.42	41.28	0.00	0.0	0.00	55.656	0.00	0.94
1.4D+1.7(CL4)+0.5Lr	0.03	41.30	0.00	0.0	0.00	55.680	0.00	0.07
1.4D+1.7(CL5)+0.5Lr	-0.24	36.62	0.00	0.0	0.00	51.003	0.00	0.54
1.4D+1.7(CL6)+0.5Lr	-0.24	36.62	0.00	0.0	0.00	51.001	0.00	0.54
1.4D+1.7(CL5)+0.5Lr	0.22	41.29	0.00	0.0	0.00	55.669	0.00	0.51
1.4D+1.7(CL6)+0.5Lr	0.22	41.29	0.00	0.0	0.00	55.667	0.00	0.51
1.2D+1.6Lr+1.0(CL1)	-0.29	38.02	0.00	0.0	0.00	52.401	0.00	0.66
1.2D+1.6Lr+1.0(CL2)	-0.29	38.02	0.00	0.0	0.00	52.401	0.00	0.66
1.2D+1.6Lr+1.0(CL3)	0.09	40.76	0.00	0.0	0.00	55.140	0.00	0.21
1.2D+1.6Lr+1.0(CL4)	-0.13	40.77	0.00	0.0	0.00	55.154	0.00	0.30
1.2D+1.6Lr+1.0(CL5)	-0.29	38.02	0.00	0.0	0.00	52.402	0.00	0.66
1.2D+1.6Lr+1.0(CL6)	-0.29	38.02	0.00	0.0	0.00	52.401	0.00	0.66
1.2D+1.6Lr+1.0(CL7)	-0.02	40.77	0.00	0.0	0.00	55.147	0.00	0.04
1.2D+1.6Lr+1.0(CL8)	-0.02	40.77	0.00	0.0	0.00	55.146	0.00	0.04
1.2D + 1.6WL1+1.0L+0.5Lr	0.42	26.83	0.00	0.0	0.00	41.207	0.00	0.94
1.2D + 1.6WL2+1.0L+0.5Lr	2.22	23.95	0.00	0.0	0.00	38.327	0.00	4.99
1.2D + 1.6WL3+1.0L+0.5Lr	-3.68	24.46	0.00	0.0	0.00	38.843	0.00	8.28
1.2D + 1.6WL4+1.0L+0.5Lr	-1.98	21.97	0.00	0.0	0.00	36.353	0.00	4.46
1.2D + 1.6WL5+1.0L+0.5Lr	0.92	24.86	0.00	0.0	0.00	39.243	0.00	2.07
1.2D + 1.6WL6+1.0L+0.5Lr	2.62	21.28	0.00	0.0	0.00	35.665	0.00	5.89
1.2D + 1.6WL7+1.0L+0.5Lr	0.92	24.86	0.00	0.0	0.00	39.239	0.00	2.07
1.2D + 1.6WL8+1.0L+0.5Lr	2.62	21.74	0.00	0.0	0.00	36.119	0.00	5.89
0.9D+1.6WL1	0.54	17.15	0.00	0.0	0.00	31.534	0.00	1.21
0.9D+1.6WL2	2.34	14.27	0.00	0.0	0.00	28.654	0.00	5.26
0.9D+1.6WL3	-3.56	14.79	0.00	0.0	0.00	29.170	0.00	8.01
0.9D+1.6WL4	-1.86	12.30	0.00	0.0	0.00	26.680	0.00	4.19
0.9D+1.6WL5	1.04	15.19	0.00	0.0	0.00	29.570	0.00	2.34
0.9D+1.6WL6	2.74	11.61	0.00	0.0	0.00	25.992	0.00	6.16
0.9D+1.6WL7	1.04	15.19	0.00	0.0	0.00	29.566	0.00	2.34
0.9D+1.6WL8	2.74	12.07	0.00	0.0	0.00	26.446	0.00	6.16

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	C01	2
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Check for Bearing Pressure

Load Combination & Calculation of Base pressure

Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

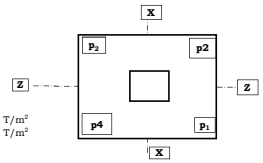
Appox. Surcharge load = 2 T/m^2

Net SBC of soil for DL, LL= 50 T/m^2 (BH28)

Gross SBC of soil for DL, LL= 54.05 T/m^2

For WL/SL case = 62.5 T/m^2

For WL/SL case = 66.55 T/m^2



Load Case	Design Load/moments			Base Pressure (T/m^2)								
	V (T)	Mz (T-m)	Mx (T-m)	V/ (Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	Remarks
DL+Lr	42.35	0.90	0.00	10.59	0.68	0.00	13.27	13.27	11.91	11.91	10.12	O.K
DL+CL1	37.85	0.48	0.00	9.46	0.36	0.00	11.82	11.82	11.10	11.10	8.67	O.K
DL+CL2	37.85	0.48	0.00	9.46	0.36	0.00	11.82	11.82	11.10	11.10	8.67	O.K
DL+CL3	40.59	0.39	0.00	10.15	0.29	0.00	12.44	12.44	11.85	11.85	9.29	O.K
DL+CL4	40.60	0.12	0.00	10.15	0.09	0.00	12.24	12.24	12.06	12.06	9.09	O.K
DL+CL5	37.85	0.48	0.00	9.46	0.36	0.00	11.82	11.82	11.10	11.10	8.67	O.K
DL+CL6	37.85	0.48	0.00	9.46	0.36	0.00	11.82	11.82	11.10	11.10	8.67	O.K
DL+CL5	40.59	0.14	0.00	10.15	0.10	0.00	12.25	12.25	12.04	12.04	9.10	O.K
DL+CL6	40.59	0.14	0.00	10.15	0.10	0.00	12.25	12.25	12.04	12.04	9.10	O.K
DL+0.75*CL1+0.75*Lr	42.17	0.58	0.00	10.54	0.44	0.00	12.98	12.98	12.11	12.11	9.83	O.K
DL+0.75*CL2+0.75*Lr	42.17	0.58	0.00	10.54	0.44	0.00	12.98	12.98	12.11	12.11	9.83	O.K
DL+0.75*CL3+0.75*Lr	44.23	0.07	0.00	11.06	0.05	0.00	13.11	13.11	13.01	13.01	9.96	O.K
DL+0.75*CL4+0.75*Lr	44.24	0.31	0.00	11.06	0.23	0.00	13.29	13.29	12.83	12.83	10.14	O.K
DL+0.75*CL5+0.75*Lr	42.17	0.58	0.00	10.54	0.44	0.00	12.98	12.98	12.11	12.11	9.83	O.K
DL+0.75*CL6+0.75*Lr	42.17	0.58	0.00	10.54	0.44	0.00	12.98	12.98	12.11	12.11	9.83	O.K
DL+0.75*CL7+0.75*Lr	44.23	0.12	0.00	11.06	0.09	0.00	13.15	13.15	12.97	12.97	10.00	O.K
DL+0.75*CL8+0.75*Lr	44.23	0.12	0.00	11.06	0.09	0.00	13.15	13.15	12.97	12.97	10.00	O.K
DL+0.75*Lr+0.75*WL1	39.69	0.05	0.00	9.92	0.03	0.00	11.96	11.96	11.89	11.89	8.81	O.K
DL+0.75*Lr+0.75*WL2	38.34	1.94	0.00	9.59	1.46	0.00	13.04	13.04	10.13	10.13	9.89	O.K
DL+0.75*Lr+0.75*WL3	38.59	4.28	0.00	9.65	3.21	0.00	14.85	14.85	8.44	8.44	11.70	O.K
DL+0.75*Lr+0.75*WL4	37.42	2.48	0.00	9.35	1.86	0.00	13.22	13.22	9.49	9.49	10.07	O.K
DL+0.75*Lr+0.75*WL5	38.77	0.57	0.00	9.69	0.43	0.00	12.12	12.12	11.26	11.26	8.97	O.K
DL+0.75*Lr+0.75*WL6	37.10	2.37	0.00	9.27	1.78	0.00	13.05	13.05	9.50	9.50	9.90	O.K
DL+0.75*Lr+0.75*WL7	38.77	0.57	0.00	9.69	0.43	0.00	12.12	12.12	11.26	11.26	8.97	O.K
DL+0.75*Lr+0.75*WL8	37.31	2.37	0.00	9.33	1.78	0.00	13.10	13.10	9.55	9.55	9.95	O.K
0.6*DL+WL1	25.91	0.72	0.00	6.48	0.54	0.00	9.02	9.02	7.93	7.93	5.87	O.K
0.6*DL+WL2	24.11	3.26	0.00	6.03	2.44	0.00	10.47	10.47	5.59	5.59	7.32	O.K
0.6*DL+WL3	24.43	5.04	0.00	6.11	3.78	0.00	11.89	11.89	4.33	4.33	8.74	O.K
0.6*DL+WL4	22.88	2.65	0.00	5.72	1.99	0.00	9.71	9.71	5.73	5.73	6.56	O.K
0.6*DL+WL5	24.68	1.43	0.00	6.17	1.07	0.00	9.24	9.24	7.10	7.10	6.09	O.K
0.6*DL+WL6	22.45	3.82	0.00	5.61	2.86	0.00	10.47	10.47	4.75	4.75	7.32	O.K
0.6*DL+WL7	24.68	1.43	0.00	6.17	1.07	0.00	9.24	9.24	7.10	7.10	6.09	O.K
0.6*DL+WL8	22.73	3.82	0.00	5.68	2.86	0.00	10.55	10.55	4.82	4.82	7.40	O.K

MA'ADEN معادن Saudi Arabian Mining Company	Outotec	LARSEN & TOUBRO	DESIGN DOCUMENT		PAGE
			Ma'aden Document Number:	MD-216-49PR-EG-CV-CAL-1600	9
			Outotec Document Number:	900707_GAU01_WG01_CED01_45001_OU100681407	of 93
			L&T Document Number:	O19045_008_C_K_24296_001	

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
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Ma'aden Revision No:	A01	C01	D01
Date	29-04-2020	01-06-2020	26.10.2023
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Outotec / L&T Revision No.:	0	C01	2
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Calculation of Design Pressure

Load Combination & Calculation of Base pressure

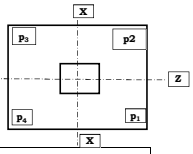
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 2 T/m^2

Net SBC of soil for DL, LL= 50 T/m^2 For WL/SL case = 62.5 T/m^2

Gross SBC of soil for DL, LL= 54.05 T/m^2 For WL/SL case = 66.55 T/m^2



Load Case	Design Load/moments V (T)	Mz (T-m)	Mx (T-m)	V/ (Lx.Lz)	6.Mz/ (Lx)².Lz	6.Mx/ Lx.(Lz)²	p1 (p_{max})	p2	p4	p3 (p_{min})	(p_{max}-γh)
1.4D	44.6	1.26	0.00	11.15	0.95	0.00	14.09	14.09	12.20	12.20	10.94
1.4D+1.7(CL1)+0.5Lr	51.0	0.54	0.00	12.75	0.41	0.00	15.16	13.16	12.34	12.34	12.64
1.4D+1.7(CL2)+0.5Lr	51.0	0.54	0.00	12.75	0.41	0.00	15.16	13.16	12.34	12.34	12.64
1.4D+1.7(CL3)+0.5Lr	55.7	0.94	0.00	13.91	0.70	0.00	16.62	14.62	13.21	13.21	14.10
1.4D+1.7(CL4)+0.5Lr	55.7	0.07	0.00	13.92	0.06	0.00	15.98	13.98	13.86	13.86	13.46
1.4D+1.7(CL5)+0.5Lr	51.0	0.54	0.00	12.75	0.41	0.00	15.16	13.16	12.34	12.34	12.64
1.4D+1.7(CL6)+0.5Lr	51.0	0.54	0.00	12.75	0.41	0.00	15.16	13.16	12.34	12.34	12.64
1.4D+1.7(CL5)+0.5Lr	55.7	0.51	0.00	13.92	0.38	0.00	16.30	14.30	13.54	13.54	13.78
1.4D+1.7(CL6)+0.5Lr	55.7	0.51	0.00	13.92	0.38	0.00	16.30	14.30	13.54	13.54	13.78
1.2D+1.6Lr+1.0(CL1)	52.4	0.66	0.00	13.10	0.49	0.00	15.59	13.59	12.61	12.61	13.07
1.2D+1.6Lr+1.0(CL2)	52.4	0.66	0.00	13.10	0.49	0.00	15.59	13.59	12.61	12.61	13.07
1.2D+1.6Lr+1.0(CL3)	55.1	0.21	0.00	13.78	0.16	0.00	15.94	13.94	13.63	13.63	13.42
1.2D+1.6Lr+1.0(CL4)	55.2	0.30	0.00	13.79	0.22	0.00	16.01	14.01	13.57	13.57	13.49
1.2D+1.6Lr+1.0(CL5)	52.4	0.66	0.00	13.10	0.49	0.00	15.59	13.59	12.61	12.61	13.07
1.2D+1.6Lr+1.0(CL6)	52.4	0.66	0.00	13.10	0.49	0.00	15.59	13.59	12.61	12.61	13.07
1.2D+1.6Lr+1.0(CL7)	55.1	0.04	0.00	13.79	0.03	0.00	15.82	13.82	13.75	13.75	13.30
1.2D+1.6Lr+1.0(CL8)	55.1	0.04	0.00	13.79	0.03	0.00	15.82	13.82	13.75	13.75	13.30
1.2D + 1.6WL1+1.0L+0.5Lr	41.2	0.94	0.00	10.30	0.71	0.00	13.01	11.01	9.60	9.60	10.49
1.2D + 1.6WL2+1.0L+0.5Lr	38.3	4.99	0.00	9.58	3.74	0.00	15.33	13.33	5.84	5.84	12.81
1.2D + 1.6WL3+1.0L+0.5Lr	38.8	8.28	0.00	9.71	6.21	0.00	17.92	15.92	3.50	3.50	15.40
1.2D + 1.6WL4+1.0L+0.5Lr	36.4	4.46	0.00	9.09	3.34	0.00	14.43	12.43	5.74	5.74	11.91
1.2D + 1.6WL5+1.0L+0.5Lr	39.2	2.07	0.00	9.81	1.55	0.00	13.36	11.36	8.26	8.26	10.84
1.2D + 1.6WL6+1.0L+0.5Lr	35.7	5.89	0.00	8.92	4.42	0.00	15.33	13.33	4.50	4.50	12.81
1.2D + 1.6WL7+1.0L+0.5Lr	39.2	2.07	0.00	9.81	1.55	0.00	13.36	11.36	8.26	8.26	10.84
1.2D + 1.6WL8+1.0L+0.5Lr	36.1	5.89	0.00	9.03	4.42	0.00	15.45	13.45	4.61	4.61	12.93
0.9D+1.6WL1	31.5	1.21	0.00	7.88	0.91	0.00	10.79	8.79	6.97	6.97	8.27
0.9D+1.6WL2	28.7	5.26	0.00	7.16	3.95	0.00	13.11	11.11	3.22	3.22	10.59
0.9D+1.6WL3	29.2	8.01	0.00	7.29	6.01	0.00	15.30	13.30	1.28	1.28	12.78
0.9D+1.6WL4	26.7	4.19	0.00	6.67	3.14	0.00	11.81	9.81	3.53	3.53	9.29
0.9D+1.6WL5	29.6	2.34	0.00	7.39	1.75	0.00	11.15	9.15	5.64	5.64	8.63
0.9D+1.6WL6	26.0	6.16	0.00	6.50	4.62	0.00	13.12	11.12	1.88	1.88	10.60
0.9D+1.6WL7	29.6	2.34	0.00	7.39	1.75	0.00	11.14	9.14	5.64	5.64	8.62
0.9D+1.6WL8	26.4	6.16	0.00	6.61	4.62	0.00	13.23	11.23	1.99	1.99	10.71

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED 001	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	C01	2
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED 001	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Calculation of Bending Moment and Shear Force in X direction :

BM_{1x} =Moment at the face of column at section 1x-1x in T-m/m.

BM_{1z} =Moment at the face of column 1z-1z in T-m/m.

SF (One way)= Oneway shear force at a distance 'd' from the face of column in T/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in T/m.

Punching perimeter= ((0.6+2*0.5*0.251)+(0.6+2*0.5*0.251))*2

= 3.404 m

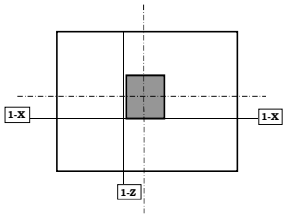
Punching Area= ((0.6+2*0.5*0.251)*(0.6+2*0.5*0.251))

= 0.72 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.35*2.5+1.4*1.8)

= 3.40 T/m^2

Area of footing, A_f= 4.00 m^2



Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 50.51 T

Strength of Concrete in shear Vc1 = 182.73 T — **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 208.49 T — **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 168.49 T — **Table 22.6.5.2_ (c)**

Design Vc= 168.49 T

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (50.51*10000)/(3404*251)

= 0.59 N/mm^2

No shear reinforcement reqd.

Load Case	p_{1x} (T/m^2)	B.M_{1x} T™	S.$F_{(oneway)}$ (T)	B.M_{1z}	S.$F_{(Oneway)}$ (T)	S.$F_{(punching)}$	V_u/b_od	Punching check (Remarks)	Allowable shear (0.75*√(fc)/6 (N/mm^2)	Actual Shear	Remarks	Mu/bd2
Ultimate Conditions												
1.4D	10.94	2.68	4.74	2.68	4.91	35.88	0.42	OK	0.74	0.196	OK	0.43
1.4D+1.7(CL1)+0.5Lr	12.64	3.10	5.47	3.10	5.68	41.46	0.49	OK	0.74	0.226	OK	0.49
1.4D+1.7(CL2)+0.5Lr	12.64	3.10	5.47	3.10	5.68	41.46	0.49	OK	0.74	0.226	OK	0.49
1.4D+1.7(CL3)+0.5Lr	14.10	3.45	6.11	3.45	6.33	46.25	0.54	OK	0.74	0.252	OK	0.55
1.4D+1.7(CL4)+0.5Lr	13.46	3.30	5.83	3.30	6.04	44.15	0.52	OK	0.74	0.241	OK	0.52
1.4D+1.7(CL5)+0.5Lr	12.64	3.10	5.47	3.10	5.68	41.46	0.49	OK	0.74	0.226	OK	0.49
1.4D+1.7(CL6)+0.5Lr	12.64	3.10	5.47	3.10	5.68	41.46	0.49	OK	0.74	0.226	OK	0.49
1.4D+1.7(CL5)+0.5Lr	13.78	3.38	5.97	3.38	6.19	45.20	0.53	OK	0.74	0.247	OK	0.54
1.4D+1.7(CL6)+0.5Lr	13.78	3.38	5.97	3.38	6.19	45.20	0.53	OK	0.74	0.247	OK	0.54
1.2D+1.6Lr+1.0(CL1)	13.07	3.20	5.66	3.20	5.87	42.87	0.5	OK	0.74	0.234	OK	0.51
1.2D+1.6Lr+1.0(CL2)	13.07	3.20	5.66	3.20	5.87	42.87	0.5	OK	0.74	0.234	OK	0.51
1.2D+1.6Lr+1.0(CL3)	13.42	3.29	5.81	3.29	6.03	44.02	0.52	OK	0.74	0.24	OK	0.52
1.2D+1.6Lr+1.0(CL4)	13.49	3.31	5.84	3.31	6.06	44.25	0.52	OK	0.74	0.241	OK	0.52
1.2D+1.6Lr+1.0(CL5)	13.07	3.20	5.66	3.20	5.87	42.87	0.5	OK	0.74	0.234	OK	0.51
1.2D+1.6Lr+1.0(CL6)	13.07	3.20	5.66	3.20	5.87	42.87	0.5	OK	0.74	0.234	OK	0.51
1.2D+1.6Lr+1.0(CL7)	13.30	3.26	5.76	3.26	5.97	43.62	0.51	OK	0.74	0.238	OK	0.52
1.2D+1.6Lr+1.0(CL8)	13.30	3.26	5.76	3.26	5.97	43.62	0.51	OK	0.74	0.238	OK	0.52
1.2D + 1.6WL1+1.0L+0.5Lr	10.49	2.57	4.54	2.57	4.71	34.41	0.4	OK	0.74	0.188	OK	0.41
1.2D + 1.6WL2+1.0L+0.5Lr	12.81	3.14	5.55	3.14	5.75	42.02	0.49	OK	0.74	0.229	OK	0.50
1.2D + 1.6WL3+1.0L+0.5Lr	15.40	3.77	6.67	3.77	6.91	50.51	0.59	OK	0.74	0.275	OK	0.60
1.2D + 1.6WL4+1.0L+0.5Lr	11.91	2.92	5.16	2.92	5.35	39.06	0.46	OK	0.74	0.213	OK	0.46
1.2D + 1.6WL5+1.0L+0.5Lr	10.84	2.66	4.69	2.66	4.87	35.56	0.42	OK	0.74	0.194	OK	0.42
1.2D + 1.6WL6+1.0L+0.5Lr	12.81	3.14	5.55	3.14	5.75	42.02	0.49	OK	0.74	0.229	OK	0.50
1.2D + 1.6WL7+1.0L+0.5Lr	10.84	2.66	4.69	2.66	4.87	35.56	0.42	OK	0.74	0.194	OK	0.42
1.2D + 1.6WL8+1.0L+0.5Lr	12.93	3.17	5.60	3.17	5.81	42.41	0.5	OK	0.74	0.231	OK	0.50
0.9D+1.6WL1	8.27	2.03	3.58	2.03	3.71	27.13	0.32	OK	0.74	0.148	OK	0.32
0.9D+1.6WL2	10.59	2.59	4.59	2.59	4.75	34.74	0.41	OK	0.74	0.189	OK	0.41
0.9D+1.6WL3	12.78	3.13	5.53	3.13	5.74	41.92	0.49	OK	0.74	0.229	OK	0.50
0.9D+1.6WL4	9.29	2.28	4.02	2.28	4.17	30.47	0.36	OK	0.74	0.166	OK	0.36
0.9D+1.6WL5	8.63	2.11	3.74	2.11	3.87	28.31	0.33	OK	0.74	0.154	OK	0.34
0.9D+1.6WL6	10.60	2.60	4.59	2.60	4.76	34.77	0.41	OK	0.74	0.19	OK	0.41
0.9D+1.6WL7	8.62	2.11	3.73	2.11	3.87	28.27	0.33	OK	0.74	0.154	OK	0.34
0.9D+1.6WL8	10.71	2.62	4.64	2.62	4.81	35.13	0.41	OK	0.74	0.192	OK	0.42

Load Case	kn (Ø=0.75)	p =	A_{st}
Ultimate Conditions			
1.4D	0.57	0.001	267
1.4D+1.7(CL1)+0.5Lr	0.66	0.002	534
1.4D+1.7(CL2)+0.5Lr	0.66	0.002	534
1.4D+1.7(CL3)+0.5Lr	0.73	0.002	534
1.4D+1.7(CL4)+0.5Lr	0.7	0.002	534
1.4D+1.7(CL5)+0.5Lr	0.66	0.002	534
1.4D+1.7(CL6)+0.5Lr	0.66	0.002	534
1.4D+1.7(CL5)+0.5Lr	0.71	0.002	534
1.4D+1.7(CL6)+0.5Lr	0.71	0.002	534
1.2D+1.6Lr+1.0(CL1)	0.68	0.002	534
1.2D+1.6Lr+1.0(CL2)	0.68	0.002	534
1.2D+1.6Lr+1.0(CL3)	0.7	0.002	534
1.2D+1.6Lr+1.0(CL4)	0.7	0.002	534
1.2D+1.6Lr+1.0(CL5)	0.68	0.002	534
1.2D+1.6Lr+1.0(CL6)	0.68	0.002	534
1.2D+1.6Lr+1.0(CL7)	0.69	0.002	534
1.2D+1.6Lr+1.0(CL8)	0.69	0.002	534
1.2D + 1.6WL1+1.0L+0.5Lr	0.54	0.001	267
1.2D + 1.6WL2+1.0L+0.5Lr	0.66	0.002	534
1.2D + 1.6WL3+1.0L+0.5Lr	0.8	0.002	534
1.2D + 1.6WL4+1.0L+0.5Lr	0.62	0.001	267
1.2D + 1.6WL5+1.0L+0.5Lr	0.56	0.001	267
1.2D + 1.6WL6+1.0L+0.5Lr	0.66	0.002	534
1.2D + 1.6WL7+1.0L+0.5Lr	0.56	0.001	267
1.2D + 1.6WL8+1.0L+0.5Lr	0.67	0.002	534
0.9D+1.6WL1	0.43	0.001	267
0.9D+1.6WL2	0.55	0.001	267
0.9D+1.6WL3	0.66	0.002	534
0.9D+1.6WL4	0.48	0.001	267
0.9D+1.6WL5	0.45	0.001	267
0.9D+1.6WL6	0.55	0.001	267
0.9D+1.6WL7	0.45	0.001	267
0.9D+1.6WL8	0.56	0.001	267

Location	Mux (T-m)	ϕ Kn = M_{u1}/bd^2	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing	Spacing provd. (mm)
At bottom	3.45	0.55	0.73	0.002	534.00	480.60	12ϕT	211.79	300	200
At top	0.00	0.00	0.00	0.0018	451.80	451.80	12ϕT	250.33	300	200

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :	At Bottom	**12ϕT**	**200**	c/c	At Top	**12ϕT**	**200**	c/c
Along Z Direction :	At Bottom	**12ϕT**	**200**	c/c	At Top	**12ϕT**	**200**	c/c

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	C01	2
Date	29-04-2020	01-06-2020	26.10.2023
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Design of Foundation mkd. F1A (GRID B)

	B grid Line-0,1,2,3,4,5,6		
LOAD DEFINATION	Fx Ton	Fy Ton	Fz Ton
1 DL	0.59	24.02	0.00
2 LLROOF	0.03	7.60	0.00
3 CL1(L+XS)	0.23	6.23	0.00
4 CL2(L-XS)	-0.52	4.60	0.00
5 CL3(R+XS)	-0.20	1.75	0.00
6 CL4(R-XS)	-0.72	0.05	0.00
7 CL5(L+ZT)	-0.14	5.42	0.00
8 CL6(L-ZT)	-0.15	5.41	0.00
9 CL7(R+ZT)	-0.46	0.91	0.00
10 CL8(R-ZT)	-0.46	0.89	0.00
21 WL1(+X)S	3.50	4.12	0.00
22 WL2(+X)P	2.90	2.77	0.00
23 WL3(-X)S	-2.41	-8.73	0.00
24 WL4(-X)P	-3.15	-10.39	0.00
25 WL5(+Z)S	-0.63	-3.03	0.00
26 WL6(+Z)P	-1.39	-5.80	0.00
27 WL7(-Z)S	-0.55	-2.89	0.00
28 WL8(-Z)P	-1.23	-4.88	0.00

STRENGTH DESIGN	GRID B 0 TO 6		
Load Combinations	FX (T)	FY (T)	FZ (T)
1.4D	0.83	33.63	0.00
1.4D+1.7(CL1)+0.5Lr	1.23	48.02	0.00
1.4D+1.7(CL2)+0.5Lr	-0.04	45.24	0.00
1.4D+1.7(CL3)+0.5Lr	0.50	40.40	0.00
1.4D+1.7(CL4)+0.5Lr	-0.38	37.51	0.00
1.4D+1.7(CL5)+0.5Lr	0.60	46.64	0.00
1.4D+1.7(CL6)+0.5Lr	0.59	46.62	0.00
1.4D+1.7(CL7)+0.5Lr	0.07	38.97	0.00
1.4D+1.7(CL8)+0.5Lr	0.06	38.94	0.00
1.2D+1.6Lr+1.0(CL1)	0.99	47.22	0.00
1.2D+1.6Lr+1.0(CL2)	0.24	45.58	0.00
1.2D+1.6Lr+1.0(CL3)	0.56	42.73	0.00
1.2D+1.6Lr+1.0(CL4)	0.04	41.03	0.00
1.2D+1.6Lr+1.0(CL5)	0.61	46.41	0.00
1.2D+1.6Lr+1.0(CL6)	0.61	46.39	0.00
1.2D+1.6Lr+1.0(CL7)	0.30	41.89	0.00
1.2D+1.6Lr+1.0(CL8)	0.30	41.87	0.00
1.2D + 1.6WL1+1.0L+0.5Lr	6.33	39.22	0.00
1.2D + 1.6WL2+1.0L+0.5Lr	5.37	37.06	0.00
1.2D + 1.6WL3+1.0L+0.5Lr	-3.13	18.66	0.00
1.2D + 1.6WL4+1.0L+0.5Lr	-4.31	16.01	0.00
1.2D + 1.6WL5+1.0L+0.5Lr	-0.28	27.78	0.00
1.2D + 1.6WL6+1.0L+0.5Lr	-1.50	23.35	0.00
1.2D + 1.6WL7+1.0L+0.5Lr	-0.15	28.01	0.00
1.2D + 1.6WL8+1.0L+0.5Lr	-1.24	24.82	0.00
0.9D+1.6WL1	6.13	28.21	0.00
0.9D+1.6WL2	5.18	26.05	0.00
0.9D+1.6WL3	-3.33	7.66	0.00
0.9D+1.6WL4	-4.51	5.00	0.00
0.9D+1.6WL5	-0.47	16.78	0.00
0.9D+1.6WL6	-1.70	12.35	0.00
0.9D+1.6WL7	-0.35	17.00	0.00
0.9D+1.6WL8	-1.43	13.82	0.00

Global Project ID	900707
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Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	29-04-2020	01-06-2020	26.10.2023
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DESIGN OF FOUNDATION MKD.F1A

	ABOUT-X	ABOUT-Z
Size of column :	0.60m	0.60m
Size of fdn. raft :	2.00m	2.00m
Thickness of raft :	0.35 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-1.75 m	
U/S level Base Plate :	0.50 m	
Unit weight of conc :	2.5 T/m^3	
Unit weight of soil :	1.8 T/m^3	

SYMBOLS AND NOTATIONS

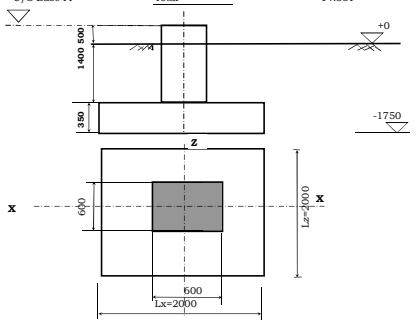
Dead Load	D
live load	LL
Roof live load	Lr
Wind Load	W
Crane Load	CL

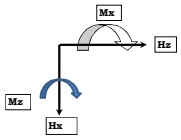
Calculation of Weight of Foundation :

Weight of raft & Col.= (2*2*0.35*2.5)+(0.6*0.6*1.9*2.5) = 5.21T

Weight of soil = ((2*2)-(0.6*0.6))*1.4*1.8 = 9.17T

U/S Base Pl Total = 14.38T





Design Input:

Characteristic strength of concrete, f'$_c$=	35 MPa
Yield stress of steel reinforcement, f$_y$=	420 MPa
Clear cover for main reinforcement bars =	75mm
Dia. of bars provd. along X direction at bottom =	20mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	20mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.12%
Mininum percentage of reinforcement at top=	0.06%

Effective depth of slab along X direction = 265mm = 0.265m

Effective depth of slab along Z direction = 245mm = 0.245m

Clear projection along Z direction = 0.7m

Clear projection along X direction = 0.7m

WORKING STRESS DESIGN	Load at U/S Of Base Plt					Load at bottom of foundation				
Load Case	FX (T)	FY(T)	FZ(T)	M_X	M_Z	V (T)	Hx (T)	MX (T-m)	Hz (T)	MZ (T-m)
DL+Lr	0.62	31.62	0.00	0.0	0.00	46.00	0.62	0.00	0.00	1.40
DL+CL1	0.82	30.25	0.00	0.0	0.00	44.63	0.82	0.00	0.00	1.84
DL+CL2	0.07	28.62	0.00	0.0	0.00	43.00	0.07	0.00	0.00	0.16
DL+CL3	0.39	25.77	0.00	0.0	0.00	40.15	0.39	0.00	0.00	0.87
DL+CL4	-0.13	24.07	0.00	0.0	0.00	38.45	-0.13	0.00	0.00	0.29
DL+CL5	0.45	29.44	0.00	0.0	0.00	43.82	0.45	0.00	0.00	1.00
DL+CL6	0.44	29.43	0.00	0.0	0.00	43.81	0.44	0.00	0.00	0.99
DL+CL5	0.13	24.93	0.00	0.0	0.00	39.31	0.13	0.00	0.00	0.30
DL+CL6	0.13	24.91	0.00	0.0	0.00	39.29	0.13	0.00	0.00	0.29
DL+0.75*CL1+0.75*Lr	0.78	34.39	0.00	0.0	0.00	48.77	0.78	0.00	0.00	1.76
DL+0.75*CL2+0.75*Lr	0.22	33.17	0.00	0.0	0.00	47.55	0.22	0.00	0.00	0.50
DL+0.75*CL3+0.75*Lr	0.46	31.03	0.00	0.0	0.00	45.41	0.46	0.00	0.00	1.04
DL+0.75*CL4+0.75*Lr	0.07	29.76	0.00	0.0	0.00	44.14	0.07	0.00	0.00	0.17
DL+0.75*CL5+0.75*Lr	0.51	33.79	0.00	0.0	0.00	48.17	0.51	0.00	0.00	1.14
DL+0.75*CL6+0.75*Lr	0.50	33.77	0.00	0.0	0.00	48.15	0.50	0.00	0.00	1.13
DL+0.75*CL7+0.75*Lr	0.27	30.40	0.00	0.0	0.00	44.78	0.27	0.00	0.00	0.61
DL+0.75*CL8+0.75*Lr	0.27	30.39	0.00	0.0	0.00	44.77	0.27	0.00	0.00	0.60
DL+0.75*Lr+0.75*WL1	3.24	32.81	0.00	0.0	0.00	47.19	3.24	0.00	0.00	7.29
DL+0.75*Lr+0.75*WL2	2.79	31.80	0.00	0.0	0.00	46.18	2.79	0.00	0.00	6.28
DL+0.75*Lr+0.75*WL3	-1.20	23.18	0.00	0.0	0.00	37.56	-1.20	0.00	0.00	2.69
DL+0.75*Lr+0.75*WL4	-1.75	21.93	0.00	0.0	0.00	36.31	-1.75	0.00	0.00	3.93
DL+0.75*Lr+0.75*WL5	0.14	27.45	0.00	0.0	0.00	41.83	0.14	0.00	0.00	0.33
DL+0.75*Lr+0.75*WL6	-0.43	25.37	0.00	0.0	0.00	39.75	-0.43	0.00	0.00	0.97
DL+0.75*Lr+0.75*WL7	0.20	27.56	0.00	0.0	0.00	41.94	0.20	0.00	0.00	0.45
DL+0.75*Lr+0.75*WL8	-0.31	26.06	0.00	0.0	0.00	40.44	-0.31	0.00	0.00	0.69
0.6*DL+WL1	3.86	18.53	0.00	0.0	0.00	32.91	3.86	0.00	0.00	8.67
0.6*DL+WL2	3.26	17.18	0.00	0.0	0.00	31.56	3.26	0.00	0.00	7.33
0.6*DL+WL3	-2.06	5.69	0.00	0.0	0.00	20.07	-2.06	0.00	0.00	4.63
0.6*DL+WL4	-2.79	4.03	0.00	0.0	0.00	18.41	-2.79	0.00	0.00	6.29
0.6*DL+WL5	-0.27	11.39	0.00	0.0	0.00	25.77	-0.27	0.00	0.00	0.61
0.6*DL+WL6	-1.04	8.62	0.00	0.0	0.00	23.00	-1.04	0.00	0.00	2.34
0.6*DL+WL7	-0.19	11.53	0.00	0.0	0.00	25.91	-0.19	0.00	0.00	0.44
0.6*DL+WL8	-0.87	9.54	0.00	0.0	0.00	23.92	-0.87	0.00	0.00	1.96

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	C01	2
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
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Check for overturning and sliding:

	Load at Foundation Base					Sliding Coeffecient, μ = 0.4			
	V (T)	Hx (T)	Mz (Tm)	Hz (T)	Mx (Tm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
Load Case									
DL+Lr	46.00	0.62	1.40	0.00	0.00	32.91	NA	29.62	NA
DL+CL1	44.63	0.82	1.84	0.00	0.00	24.26	NA	21.84	NA
DL+CL2	43.00	0.07	0.16	0.00	0.00	272.99	NA	245.69	NA
DL+CL3	40.15	0.39	0.87	0.00	0.00	45.90	NA	41.31	NA
DL+CL4	38.45	-0.13	0.29	0.00	0.00	132.72	NA	119.45	NA
DL+CL5	43.82	0.45	1.00	0.00	0.00	43.64	NA	39.28	NA
DL+CL6	43.81	0.44	0.99	0.00	0.00	44.12	NA	39.71	NA
DL+CL5	39.31	0.13	0.30	0.00	0.00	130.61	NA	117.55	NA
DL+CL6	39.29	0.13	0.29	0.00	0.00	136.96	NA	123.26	NA
DL+0.75*CL1+0.75*Lr	48.77	0.78	1.76	0.00	0.00	27.65	NA	24.88	NA
DL+0.75*CL2+0.75*Lr	47.55	0.22	0.50	0.00	0.00	94.58	NA	85.12	NA
DL+0.75*CL3+0.75*Lr	45.41	0.46	1.04	0.00	0.00	43.64	NA	39.27	NA
DL+0.75*CL4+0.75*Lr	44.14	0.07	0.17	0.00	0.00	263.75	NA	237.37	NA
DL+0.75*CL5+0.75*Lr	48.17	0.51	1.14	0.00	0.00	42.34	NA	38.10	NA
DL+0.75*CL6+0.75*Lr	48.15	0.50	1.13	0.00	0.00	42.64	NA	38.38	NA
DL+0.75*CL7+0.75*Lr	44.78	0.27	0.61	0.00	0.00	73.37	NA	66.03	NA
DL+0.75*CL8+0.75*Lr	44.77	0.27	0.60	0.00	0.00	74.64	NA	67.18	NA
DL+0.75*Lr+0.75*WL1	47.19	3.24	7.29	0.00	0.00	6.47	NA	5.83	NA
DL+0.75*Lr+0.75*WL2	46.18	2.79	6.28	0.00	0.00	7.35	NA	6.62	NA
DL+0.75*Lr+0.75*WL3	37.56	-1.20	2.69	0.00	0.00	13.97	NA	12.57	NA
DL+0.75*Lr+0.75*WL4	36.31	-1.75	3.93	0.00	0.00	9.24	NA	8.31	NA
DL+0.75*Lr+0.75*WL5	41.83	0.14	0.33	0.00	0.00	128.49	NA	115.64	NA
DL+0.75*Lr+0.75*WL6	39.75	-0.43	0.97	0.00	0.00	41.00	NA	36.90	NA
DL+0.75*Lr+0.75*WL7	41.94	0.20	0.45	0.00	0.00	92.33	NA	83.09	NA
DL+0.75*Lr+0.75*WL8	40.44	-0.31	0.69	0.00	0.00	58.87	NA	52.99	NA
0.6*DL+WL1	32.91	3.86	8.67	0.00	0.00	3.79	NA	3.42	NA
0.6*DL+WL2	31.56	3.26	7.33	0.00	0.00	4.31	NA	3.88	NA
0.6*DL+WL3	20.07	-2.06	4.63	0.00	0.00	4.34	NA	3.90	NA
0.6*DL+WL4	18.41	-2.79	6.29	0.00	0.00	2.93	NA	2.64	NA
0.6*DL+WL5	25.77	-0.27	0.61	0.00	0.00	42.26	NA	38.03	NA
0.6*DL+WL6	23.00	-1.04	2.34	0.00	0.00	9.84	NA	8.86	NA
0.6*DL+WL7	25.91	-0.19	0.44	0.00	0.00	59.12	NA	53.21	NA
0.6*DL+WL8	23.92	-0.87	1.96	0.00	0.00	12.20	NA	10.98	NA

ULTIMATE STRESS DESIGN	Load at U/S Of Base Pit					Load at bottom of foundation		
Load Case	FX (T)	FY(T)	FZ(T)	M_x	M_z	V (T)	Mx (T-m)	Mz (T-m)
1.4D	0.83	33.63	0.00	0.0	0.00	48.008	0.00	1.86
1.4D+1.7(CL1)+0.5Lr	1.23	48.02	0.00	0.0	0.00	62.401	0.00	2.76
1.4D+1.7(CL2)+0.5Lr	-0.04	45.24	0.00	0.0	0.00	59.622	0.00	0.10
1.4D+1.7(CL3)+0.5Lr	0.50	40.40	0.00	0.0	0.00	54.777	0.00	1.12
1.4D+1.7(CL4)+0.5Lr	-0.38	37.51	0.00	0.0	0.00	51.891	0.00	0.86
1.4D+1.7(CL5)+0.5Lr	0.60	46.64	0.00	0.0	0.00	61.024	0.00	1.34
1.4D+1.7(CL6)+0.5Lr	0.59	46.62	0.00	0.0	0.00	60.997	0.00	1.32
1.4D+1.7(CL7)+0.5Lr	0.07	38.97	0.00	0.0	0.00	53.349	0.00	0.15
1.4D+1.7(CL8)+0.5Lr	0.06	38.94	0.00	0.0	0.00	53.321	0.00	0.12
1.2D+1.6Lr+1.0(CL1)	0.99	47.22	0.00	0.0	0.00	61.595	0.00	2.22
1.2D+1.6Lr+1.0(CL2)	0.24	45.58	0.00	0.0	0.00	59.960	0.00	0.54
1.2D+1.6Lr+1.0(CL3)	0.56	42.73	0.00	0.0	0.00	57.110	0.00	1.25
1.2D+1.6Lr+1.0(CL4)	0.04	41.03	0.00	0.0	0.00	55.413	0.00	0.09
1.2D+1.6Lr+1.0(CL5)	0.61	46.41	0.00	0.0	0.00	60.785	0.00	1.38
1.2D+1.6Lr+1.0(CL6)	0.61	46.39	0.00	0.0	0.00	60.769	0.00	1.37
1.2D+1.6Lr+1.0(CL7)	0.30	41.89	0.00	0.0	0.00	56.270	0.00	0.68
1.2D+1.6Lr+1.0(CL8)	0.30	41.87	0.00	0.0	0.00	56.254	0.00	0.66
1.2D + 1.6WL1+1.0L+0.5Lr	6.33	39.22	0.00	0.0	0.00	53.600	0.00	14.23
1.2D + 1.6WL2+1.0L+0.5Lr	5.37	37.06	0.00	0.0	0.00	51.438	0.00	12.08
1.2D + 1.6WL3+1.0L+0.5Lr	-3.13	18.66	0.00	0.0	0.00	33.044	0.00	7.05
1.2D + 1.6WL4+1.0L+0.5Lr	-4.31	16.01	0.00	0.0	0.00	30.386	0.00	9.70
1.2D + 1.6WL5+1.0L+0.5Lr	-0.28	27.78	0.00	0.0	0.00	42.162	0.00	0.62
1.2D + 1.6WL6+1.0L+0.5Lr	-1.50	23.35	0.00	0.0	0.00	37.732	0.00	3.38
1.2D + 1.6WL7+1.0L+0.5Lr	-0.15	28.01	0.00	0.0	0.00	42.388	0.00	0.35
1.2D + 1.6WL8+1.0L+0.5Lr	-1.24	24.82	0.00	0.0	0.00	39.204	0.00	2.78
0.9D+1.6WL1	6.13	28.21	0.00	0.0	0.00	42.594	0.00	13.80
0.9D+1.6WL2	5.18	26.05	0.00	0.0	0.00	40.432	0.00	11.65
0.9D+1.6WL3	-3.33	7.66	0.00	0.0	0.00	22.038	0.00	7.49
0.9D+1.6WL4	-4.51	5.00	0.00	0.0	0.00	19.380	0.00	10.14
0.9D+1.6WL5	-0.47	16.78	0.00	0.0	0.00	31.156	0.00	1.06
0.9D+1.6WL6	-1.70	12.35	0.00	0.0	0.00	26.726	0.00	3.82
0.9D+1.6WL7	-0.35	17.00	0.00	0.0	0.00	31.382	0.00	0.78
0.9D+1.6WL8	-1.43	13.82	0.00	0.0	0.00	28.198	0.00	3.22

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	C01	2
Date	29-04-2020	01-06-2020	26.10.2023
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CHECK FOR BEARING PRESSURE

Load Combination & Calculation of Base pressure

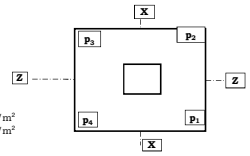
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Appox. Surcharge load = 2 T/m^2

Net SBC of soil for DL, LL= 50 T/m^2 For WL/SL case = 62.5 T/m^2

Gross SBC of soil for DL, LL= 54.05 T/m^2 For WL/SL case = 66.55 T/m^2



Load Case	Design Load/moments			Base Pressure (T/m^2)								
	V (T)	Mz (T-m)	Mx (T-m)	V/(Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	Remarks
DL+Lr	46.00	1.40	0.00	11.50	1.05	0.00	14.55	14.55	12.45	12.45	11.40	O.K
DL+CL1	44.63	1.84	0.00	11.16	1.38	0.00	14.54	14.54	11.78	11.78	11.39	O.K
DL+CL2	43.00	0.16	0.00	10.75	0.12	0.00	12.87	12.87	12.63	12.63	9.72	O.K
DL+CL3	40.15	0.87	0.00	10.04	0.66	0.00	12.69	12.69	11.38	11.38	9.54	O.K
DL+CL4	38.45	0.29	0.00	9.61	0.22	0.00	11.83	11.83	11.39	11.39	8.68	O.K
DL+CL5	43.82	1.00	0.00	10.96	0.75	0.00	13.71	13.71	12.20	12.20	10.56	O.K
DL+CL6	43.81	0.99	0.00	10.95	0.74	0.00	13.70	13.70	12.21	12.21	10.55	O.K
DL+CL5	39.31	0.30	0.00	9.83	0.23	0.00	12.05	12.05	11.60	11.60	8.90	O.K
DL+CL6	39.29	0.29	0.00	9.82	0.22	0.00	12.04	12.04	11.61	11.61	8.89	O.K
DL+0.75*CL1+0.75*Lr	48.77	1.76	0.00	12.19	1.32	0.00	15.52	15.52	12.87	12.87	12.37	O.K
DL+0.75*CL2+0.75*Lr	47.55	0.50	0.00	11.89	0.38	0.00	14.26	14.26	13.51	13.51	11.11	O.K
DL+0.75*CL3+0.75*Lr	45.41	1.04	0.00	11.35	0.78	0.00	14.13	14.13	12.57	12.57	10.98	O.K
DL+0.75*CL4+0.75*Lr	44.14	0.17	0.00	11.03	0.13	0.00	13.16	13.16	12.91	12.91	10.01	O.K
DL+0.75*CL5+0.75*Lr	48.17	1.14	0.00	12.04	0.85	0.00	14.89	14.89	13.19	13.19	11.74	O.K
DL+0.75*CL6+0.75*Lr	48.15	1.13	0.00	12.04	0.85	0.00	14.89	14.89	13.19	13.19	11.74	O.K
DL+0.75*CL7+0.75*Lr	44.78	0.61	0.00	11.19	0.46	0.00	13.65	13.65	12.74	12.74	10.50	O.K
DL+0.75*CL8+0.75*Lr	44.77	0.60	0.00	11.19	0.45	0.00	13.64	13.64	12.74	12.74	10.49	O.K
DL+0.75*Lr+0.75*WL1	47.19	7.29	0.00	11.80	5.47	0.00	19.26	19.26	8.33	8.33	16.11	O.K
DL+0.75*Lr+0.75*WL2	46.18	6.28	0.00	11.54	4.71	0.00	18.25	18.25	8.83	8.83	15.10	O.K
DL+0.75*Lr+0.75*WL3	37.56	2.69	0.00	9.39	2.02	0.00	13.41	13.41	9.37	9.37	10.26	O.K
DL+0.75*Lr+0.75*WL4	36.31	3.93	0.00	9.08	2.95	0.00	14.03	14.03	8.13	8.13	10.88	O.K
DL+0.75*Lr+0.75*WL5	41.83	0.33	0.00	10.46	0.24	0.00	12.70	12.70	12.21	12.21	9.55	O.K
DL+0.75*Lr+0.75*WL6	39.75	0.97	0.00	9.94	0.73	0.00	12.67	12.67	11.21	11.21	9.52	O.K
DL+0.75*Lr+0.75*WL7	41.94	0.45	0.00	10.48	0.34	0.00	12.82	12.82	12.14	12.14	9.67	O.K
DL+0.75*Lr+0.75*WL8	40.44	0.69	0.00	10.11	0.52	0.00	12.63	12.63	11.60	11.60	9.48	O.K
0.6*DL+WL1	32.91	8.67	0.00	8.23	6.51	0.00	16.73	16.73	3.72	3.72	13.58	O.K
0.6*DL+WL2	31.56	7.33	0.00	7.89	5.50	0.00	15.39	15.39	4.39	4.39	12.24	O.K
0.6*DL+WL3	20.07	4.63	0.00	5.02	3.47	0.00	10.49	10.49	3.55	3.55	7.34	O.K
0.6*DL+WL4	18.41	6.29	0.00	4.60	4.71	0.00	11.32	11.32	1.89	1.89	8.17	O.K
0.6*DL+WL5	25.77	0.61	0.00	6.44	0.46	0.00	8.90	8.90	7.98	7.98	5.75	O.K
0.6*DL+WL6	23.00	2.34	0.00	5.75	1.75	0.00	9.50	9.50	6.00	6.00	6.35	O.K
0.6*DL+WL7	25.91	0.44	0.00	6.48	0.33	0.00	8.81	8.81	8.15	8.15	5.66	O.K
0.6*DL+WL8	23.92	1.96	0.00	5.98	1.47	0.00	9.45	9.45	6.51	6.51	6.30	O.K

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Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

CALCULATION OF DESIGN PRESSURE

Load Combination & Calculation of Base pressure

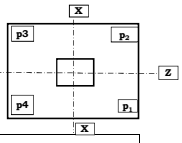
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 2 T/m^2

Net SBC of soil for DL, LL= 50 T/m^2 For WL/SL case = 62.5 T/m^2

Gross SBC of soil for DL, LL= 54.05 T/m^2 For WL/SL case = 66.55 T/m^2



Load Case	Design Load/moments			Base Pressure (T/m^2)							
	V (T)	Mz (T-m)	Mx (T-m)	V/ (Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-(h)
1.4D	48.0	1.86	0.00	12.00	1.39	0.00	15.40	15.40	12.61	12.61	12.25
1.4D+1.7(CL1)+0.5Lr	62.4	2.76	0.00	15.60	2.07	0.00	19.67	19.67	15.53	15.53	16.52
1.4D+1.7(CL2)+0.5Lr	59.6	0.10	0.00	14.91	0.07	0.00	16.98	16.98	16.83	16.83	13.83
1.4D+1.7(CL3)+0.5Lr	54.8	1.12	0.00	13.69	0.84	0.00	16.54	16.54	14.85	14.85	13.39
1.4D+1.7(CL4)+0.5Lr	51.9	0.86	0.00	12.97	0.64	0.00	15.61	15.61	14.33	14.33	12.46
1.4D+1.7(CL5)+0.5Lr	61.0	1.34	0.00	15.26	1.01	0.00	18.26	18.26	16.25	16.25	15.11
1.4D+1.7(CL6)+0.5Lr	61.0	1.32	0.00	15.25	0.99	0.00	18.24	18.24	16.26	16.26	15.09
1.4D+1.7(CL7)+0.5Lr	53.3	0.15	0.00	13.34	0.11	0.00	15.45	15.45	15.23	15.23	12.30
1.4D+1.7(CL8)+0.5Lr	53.3	0.12	0.00	13.33	0.09	0.00	15.42	15.42	15.24	15.24	12.27
1.2D+1.6Lr+1.0(CL1)	61.6	2.22	0.00	15.40	1.66	0.00	19.06	19.06	15.74	15.74	15.91
1.2D+1.6Lr+1.0(CL2)	60.0	0.54	0.00	14.99	0.40	0.00	17.39	17.39	16.59	16.59	14.24
1.2D+1.6Lr+1.0(CL3)	57.1	1.25	0.00	14.28	0.94	0.00	17.22	17.22	15.34	15.34	14.07
1.2D+1.6Lr+1.0(CL4)	55.4	0.09	0.00	13.85	0.07	0.00	15.92	15.92	15.79	15.79	12.77
1.2D+1.6Lr+1.0(CL5)	60.8	1.38	0.00	15.20	1.04	0.00	18.23	18.23	16.16	16.16	15.08
1.2D+1.6Lr+1.0(CL6)	60.8	1.37	0.00	15.19	1.03	0.00	18.22	18.22	16.16	16.16	15.07
1.2D+1.6Lr+1.0(CL7)	56.3	0.68	0.00	14.07	0.51	0.00	16.58	16.58	15.56	15.56	13.43
1.2D+1.6Lr+1.0(CL8)	56.3	0.66	0.00	14.06	0.50	0.00	16.56	16.56	15.56	15.56	13.41
1.2D + 1.6WL1+1.0L+0.5Lr	53.6	14.23	0.00	13.40	10.67	0.00	26.07	26.07	4.73	4.73	22.92
1.2D + 1.6WL2+1.0L+0.5Lr	51.4	12.08	0.00	12.86	9.06	0.00	23.92	23.92	5.80	5.80	20.77
1.2D + 1.6WL3+1.0L+0.5Lr	33.0	7.05	0.00	8.26	5.29	0.00	15.55	15.55	4.97	4.97	12.40
1.2D + 1.6WL4+1.0L+0.5Lr	30.4	9.70	0.00	7.60	7.28	0.00	16.87	16.87	2.32	2.32	13.72
1.2D + 1.6WL5+1.0L+0.5Lr	42.2	0.62	0.00	10.54	0.47	0.00	13.01	13.01	12.07	12.07	9.86
1.2D + 1.6WL6+1.0L+0.5Lr	37.7	3.38	0.00	9.43	2.54	0.00	13.97	13.97	8.89	8.89	10.82
1.2D + 1.6WL7+1.0L+0.5Lr	42.4	0.35	0.00	10.60	0.26	0.00	12.86	12.86	12.34	12.34	9.71
1.2D + 1.6WL8+1.0L+0.5Lr	39.2	2.78	0.00	9.80	2.09	0.00	13.89	13.89	9.71	9.71	10.74
0.9D+1.6WL1	42.6	13.80	0.00	10.65	10.35	0.00	23.00	23.00	2.30	2.30	19.85
0.9D+1.6WL2	40.4	11.65	0.00	10.11	8.74	0.00	20.84	20.84	3.37	3.37	17.69
0.9D+1.6WL3	22.0	7.49	0.00	5.51	5.61	0.00	13.12	13.12	1.90	1.90	9.97
0.9D+1.6WL4	19.4	10.14	0.00	4.85	7.60	0.00	14.45	14.45	-0.76	-0.76	11.30
0.9D+1.6WL5	31.2	1.06	0.00	7.79	0.79	0.00	10.58	10.58	9.00	9.00	7.43
0.9D+1.6WL6	26.7	3.82	0.00	6.68	2.86	0.00	11.55	11.55	5.82	5.82	8.40
0.9D+1.6WL7	31.4	0.78	0.00	7.85	0.59	0.00	10.43	10.43	9.26	9.26	7.28
0.9D+1.6WL8	28.2	3.22	0.00	7.05	2.41	0.00	11.46	11.46	6.64	6.64	8.31

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	29-04-2020	01-06-2020	26.10.2023
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Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	C01	2
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Prepared By	SHD	SHD	SHD
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Calculation of Bending Moment and Shear Force in X direction :

BM_{1x} =Moment at the face of column at section 1x-1x in T-m/m.

BM1z =Moment at the face of column 1z-1Z in T-m/m.

SF(oneway) = Oneway shear force at a distance 'd' from the face of column in T/m.

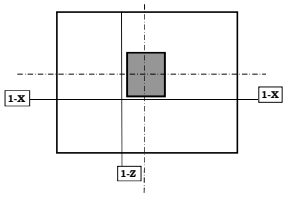
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in T/m.

Punching perimeter= ((0.6+2*0.5*0.251)+(0.6+2*0.5*0.251))*2
= 3.38 m

Punching Area= ((0.6+2*0.5*0.245)*(0.6+2*0.5*0.245))
= 0.71 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.35*2.5+1.4*1.8)
= 3.40 T/m^2

Area of footing, A_f= 4.00 m^2



Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 75.41 T

Strength of Concrete in shear Vc1 = 177.11 T **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 200.02 T **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 163.30 T **Table 22.6.5.2_ (c)**

Design Vc= 163.30 T

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ= (75.41*10000)/(3380*245)
= 0.91 N/mm^2

No shear reinforcement reqd.

Load Case	p_{1x} (T/m^2)	$B.M_{1x}$ T™	$S.F_{(oneway)}$ (T)	$B.M_{1z}$ T™	$S.F_{(oneway)}$ (T)	$S.F_{(punching)}$ (T)	V_u/b_0d	Punching check (Remarks)	Allowable shear $(0.75*\sqrt{(fc)}/6$ (N/mm^2)	Actual Shear	Remarks	Mu/bd2
Ultimate Conditions												
1.4D	12.25	3.00	5.33	3.00	5.57	40.30	0.49	OK	0.74	0.228	OK	0.50
1.4D+1.7(CL1)+0.5Lr	16.52	4.05	7.19	4.05	7.52	54.35	0.66	OK	0.74	0.307	OK	0.67
1.4D+1.7(CL2)+0.5Lr	13.83	3.39	6.02	3.39	6.29	45.50	0.55	OK	0.74	0.257	OK	0.56
1.4D+1.7(CL3)+0.5Lr	13.39	3.28	5.82	3.28	6.09	44.05	0.53	OK	0.74	0.249	OK	0.55
1.4D+1.7(CL4)+0.5Lr	12.46	3.05	5.42	3.05	5.67	40.99	0.5	OK	0.74	0.231	OK	0.51
1.4D+1.7(CL5)+0.5Lr	15.11	3.70	6.57	3.70	6.88	49.71	0.6	OK	0.74	0.281	OK	0.62
1.4D+1.7(CL6)+0.5Lr	15.09	3.70	6.56	3.70	6.87	49.65	0.6	OK	0.74	0.28	OK	0.62
1.4D+1.7(CL7)+0.5Lr	12.30	3.01	5.35	3.01	5.60	40.47	0.49	OK	0.74	0.228	OK	0.50
1.4D+1.7(CL8)+0.5Lr	12.27	3.01	5.34	3.01	5.58	40.37	0.49	OK	0.74	0.228	OK	0.50
1.2D+1.6Lr+1.0(CL1)	15.91	3.90	6.92	3.90	7.24	52.34	0.63	OK	0.74	0.295	OK	0.65
1.2D+1.6Lr+1.0(CL2)	14.24	3.49	6.19	3.49	6.48	46.85	0.57	OK	0.74	0.264	OK	0.58
1.2D+1.6Lr+1.0(CL3)	14.07	3.45	6.12	3.45	6.40	46.29	0.56	OK	0.74	0.261	OK	0.57
1.2D+1.6Lr+1.0(CL4)	12.77	3.13	5.55	3.13	5.81	42.01	0.51	OK	0.74	0.237	OK	0.52
1.2D+1.6Lr+1.0(CL5)	15.08	3.69	6.56	3.69	6.86	49.61	0.6	OK	0.74	0.28	OK	0.62
1.2D+1.6Lr+1.0(CL6)	15.07	3.69	6.56	3.69	6.86	49.58	0.6	OK	0.74	0.28	OK	0.62
1.2D+1.6Lr+1.0(CL7)	13.43	3.29	5.84	3.29	6.11	44.18	0.53	OK	0.74	0.249	OK	0.55
1.2D+1.6Lr+1.0(CL8)	13.41	3.29	5.83	3.29	6.10	44.12	0.53	OK	0.74	0.249	OK	0.55
1.2D + 1.6WL1+1.0L+0.5Lr	22.92	5.62	9.97	5.62	10.43	75.41	0.91	OK	0.74	0.426	OK	0.94
1.2D + 1.6WL2+1.0L+0.5Lr	20.77	5.09	9.03	5.09	9.45	68.33	0.83	OK	0.74	0.386	OK	0.85
1.2D + 1.6WL3+1.0L+0.5Lr	12.40	3.04	5.39	3.04	5.64	40.80	0.49	OK	0.74	0.23	OK	0.51
1.2D + 1.6WL4+1.0L+0.5Lr	13.72	3.36	5.97	3.36	6.24	45.14	0.55	OK	0.74	0.255	OK	0.56
1.2D + 1.6WL5+1.0L+0.5Lr	9.86	2.42	4.29	2.42	4.49	32.44	0.39	OK	0.74	0.183	OK	0.40
1.2D + 1.6WL6+1.0L+0.5Lr	10.82	2.65	4.71	2.65	4.92	35.60	0.43	OK	0.74	0.201	OK	0.44
1.2D + 1.6WL7+1.0L+0.5Lr	9.71	2.38	4.22	2.38	4.42	31.95	0.39	OK	0.74	0.18	OK	0.40
1.2D + 1.6WL8+1.0L+0.5Lr	10.74	2.63	4.67	2.63	4.89	35.33	0.43	OK	0.74	0.199	OK	0.44
0.9D+1.6WL1	19.85	4.86	8.63	4.86	9.03	65.31	0.79	OK	0.74	0.369	OK	0.81
0.9D+1.6WL2	17.69	4.33	7.70	4.33	8.05	58.20	0.7	OK	0.74	0.329	OK	0.72
0.9D+1.6WL3	9.97	2.44	4.34	2.44	4.54	32.80	0.4	OK	0.74	0.185	OK	0.41
0.9D+1.6WL4	11.30	2.77	4.92	2.77	5.14	37.18	0.45	OK	0.74	0.21	OK	0.46
0.9D+1.6WL5	7.43	1.82	3.23	1.82	3.38	24.44	0.3	OK	0.74	0.138	OK	0.30
0.9D+1.6WL6	8.40	2.06	3.65	2.06	3.82	27.64	0.33	OK	0.74	0.156	OK	0.34
0.9D+1.6WL7	7.28	1.78	3.17	1.78	3.31	23.95	0.29	OK	0.74	0.135	OK	0.30
0.9D+1.6WL8	8.31	2.04	3.61	2.04	3.78	27.34	0.33	OK	0.74	0.154	OK	0.34

Load Case	kn (Ø=0.75)	p =	A_{st}
Ultimate Conditions			
1.4D	0.67	0.002	530
1.4D+1.7(CL1)+0.5Lr	0.9	0.002	530
1.4D+1.7(CL2)+0.5Lr	0.75	0.002	530
1.4D+1.7(CL3)+0.5Lr	0.73	0.002	530
1.4D+1.7(CL4)+0.5Lr	0.68	0.002	530
1.4D+1.7(CL5)+0.5Lr	0.82	0.002	530
1.4D+1.7(CL6)+0.5Lr	0.82	0.002	530
1.4D+1.7(CL7)+0.5Lr	0.67	0.002	530
1.4D+1.7(CL8)+0.5Lr	0.67	0.002	530
1.2D+1.6Lr+1.0(CL1)	0.87	0.002	530
1.2D+1.6Lr+1.0(CL2)	0.77	0.002	530
1.2D+1.6Lr+1.0(CL3)	0.77	0.002	530
1.2D+1.6Lr+1.0(CL4)	0.69	0.002	530
1.2D+1.6Lr+1.0(CL5)	0.82	0.002	530
1.2D+1.6Lr+1.0(CL6)	0.82	0.002	530
1.2D+1.6Lr+1.0(CL7)	0.73	0.002	530
1.2D+1.6Lr+1.0(CL8)	0.73	0.002	530
1.2D + 1.6WL1+1.0L+0.5Lr	1.25	0.003	795
1.2D + 1.6WL2+1.0L+0.5Lr	1.13	0.003	795
1.2D + 1.6WL3+1.0L+0.5Lr	0.67	0.002	530
1.2D + 1.6WL4+1.0L+0.5Lr	0.75	0.002	530
1.2D + 1.6WL5+1.0L+0.5Lr	0.54	0.001	265
1.2D + 1.6WL6+1.0L+0.5Lr	0.59	0.001	265
1.2D + 1.6WL7+1.0L+0.5Lr	0.53	0.001	265
1.2D + 1.6WL8+1.0L+0.5Lr	0.58	0.001	265
0.9D+1.6WL1	1.08	0.003	795
0.9D+1.6WL2	0.96	0.002	530
0.9D+1.6WL3	0.54	0.001	265
0.9D+1.6WL4	0.61	0.001	265
0.9D+1.6WL5	0.4	0.001	265
0.9D+1.6WL6	0.46	0.001	265
0.9D+1.6WL7	0.4	0.001	265
0.9D+1.6WL8	0.45	0.001	265

Location	Mux (T-m)	ϕ Kn = M_{ux}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)
At bottom	4.05	0.67	0.90	0.002	795.00	477.00	16ϕT	252.91	300	200
At top	0.00	0.00	0.00	0.0018	441.00	441.00	12ϕT	256.46	300	200

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :	At Bottom	**16ϕT**	**200**	c/c	At Top	**12ϕT**	**200**	c/c
Along Z Direction :	At Bottom	**16ϕT**	**200**	c/c	At Top	**12ϕT**	**200**	c/c

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

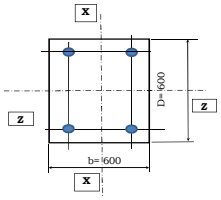
Ma'aden Revision No:	A01	C01	D01
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Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	C01	2
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Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
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DESIGN OF COLUMN MKD C1 (TYPICAL FOR F1 & F2)

Input:	
Clear cover to reinf.=	75mm
Dia. of main reinf.=	25mm
Dia. of link=	10mm
Effective cover =	97.5mm
Grade of Concrete (f_c') =	M35
Grade of Steel (f_y) =	Fy420
Grade of Reinf(links).=	Fy420
Min. Steel Ratio	0.50%
Max. Steel Ratio	4.00%
Pedestal Height	1900

Table 20.6.1.3.1



For Compression-controlled sections
Strength reduction factor ϕ= 0.65 Table 21.2.2

Unsupported length of column about X (l_{ux}) = 1900.00
Unsupported length of column about Z (l_{uz}) = 1900.00

For Columns not braced against sidesway,

Slenderness ratio, $K\,l_{ux} / r_x$ = 21.95 < 22 SECTION 6.2.5a
$K\,l_{uz} / r_z$ = 21.95 < 22 SECTION 6.2.5a

Where, Effective length factor K= 2
Radius of gyration r_x= 173
Radius of gyration r_z= 173

γ_y =	0.71
γ_z =	0.71

Load Combination	Factored Load/Moment (T)			Nominal Load/Moment			Kn = Pn/f_c'bD (N/mm^2)
	Hx	V	Hz	Mz (Tm)	V (T)	Mx (Tm)	
1.4D	0.83	33.63	0.00	2.4	51.7	0.0	#REF!
1.4D+1.7(CL1)+0.5Lr	1.23	48.02	0.00	3.6	73.9	0.0	#REF!
1.4D+1.7(CL2)+0.5Lr	-0.04	45.24	0.00	0.1	69.6	0.0	#REF!
1.4D+1.7(CL3)+0.5Lr	0.50	40.40	0.00	1.5	62.1	0.0	#REF!
1.4D+1.7(CL4)+0.5Lr	-0.38	37.51	0.00	1.1	57.7	0.0	#REF!
1.4D+1.7(CL5)+0.5Lr	0.60	46.64	0.00	1.7	71.8	0.0	#REF!
1.4D+1.7(CL6)+0.5Lr	0.59	46.62	0.00	1.7	71.7	0.0	#REF!
1.4D+1.7(CL7)+0.5Lr	0.07	38.97	0.00	0.2	60.0	0.0	#REF!
1.4D+1.7(CL8)+0.5Lr	0.06	38.94	0.00	0.2	59.9	0.0	#REF!
1.2D+1.6Lr+1.0(CL1)	0.99	47.22	0.00	2.9	72.6	0.0	#REF!
1.2D+1.6Lr+1.0(CL2)	0.24	45.58	0.00	0.7	70.1	0.0	#REF!
1.2D+1.6Lr+1.0(CL3)	0.56	42.73	0.00	1.6	65.7	0.0	#REF!
1.2D+1.6Lr+1.0(CL4)	0.04	41.03	0.00	0.1	63.1	0.0	#REF!
1.2D+1.6Lr+1.0(CL5)	0.61	46.41	0.00	1.8	71.4	0.0	#REF!
1.2D+1.6Lr+1.0(CL6)	0.61	46.39	0.00	1.8	71.4	0.0	#REF!
1.2D+1.6Lr+1.0(CL7)	0.30	41.89	0.00	0.9	64.4	0.0	#REF!
1.2D+1.6Lr+1.0(CL8)	0.30	41.87	0.00	0.9	64.4	0.0	#REF!
1.2D + 1.6WL1+1.0L+0.5Lr	6.33	39.22	0.00	18.5	60.3	0.0	#REF!
1.2D + 1.6WL2+1.0L+0.5Lr	5.37	37.06	0.00	15.7	57.0	0.0	#REF!
1.2D + 1.6WL3+1.0L+0.5Lr	-3.13	18.66	0.00	9.2	28.7	0.0	#REF!
1.2D + 1.6WL4+1.0L+0.5Lr	-4.31	16.01	0.00	12.6	24.6	0.0	#REF!
1.2D + 1.6WL5+1.0L+0.5Lr	-0.28	27.78	0.00	0.8	42.7	0.0	#REF!
1.2D + 1.6WL6+1.0L+0.5Lr	-1.50	23.35	0.00	4.4	35.9	0.0	#REF!
1.2D + 1.6WL7+1.0L+0.5Lr	-0.15	28.01	0.00	0.5	43.1	0.0	#REF!
1.2D + 1.6WL8+1.0L+0.5Lr	-1.24	24.82	0.00	3.6	38.2	0.0	#REF!
0.9D+1.6WL1	6.13	28.21	0.00	17.9	43.4	0.0	#REF!
0.9D+1.6WL2	5.18	26.05	0.00	15.1	40.1	0.0	#REF!
0.9D+1.6WL3	-3.33	7.66	0.00	9.7	11.8	0.0	#REF!
0.9D+1.6WL4	-4.51	5.00	0.00	13.2	7.7	0.0	#REF!
0.9D+1.6WL5	-0.47	16.78	0.00	1.4	25.8	0.0	#REF!
0.9D+1.6WL6	-1.70	12.35	0.00	5.0	19.0	0.0	#REF!
0.9D+1.6WL7	-0.35	17.00	0.00	1.0	26.2	0.0	#REF!
0.9D+1.6WL8	-1.43	13.82	0.00	4.2	21.3	0.0	#REF!

REINFORCEMENT

Provide,

4-Nos	25 mm Dia	1964.29 Sq mm	(At Corner)	A_{sc} pvd. = 3928.57 Sq mm
2-Nos	25 mm Dia	982.14 Sq mm	(At middle)	
2-Nos	25 mm Dia	982.14 Sq mm	(At middle)	Total A_{sc} = 1.0 %

β = P_n / P_0 (range is 0.55 to 0.70) Where, P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc} = 1224.3125 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.7		M_{nox}	M_{nox}	β	α_n = log(0.5)/ log(β)	Interaction ratio	Remarks
	R_{nx}=M_{nox} /f_c'b^{2}D	R_{nx}=M_{nox} /f_c'bD^2						
1.4D	0.06	0.06	#REF!	#REF!	0.55	1.16	#REF!	#REF!
1.4D+1.7(CL1)+0.5Lr	0.07	0.07	#REF!	#REF!	0.55	1.16	#REF!	#REF!
1.4D+1.7(CL2)+0.5Lr	0.06	0.06	#REF!	#REF!	0.55	1.16	#REF!	#REF!
1.4D+1.7(CL3)+0.5Lr	0.06	0.06	#REF!	#REF!	0.55	1.16	#REF!	#REF!
1.4D+1.7(CL4)+0.5Lr	0.06	0.06	#REF!	#REF!	0.55	1.16	#REF!	#REF!
1.4D+1.7(CL5)+0.5Lr	0.06	0.06	#REF!	#REF!	0.55	1.16	#REF!	#REF!
1.4D+1.7(CL6)+0.5Lr	0.06	0.06	#REF!	#REF!	0.55	1.16	#REF!	#REF!
1.4D+1.7(CL5)+0.5Lr	0.06	0.06	#REF!	#REF!	0.55	1.16	#REF!	#REF!
1.4D+1.7(CL6)+0.5Lr	0.06	0.06	#REF!	#REF!	0.55	1.16	#REF!	#REF!
1.2D+1.6Lr+1.0(CL1)	0.06	0.06	#REF!	#REF!	0.55	1.16	#REF!	#REF!
1.2D+1.6Lr+1.0(CL2)	0.06	0.06	#REF!	#REF!	0.55	1.16	#REF!	#REF!
1.2D+1.6Lr+1.0(CL3)	0.06	0.06	#REF!	#REF!	0.55	1.16	#REF!	#REF!
1.2D+1.6Lr+1.0(CL4)	0.06	0.06	#REF!	#REF!	0.55	1.16	#REF!	#REF!
1.2D+1.6Lr+1.0(CL5)	0.06	0.06	#REF!	#REF!	0.55	1.16	#REF!	#REF!
1.2D+1.6Lr+1.0(CL6)	0.06	0.06	#REF!	#REF!	0.55	1.16	#REF!	#REF!
1.2D+1.6Lr+1.0(CL5)	0.06	0.06	#REF!	#REF!	0.55	1.16	#REF!	#REF!
1.2D+1.6Lr+1.0(CL6)	0.06	0.06	#REF!	#REF!	0.55	1.16	#REF!	#REF!
1.2D + 1.6WL1+1.0L+0.5Lr	0.06	0.06	#REF!	#REF!	0.55	1.16	#REF!	#REF!
1.2D + 1.6WL2+1.0L+0.5Lr	0.06	0.06	#REF!	#REF!	0.55	1.16	#REF!	#REF!
1.2D + 1.6WL3+1.0L+0.5Lr	0.06	0.06	#REF!	#REF!	0.55	1.16	#REF!	#REF!
1.2D + 1.6WL4+1.0L+0.5Lr	0.06	0.06	#REF!	#REF!	0.55	1.16	#REF!	#REF!
1.2D + 1.6WL5+1.0L+0.5Lr	0.06	0.06	#REF!	#REF!	0.55	1.16	#REF!	#REF!
1.2D + 1.6WL6+1.0L+0.5Lr	0.06	0.06	#REF!	#REF!	0.55	1.16	#REF!	#REF!
1.2D + 1.6WL7+1.0L+0.5Lr	0.06	0.06	#REF!	#REF!	0.55	1.16	#REF!	#REF!
1.2D + 1.6WL8+1.0L+0.5Lr	0.06	0.06	#REF!	#REF!	0.55	1.16	#REF!	#REF!
0.9D+1.6WL1	0.06	0.06	#REF!	#REF!	0.55	1.16	#REF!	#REF!
0.9D+1.6WL2	0.06	0.06	#REF!	#REF!	0.55	1.16	#REF!	#REF!
0.9D+1.6WL3	0.06	0.06	#REF!	#REF!	0.55	1.16	#REF!	#REF!
0.9D+1.6WL4	0.06	0.06	#REF!	#REF!	0.55	1.16	#REF!	#REF!
0.9D+1.6WL5	0.05	0.05	#REF!	#REF!	0.55	1.16	#REF!	#REF!
0.9D+1.6WL6	0.05	0.05	#REF!	#REF!	0.55	1.16	#REF!	#REF!
0.9D+1.6WL7	0.05	0.05	#REF!	#REF!	0.55	1.16	#REF!	#REF!
0.9D+1.6WL8	0.05	0.05	#REF!	#REF!	0.55	1.16	#REF!	#REF!

Links, **10mm dia @ 300 C/C**

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	C01	2
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

3.2 Design of Foundation mkd. F2 (GRID A (0 to 6) & GABLE END COLUMN mkd. C0,C8,D0,D8,E0,E8)

	GRID A FROM 0 TO 6		
Load Cases	FX (T)	FY (T)	FZ (T)
1 DL	-0.1	19.6	0.0
2 LLROOF	0.0	1.2	0.0
3 CL1(L+XS)	-0.1	-1.6	0.0
4 CL2(L-XS)	-0.1	0.0	0.0
5 CL3(R+XS)	0.0	0.1	0.0
6 CL4(R-XS)	0.0	1.8	0.0
7 CL5(L+ZT)	-0.1	-0.8	0.0
8 CL6(L-ZT)	-0.1	-0.8	0.0
9 CL7(R+ZT)	0.0	1.0	0.0
10 CL8(R-ZT)	0.0	1.0	0.0
21 WL1(+X)S	0.7	-7.6	0.0
22 WL2(+X)P	0.4	-8.4	0.0
23 WL3(-X)S	-0.2	6.4	0.0
24 WL4(-X)P	-0.6	6.3	0.0
25 WL5(+Z)S	-0.3	-0.5	0.0
26 WL6(+Z)P	-0.7	-0.7	0.0
27 WL7(-Z)S	-0.3	-0.7	0.0
28 WL8(-Z)P	-0.7	-1.3	0.0

		Horizontal	Vertical	Horizontal
	Load cases	Fx ton	Fy ton	Fz ton
All gable post (C0,D0,E0,C8,D8,E8)	1 DL	0	4.5	0
	2 LLROOF	0	0	0
	3 CL1(L+XS)	0	0	0
	4 CL2(L-XS)	0	0	0
	5 CL3(R+XS)	0	0	0
	6 CL4(R-XS)	0	0	0
	7 CL5(L+ZT)	0	0	0
	8 CL6(L-ZT)	0	0	0
	9 CL5(R+ZT)	0	0	0
	10 CL6(R-ZT)	0	0	0
	21 WL1(+X)S	0	0	-0.76
	22 WL2(+X)P	0	0	-1.82
	23 WL3(-X)S	0	0	-0.76
	24 WL4(-X)P	0	0	-1.82
	25 WL5(+Z)S	0	0	1.82
	26 WL6(+Z)P	0	0	0.88
	27 WL7(-Z)S	0	0	-0.59
	28 WL8(-Z)P	0	0	-1.76

Strength Design	GRID A FROM 0 TO 6		
Load Combinations	FX (T)	FY (T)	FZ (T)
1.4D	-0.14	27.49	0.00
1.4D+1.7(CL1)+0.5Lr	-0.24	25.34	0.00
1.4D+1.7(CL2)+0.5Lr	-0.24	28.12	0.00
1.4D+1.7(CL3)+0.5Lr	-0.19	28.32	0.00
1.4D+1.7(CL4)+0.5Lr	-0.18	31.19	0.00
1.4D+1.7(CL5)+0.5Lr	-0.24	26.72	0.00
1.4D+1.7(CL6)+0.5Lr	-0.24	26.75	0.00
1.4D+1.7(CL5)+0.5Lr	-0.18	29.74	0.00
1.4D+1.7(CL6)+0.5Lr	-0.18	29.77	0.00
1.2D+1.6Lr+1.0(CL1)	-0.20	23.87	0.00
1.2D+1.6Lr+1.0(CL2)	-0.20	25.50	0.00
1.2D+1.6Lr+1.0(CL3)	-0.17	25.62	0.00
1.2D+1.6Lr+1.0(CL4)	-0.17	27.31	0.00
1.2D+1.6Lr+1.0(CL5)	-0.20	24.67	0.00
1.2D+1.6Lr+1.0(CL6)	-0.20	24.69	0.00
1.2D+1.6Lr+1.0(CL7)	-0.17	26.45	0.00
1.2D+1.6Lr+1.0(CL8)	-0.17	26.47	0.00
1.2D + 1.6WL1+1.0L+0.5Lr	1.04	11.97	0.00
1.2D + 1.6WL2+1.0L+0.5Lr	0.44	10.78	0.00
1.2D + 1.6WL3+1.0L+0.5Lr	-0.43	34.34	0.00
1.2D + 1.6WL4+1.0L+0.5Lr	-1.07	34.24	0.00
1.2D + 1.6WL5+1.0L+0.5Lr	-0.61	23.41	0.00
1.2D + 1.6WL6+1.0L+0.5Lr	-1.20	22.97	0.00
1.2D + 1.6WL7+1.0L+0.5Lr	-0.61	23.01	0.00
1.2D + 1.6WL8+1.0L+0.5Lr	-1.21	22.10	0.00
0.9D+1.6WL1	1.08	5.48	0.00
0.9D+1.6WL2	0.48	4.29	0.00
0.9D+1.6WL3	-0.39	27.85	0.00
0.9D+1.6WL4	-1.03	27.75	0.00
0.9D+1.6WL5	-0.57	16.92	0.00
0.9D+1.6WL6	-1.17	16.48	0.00
0.9D+1.6WL7	-0.57	16.52	0.00
0.9D+1.6WL8	-1.17	15.61	0.00

Global Project ID	900707
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Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	C01	2
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

DESIGN OF FOUNDATION MKD. F2

	ABOUT-X	ABOUT-Z
Size of column :	0.60m	0.60m
Size of fdn. raft :	1.75m	1.75m
Thickness of raft :	0.35 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-1.75 m	
U/S level Base Plate :	0.50 m	
Unit weight of conc :	2.5 T/m^3	
Unit weight of soil :	1.8 T/m^3	

SYMBOLS AND NOTATIONS

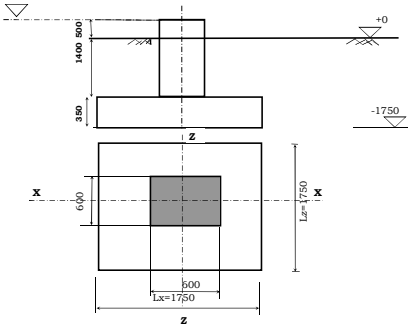
Dead Load	D
live load	LL
Roof live load	Lr
Wind Load	W
Crane Load	CL

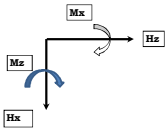
Calculation of Weight of Foundation :

Weight of raft & Col.= (1.75*1.75*0.35*2.5)+(0.6*0.6*1.9*2.5) = 4.39T

Weight of soil = ((1.75*1.75)-(0.6*0.6))*1.4*1.8 = 6.81T

U/S Base Pl Total = 11.20T





Design Input:	
Characteristic strength of concrete, f'$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	75mm
Dia. of bars provd. along X direction at bottom =	20mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	20mm
Dia. of bars provided along Z direction at top=	12mm
Minimum percentage of reinforcement at bottom=	0.12%
Mininum percentage of reinforcement at top=	0.06%

Effective depth of slab along X direction = 265mm = 0.265m

Effective depth of slab along Z direction = 245mm = 0.245m

Clear projection along Z direction = 0.575m

Clear projection along X direction = 0.575m

WORKING STRESS DESIGN	Load at U/S Of Base Plt					Load at bottom of foundation				
Load Case	FX (T)	FY(T)	FZ(T)	M_X	M_Z	V (T)	Hx (T)	MX (T-m)	Hz (T)	MZ (T-m)
DL+Lr	-0.12	20.84	0.00	0.0	0.00	32.04	-0.12	0.00	0.00	0.27
DL+CL1	-0.15	18.02	0.00	0.0	0.00	29.22	-0.15	0.00	0.00	0.34
DL+CL2	-0.15	19.66	0.00	0.0	0.00	30.86	-0.15	0.00	0.00	0.35
DL+CL3	-0.12	19.77	0.00	0.0	0.00	30.97	-0.12	0.00	0.00	0.28
DL+CL4	-0.12	21.46	0.00	0.0	0.00	32.66	-0.12	0.00	0.00	0.27
DL+CL5	-0.15	18.83	0.00	0.0	0.00	30.03	-0.15	0.00	0.00	0.34
DL+CL6	-0.15	18.85	0.00	0.0	0.00	30.05	-0.15	0.00	0.00	0.34
DL+CL7	-0.12	20.61	0.00	0.0	0.00	31.81	-0.12	0.00	0.00	0.27
DL+CL8	-0.12	20.63	0.00	0.0	0.00	31.83	-0.12	0.00	0.00	0.27
DL+0.75*CL1+0.75*Lr	-0.15	19.32	0.00	0.0	0.00	30.52	-0.15	0.00	0.00	0.34
DL+0.75*CL2+0.75*Lr	-0.15	20.55	0.00	0.0	0.00	31.75	-0.15	0.00	0.00	0.35
DL+0.75*CL3+0.75*Lr	-0.13	20.64	0.00	0.0	0.00	31.84	-0.13	0.00	0.00	0.29
DL+0.75*CL4+0.75*Lr	-0.13	21.90	0.00	0.0	0.00	33.10	-0.13	0.00	0.00	0.29
DL+0.75*CL5+0.75*Lr	-0.15	19.93	0.00	0.0	0.00	31.13	-0.15	0.00	0.00	0.34
DL+0.75*CL6+0.75*Lr	-0.15	19.94	0.00	0.0	0.00	31.14	-0.15	0.00	0.00	0.34
DL+0.75*CL7+0.75*Lr	-0.13	21.26	0.00	0.0	0.00	32.46	-0.13	0.00	0.00	0.29
DL+0.75*CL8+0.75*Lr	-0.13	21.28	0.00	0.0	0.00	32.48	-0.13	0.00	0.00	0.29
DL+0.75*Lr+0.75*WL1	0.43	14.82	0.00	0.0	0.00	26.02	0.43	0.00	0.00	0.98
DL+0.75*Lr+0.75*WL2	0.15	14.26	0.00	0.0	0.00	25.46	0.15	0.00	0.00	0.34
DL+0.75*Lr+0.75*WL3	-0.26	25.31	0.00	0.0	0.00	36.51	-0.26	0.00	0.00	0.58
DL+0.75*Lr+0.75*WL4	-0.55	25.26	0.00	0.0	0.00	36.46	-0.55	0.00	0.00	1.25
DL+0.75*Lr+0.75*WL5	-0.34	20.18	0.00	0.0	0.00	31.38	-0.34	0.00	0.00	0.77
DL+0.75*Lr+0.75*WL6	-0.62	19.98	0.00	0.0	0.00	31.18	-0.62	0.00	0.00	1.39
DL+0.75*Lr+0.75*WL7	-0.34	20.00	0.00	0.0	0.00	31.20	-0.34	0.00	0.00	0.77
DL+0.75*Lr+0.75*WL8	-0.62	19.57	0.00	0.0	0.00	30.77	-0.62	0.00	0.00	1.39
0.6*DL+WL1	0.67	4.16	0.00	0.0	0.00	15.36	0.67	0.00	0.00	1.51
0.6*DL+WL2	0.30	3.42	0.00	0.0	0.00	14.62	0.30	0.00	0.00	0.66
0.6*DL+WL3	-0.25	18.14	0.00	0.0	0.00	29.34	-0.25	0.00	0.00	0.56
0.6*DL+WL4	-0.65	18.08	0.00	0.0	0.00	29.28	-0.65	0.00	0.00	1.46
0.6*DL+WL5	-0.36	11.31	0.00	0.0	0.00	22.51	-0.36	0.00	0.00	0.81
0.6*DL+WL6	-0.73	11.03	0.00	0.0	0.00	22.23	-0.73	0.00	0.00	1.65
0.6*DL+WL7	-0.36	11.06	0.00	0.0	0.00	22.26	-0.36	0.00	0.00	0.81
0.6*DL+WL8	-0.73	10.49	0.00	0.0	0.00	21.69	-0.73	0.00	0.00	1.65

Check for overturning and sliding:

	Load at Foundation Base					Sliding Coeffecient, μ = 0.4			
	V (T)	Hx (T)	Mz (Tm)	Hz (T)	Mx (Tm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
Load Case									
DL+Lr	32.04	-0.12	0.27	0.00	0.00	104.91	NA	107.91	NA
DL+CL1	29.22	-0.15	0.34	0.00	0.00	74.51	NA	76.64	NA
DL+CL2	30.86	-0.15	0.35	0.00	0.00	78.04	NA	80.27	NA
DL+CL3	30.97	-0.12	0.28	0.00	0.00	98.32	NA	101.13	NA
DL+CL4	32.66	-0.12	0.27	0.00	0.00	104.75	NA	107.74	NA
DL+CL5	30.03	-0.15	0.34	0.00	0.00	76.57	NA	78.76	NA
DL+CL6	30.05	-0.15	0.34	0.00	0.00	76.62	NA	78.81	NA
DL+CL7	31.81	-0.12	0.27	0.00	0.00	102.01	NA	104.93	NA
DL+CL8	31.83	-0.12	0.27	0.00	0.00	102.07	NA	104.99	NA
DL+0.75*CL1+0.75*Lr	30.52	-0.15	0.34	0.00	0.00	77.68	NA	79.90	NA
DL+0.75*CL2+0.75*Lr	31.75	-0.15	0.35	0.00	0.00	80.31	NA	82.60	NA
DL+0.75*CL3+0.75*Lr	31.84	-0.13	0.29	0.00	0.00	95.01	NA	97.73	NA
DL+0.75*CL4+0.75*Lr	33.10	-0.13	0.29	0.00	0.00	99.51	NA	102.35	NA
DL+0.75*CL5+0.75*Lr	31.13	-0.15	0.34	0.00	0.00	79.22	NA	81.48	NA
DL+0.75*CL6+0.75*Lr	31.14	-0.15	0.34	0.00	0.00	79.26	NA	81.52	NA
DL+0.75*CL7+0.75*Lr	32.46	-0.13	0.29	0.00	0.00	97.58	NA	100.37	NA
DL+0.75*CL8+0.75*Lr	32.48	-0.13	0.29	0.00	0.00	97.62	NA	100.41	NA
DL+0.75*Lr+0.75*WL1	26.02	0.43	0.98	0.00	0.00	23.31	NA	23.98	NA
DL+0.75*Lr+0.75*WL2	25.46	0.15	0.34	0.00	0.00	64.80	NA	66.65	NA
DL+0.75*Lr+0.75*WL3	36.51	-0.26	0.58	0.00	0.00	55.47	NA	57.05	NA
DL+0.75*Lr+0.75*WL4	36.46	-0.55	1.25	0.00	0.00	25.59	NA	26.32	NA
DL+0.75*Lr+0.75*WL5	31.38	-0.34	0.77	0.00	0.00	35.86	NA	36.88	NA
DL+0.75*Lr+0.75*WL6	31.18	-0.62	1.39	0.00	0.00	19.62	NA	20.18	NA
DL+0.75*Lr+0.75*WL7	31.20	-0.34	0.77	0.00	0.00	35.65	NA	36.67	NA
DL+0.75*Lr+0.75*WL8	30.77	-0.62	1.39	0.00	0.00	19.31	NA	19.86	NA
0.6*DL+WL1	15.36	0.67	1.51	0.00	0.00	8.91	NA	9.16	NA
0.6*DL+WL2	14.62	0.30	0.66	0.00	0.00	19.24	NA	19.79	NA
0.6*DL+WL3	29.34	-0.25	0.56	0.00	0.00	45.73	NA	47.04	NA
0.6*DL+WL4	29.28	-0.65	1.46	0.00	0.00	17.60	NA	18.10	NA
0.6*DL+WL5	22.51	-0.36	0.81	0.00	0.00	24.18	NA	24.87	NA
0.6*DL+WL6	22.23	-0.73	1.65	0.00	0.00	11.81	NA	12.15	NA
0.6*DL+WL7	22.26	-0.36	0.81	0.00	0.00	23.91	NA	24.60	NA
0.6*DL+WL8	21.69	-0.73	1.65	0.00	0.00	11.49	NA	11.81	NA

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	C01	2
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

ULTIMATE STRESS DESIGN	Load at U/S Of Base Pit					Load at bottom of foundation		
Load Case	FX (T)	FY(T)	FZ(T)	M_x	M_z	V (T)	Mx (T-m)	Mz (T-m)
1.4D	-0.14	27.49	0.00	0.0	0.00	38.689	0.00	0.32
1.4D+1.7(CL1)+0.5Lr	-0.24	25.34	0.00	0.0	0.00	36.544	0.00	0.53
1.4D+1.7(CL2)+0.5Lr	-0.24	28.12	0.00	0.0	0.00	39.323	0.00	0.54
1.4D+1.7(CL3)+0.5Lr	-0.19	28.32	0.00	0.0	0.00	39.521	0.00	0.42
1.4D+1.7(CL4)+0.5Lr	-0.18	31.19	0.00	0.0	0.00	42.392	0.00	0.42
1.4D+1.7(CL5)+0.5Lr	-0.24	26.72	0.00	0.0	0.00	37.916	0.00	0.53
1.4D+1.7(CL6)+0.5Lr	-0.24	26.75	0.00	0.0	0.00	37.948	0.00	0.53
1.4D+1.7(CL5)+0.5Lr	-0.18	29.74	0.00	0.0	0.00	40.940	0.00	0.42
1.4D+1.7(CL6)+0.5Lr	-0.18	29.77	0.00	0.0	0.00	40.972	0.00	0.42
1.2D+1.6Lr+1.0(CL1)	-0.20	23.87	0.00	0.0	0.00	35.067	0.00	0.45
1.2D+1.6Lr+1.0(CL2)	-0.20	25.50	0.00	0.0	0.00	36.702	0.00	0.45
1.2D+1.6Lr+1.0(CL3)	-0.17	25.62	0.00	0.0	0.00	36.818	0.00	0.38
1.2D+1.6Lr+1.0(CL4)	-0.17	27.31	0.00	0.0	0.00	38.507	0.00	0.38
1.2D+1.6Lr+1.0(CL5)	-0.20	24.67	0.00	0.0	0.00	35.875	0.00	0.45
1.2D+1.6Lr+1.0(CL6)	-0.20	24.69	0.00	0.0	0.00	35.893	0.00	0.45
1.2D+1.6Lr+1.0(CL7)	-0.17	26.45	0.00	0.0	0.00	37.653	0.00	0.38
1.2D+1.6Lr+1.0(CL8)	-0.17	26.47	0.00	0.0	0.00	37.672	0.00	0.38
1.2D + 1.6WL1+1.0L+0.5Lr	1.04	11.97	0.00	0.0	0.00	23.166	0.00	2.34
1.2D + 1.6WL2+1.0L+0.5Lr	0.44	10.78	0.00	0.0	0.00	21.984	0.00	0.99
1.2D + 1.6WL3+1.0L+0.5Lr	-0.43	34.34	0.00	0.0	0.00	45.538	0.00	0.97
1.2D + 1.6WL4+1.0L+0.5Lr	-1.07	34.24	0.00	0.0	0.00	45.444	0.00	2.40
1.2D + 1.6WL5+1.0L+0.5Lr	-0.61	23.41	0.00	0.0	0.00	34.606	0.00	1.38
1.2D + 1.6WL6+1.0L+0.5Lr	-1.20	22.97	0.00	0.0	0.00	34.168	0.00	2.71
1.2D + 1.6WL7+1.0L+0.5Lr	-0.61	23.01	0.00	0.0	0.00	34.210	0.00	1.38
1.2D + 1.6WL8+1.0L+0.5Lr	-1.21	22.10	0.00	0.0	0.00	33.302	0.00	2.72
0.9D+1.6WL1	1.08	5.48	0.00	0.0	0.00	16.676	0.00	2.43
0.9D+1.6WL2	0.48	4.29	0.00	0.0	0.00	15.494	0.00	1.08
0.9D+1.6WL3	-0.39	27.85	0.00	0.0	0.00	39.048	0.00	0.88
0.9D+1.6WL4	-1.03	27.75	0.00	0.0	0.00	38.954	0.00	2.32
0.9D+1.6WL5	-0.57	16.92	0.00	0.0	0.00	28.116	0.00	1.29
0.9D+1.6WL6	-1.17	16.48	0.00	0.0	0.00	27.678	0.00	2.62
0.9D+1.6WL7	-0.57	16.52	0.00	0.0	0.00	27.720	0.00	1.29
0.9D+1.6WL8	-1.17	15.61	0.00	0.0	0.00	26.812	0.00	2.63

Check for Bearing Pressure

Load Combination & Calculation of Base pressure

Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

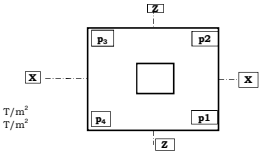
Appox. Surcharge load = 2 T/m²

Net SBC of soil for DL, LL= 50 T/m²

Gross SBC of soil for DL, LL= 54.05 T/m²

For WL/SL case = 62.5 T/m²

For WL/SL case = 66.55 T/m²



Load Case	Design Load/moments			Base Pressure (T/m²)								Remarks
	V (T)	Mz (T-m)	Mx (T-m)	V/ (Lx.Lz)	6.Mz/ (Lx)².Lz	6.Mx/ Lx.(Lz)²	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	
DL+Lr	32.04	0.27	0.00	10.46	0.30	0.00	12.76	12.76	12.16	12.16	9.61	O.K
DL+CL1	29.22	0.34	0.00	9.54	0.38	0.00	11.93	11.93	11.16	11.16	8.78	O.K
DL+CL2	30.86	0.35	0.00	10.08	0.39	0.00	12.46	12.46	11.69	11.69	9.31	O.K
DL+CL3	30.97	0.28	0.00	10.11	0.31	0.00	12.42	12.42	11.80	11.80	9.27	O.K
DL+CL4	32.66	0.27	0.00	10.66	0.31	0.00	12.97	12.97	12.36	12.36	9.82	O.K
DL+CL5	30.03	0.34	0.00	9.80	0.38	0.00	12.19	12.19	11.42	11.42	9.04	O.K
DL+CL6	30.05	0.34	0.00	9.81	0.38	0.00	12.20	12.20	11.43	11.43	9.05	O.K
DL+CL7	31.81	0.27	0.00	10.39	0.31	0.00	12.69	12.69	12.08	12.08	9.54	O.K
DL+CL8	31.83	0.27	0.00	10.39	0.31	0.00	12.70	12.70	12.09	12.09	9.55	O.K
DL+0.75*CL1+0.75*Lr	30.52	0.34	0.00	9.97	0.38	0.00	12.35	12.35	11.58	11.58	9.20	O.K
DL+0.75*CL2+0.75*Lr	31.75	0.35	0.00	10.37	0.39	0.00	12.75	12.75	11.98	11.98	9.60	O.K
DL+0.75*CL3+0.75*Lr	31.84	0.29	0.00	10.40	0.33	0.00	12.72	12.72	12.07	12.07	9.57	O.K
DL+0.75*CL4+0.75*Lr	33.10	0.29	0.00	10.81	0.33	0.00	13.14	13.14	12.48	12.48	9.99	O.K
DL+0.75*CL5+0.75*Lr	31.13	0.34	0.00	10.16	0.38	0.00	12.55	12.55	11.78	11.78	9.40	O.K
DL+0.75*CL6+0.75*Lr	31.14	0.34	0.00	10.17	0.38	0.00	12.55	12.55	11.78	11.78	9.40	O.K
DL+0.75*CL7+0.75*Lr	32.46	0.29	0.00	10.60	0.33	0.00	12.93	12.93	12.27	12.27	9.78	O.K
DL+0.75*CL8+0.75*Lr	32.48	0.29	0.00	10.60	0.33	0.00	12.93	12.93	12.28	12.28	9.78	O.K
DL+0.75*Lr+0.75*WL1	26.02	0.98	0.00	8.50	1.09	0.00	11.59	11.59	9.40	9.40	8.44	O.K
DL+0.75*Lr+0.75*WL2	25.46	0.34	0.00	8.31	0.38	0.00	10.70	10.70	9.93	9.93	7.55	O.K
DL+0.75*Lr+0.75*WL3	36.51	0.58	0.00	11.92	0.64	0.00	14.56	14.56	13.28	13.28	11.41	O.K
DL+0.75*Lr+0.75*WL4	36.46	1.25	0.00	11.91	1.40	0.00	15.30	15.30	12.51	12.51	12.15	O.K
DL+0.75*Lr+0.75*WL5	31.38	0.77	0.00	10.25	0.86	0.00	13.10	13.10	11.39	11.39	9.95	O.K
DL+0.75*Lr+0.75*WL6	31.18	1.39	0.00	10.18	1.56	0.00	13.74	13.74	10.62	10.62	10.59	O.K
DL+0.75*Lr+0.75*WL7	31.20	0.77	0.00	10.19	0.86	0.00	13.04	13.04	11.33	11.33	9.89	O.K
DL+0.75*Lr+0.75*WL8	30.77	1.39	0.00	10.05	1.56	0.00	13.61	13.61	10.49	10.49	10.46	O.K
0.6*DL+WL1	15.36	1.51	0.00	5.02	1.69	0.00	8.70	8.70	5.33	5.33	5.55	O.K
0.6*DL+WL2	14.62	0.66	0.00	4.77	0.74	0.00	7.52	7.52	6.03	6.03	4.37	O.K
0.6*DL+WL3	29.34	0.56	0.00	9.58	0.63	0.00	12.21	12.21	10.95	10.95	9.06	O.K
0.6*DL+WL4	29.28	1.46	0.00	9.56	1.63	0.00	13.19	13.19	9.93	9.93	10.04	O.K
0.6*DL+WL5	22.51	0.81	0.00	7.35	0.91	0.00	10.26	10.26	8.44	8.44	7.11	O.K
0.6*DL+WL6	22.23	1.65	0.00	7.26	1.84	0.00	11.10	11.10	7.42	7.42	7.95	O.K
0.6*DL+WL7	22.26	0.81	0.00	7.27	0.91	0.00	10.18	10.18	8.36	8.36	7.03	O.K
0.6*DL+WL8	21.69	1.65	0.00	7.08	1.85	0.00	10.93	10.93	7.23	7.23	7.78	O.K

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	C01	2
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Calculation of Design Pressure

Load Combination & Calculation of Base pressure

Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

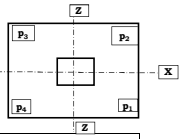
Surcharge load = 2 T/m²

Net SBC of soil for DL, LL= 50 T/m²

Gross SBC of soil for DL, LL= 54.05 T/m²

For WL/SL case = 62.5 T/m²

For WL/SL case = 66.55 T/m²



Load Case	Design Load/moments			Base Pressure (T/m²)							
	V (T)	Mz (T-m)	Mx (T-m)	V/ (Lx.Lz)	6.Mz/ (Lx)².Lz	6.Mx/ Lx.(Lz)²	p1 (p_{max})	p2	p4	p3 (p_{min})	(p_{max}-γh)
1.4D	38.7	0.32	0.00	12.63	0.36	0.00	14.99	14.99	14.28	14.28	11.84
1.4D+1.7(CL1)+0.5Lr	36.5	0.53	0.00	11.93	0.60	0.00	14.53	14.53	13.33	13.33	11.38
1.4D+1.7(CL2)+0.5Lr	39.3	0.54	0.00	12.84	0.60	0.00	15.44	15.44	14.24	14.24	12.29
1.4D+1.7(CL3)+0.5Lr	39.5	0.42	0.00	12.90	0.47	0.00	15.37	15.37	14.43	14.43	12.22
1.4D+1.7(CL4)+0.5Lr	42.4	0.42	0.00	13.84	0.46	0.00	16.31	16.31	15.38	15.38	13.16
1.4D+1.7(CL5)+0.5Lr	37.9	0.53	0.00	12.38	0.60	0.00	14.98	14.98	13.78	13.78	11.83
1.4D+1.7(CL6)+0.5Lr	37.9	0.53	0.00	12.39	0.60	0.00	14.99	14.99	13.79	13.79	11.84
1.4D+1.7(CL5)+0.5Lr	40.9	0.42	0.00	13.37	0.46	0.00	15.83	15.83	14.90	14.90	12.68
1.4D+1.7(CL6)+0.5Lr	41.0	0.42	0.00	13.38	0.46	0.00	15.84	15.84	14.91	14.91	12.69
1.2D+1.6Lr+1.0(CL1)	35.1	0.45	0.00	11.45	0.51	0.00	13.96	13.96	12.94	12.94	10.81
1.2D+1.6Lr+1.0(CL2)	36.7	0.45	0.00	11.98	0.51	0.00	14.49	14.49	13.48	13.48	11.34
1.2D+1.6Lr+1.0(CL3)	36.8	0.38	0.00	12.02	0.43	0.00	14.45	14.45	13.59	13.59	11.30
1.2D+1.6Lr+1.0(CL4)	38.5	0.38	0.00	12.57	0.43	0.00	15.00	15.00	14.15	14.15	11.85
1.2D+1.6Lr+1.0(CL5)	35.9	0.45	0.00	11.71	0.51	0.00	14.22	14.22	13.21	13.21	11.07
1.2D+1.6Lr+1.0(CL6)	35.9	0.45	0.00	11.72	0.51	0.00	14.23	14.23	13.21	13.21	11.08
1.2D+1.6Lr+1.0(CL7)	37.7	0.38	0.00	12.29	0.43	0.00	14.72	14.72	13.87	13.87	11.57
1.2D+1.6Lr+1.0(CL8)	37.7	0.38	0.00	12.30	0.43	0.00	14.73	14.73	13.87	13.87	11.58
1.2D + 1.6WL1+1.0L+0.5Lr	23.2	2.34	0.00	7.56	2.62	0.00	12.18	12.18	6.95	6.95	9.03
1.2D + 1.6WL2+1.0L+0.5Lr	22.0	0.99	0.00	7.18	1.11	0.00	10.29	10.29	8.07	8.07	7.14
1.2D + 1.6WL3+1.0L+0.5Lr	45.5	0.97	0.00	14.87	1.09	0.00	17.96	17.96	15.78	15.78	14.81
1.2D + 1.6WL4+1.0L+0.5Lr	45.4	2.40	0.00	14.84	2.69	0.00	19.53	19.53	14.15	14.15	16.38
1.2D + 1.6WL5+1.0L+0.5Lr	34.6	1.38	0.00	11.30	1.54	0.00	14.84	14.84	11.76	11.76	11.69
1.2D + 1.6WL6+1.0L+0.5Lr	34.2	2.71	0.00	11.16	3.03	0.00	16.19	16.19	10.12	10.12	13.04
1.2D + 1.6WL7+1.0L+0.5Lr	34.2	1.38	0.00	11.17	1.54	0.00	14.71	14.71	11.63	11.63	11.56
1.2D + 1.6WL8+1.0L+0.5Lr	33.3	2.72	0.00	10.87	3.04	0.00	15.92	15.92	9.83	9.83	12.77
0.9D+1.6WL1	16.7	2.43	0.00	5.45	2.72	0.00	10.16	10.16	4.73	4.73	7.01
0.9D+1.6WL2	15.5	1.08	0.00	5.06	1.21	0.00	8.27	8.27	5.85	5.85	5.12
0.9D+1.6WL3	39.0	0.88	0.00	12.75	0.99	0.00	15.74	15.74	13.76	13.76	12.59
0.9D+1.6WL4	39.0	2.32	0.00	12.72	2.59	0.00	17.31	17.31	12.13	12.13	14.16
0.9D+1.6WL5	28.1	1.29	0.00	9.18	1.44	0.00	12.62	12.62	9.74	9.74	9.47
0.9D+1.6WL6	27.7	2.62	0.00	9.04	2.93	0.00	13.97	13.97	8.10	8.10	10.82
0.9D+1.6WL7	27.7	1.29	0.00	9.05	1.44	0.00	12.49	12.49	9.61	9.61	9.34
0.9D+1.6WL8	26.8	2.63	0.00	8.75	2.94	0.00	13.70	13.70	7.81	7.81	10.55

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED 001	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	C01	2
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED 001	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Calculation of Bending Moment and Shear Force in X direction :

BM_{1x} =Moment at the face of column at section 1x-1x in T-m/m.

BM_{1z} =Moment at the face of column 1z-1z in T-m/m.

$SF_{(oneway)}$ = Oneway shear force at a distance 'd' from the face of column in T/m.

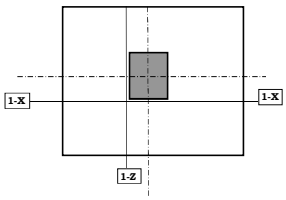
$SF_{(punching)}$ = Punching Shear Force at a distance 0.5d from the face of column in T/m.

Punching perimeter= ((0.6+2*0.5*0.251)+(0.6+2*0.5*0.251))*2
= 3.38 m

Punching Area= ((0.6+2*0.5*0.245)*(0.6+2*0.5*0.245))
= 0.71 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.35*2.5+1.4*1.8)
= 3.40 T/m^2

Area of footing, A_f= 3.06 m^2



Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 38.53 T
Strength of Concrete in shear Vc1 = 177.11 T **Table 22.6.5.2_ (a)**
Strength of Concrete in shear Vc2 = 200.02 T **Table 22.6.5.2_ (b)**
Strength of Concrete in shear Vc3 = 163.30 T **Table 22.6.5.2_ (c)**

Design Vc= 163.30 T
Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (38.53*10000)/(3380*245)
= 0.47 N/mm^2 **No shear reinforcement reqd.**

Load Case	p_{1x} (T)	$B.M_{1x}$ (T-m)	$S.F_{(Oneway)}$ (T)	$B.M_{1z}$ TM	$S.F_{(oneway)}$ (T)	$S.F_{(punching)}$ (T)	V_u/b_0d	Punching check (Remarks)	Allowable shear (0.75*√(fc)/6 (N/mm^2)	Actual Shear	Remarks	Mu/bd2
Ultimate Conditions												
1.4D	11.84	1.96	3.67	1.96	3.91	27.85	0.34	OK	0.74	0.159	OK	0.33
1.4D+1.7(CL1)+0.5Lr	11.38	1.88	3.53	1.88	3.76	26.77	0.32	OK	0.74	0.153	OK	0.31
1.4D+1.7(CL2)+0.5Lr	12.29	2.03	3.81	2.03	4.06	28.91	0.35	OK	0.74	0.166	OK	0.34
1.4D+1.7(CL3)+0.5Lr	12.22	2.02	3.79	2.02	4.03	28.75	0.35	OK	0.74	0.165	OK	0.34
1.4D+1.7(CL4)+0.5Lr	13.16	2.18	4.08	2.18	4.34	30.96	0.37	OK	0.74	0.177	OK	0.36
1.4D+1.7(CL5)+0.5Lr	11.83	1.96	3.67	1.96	3.90	27.83	0.34	OK	0.74	0.159	OK	0.33
1.4D+1.7(CL6)+0.5Lr	11.84	1.96	3.67	1.96	3.91	27.85	0.34	OK	0.74	0.159	OK	0.33
1.4D+1.7(CL5)+0.5Lr	12.68	2.10	3.93	2.10	4.18	29.83	0.36	OK	0.74	0.171	OK	0.35
1.4D+1.7(CL6)+0.5Lr	12.69	2.10	3.93	2.10	4.19	29.85	0.36	OK	0.74	0.171	OK	0.35
1.2D+1.6Lr+1.0(CL1)	10.81	1.79	3.35	1.79	3.57	25.43	0.31	OK	0.74	0.146	OK	0.30
1.2D+1.6Lr+1.0(CL2)	11.34	1.87	3.52	1.87	3.74	26.68	0.32	OK	0.74	0.153	OK	0.31
1.2D+1.6Lr+1.0(CL3)	11.30	1.87	3.50	1.87	3.73	26.58	0.32	OK	0.74	0.152	OK	0.31
1.2D+1.6Lr+1.0(CL4)	11.85	1.96	3.67	1.96	3.91	27.88	0.34	OK	0.74	0.16	OK	0.33
1.2D+1.6Lr+1.0(CL5)	11.07	1.83	3.43	1.83	3.65	26.04	0.31	OK	0.74	0.149	OK	0.30
1.2D+1.6Lr+1.0(CL6)	11.08	1.83	3.43	1.83	3.66	26.07	0.31	OK	0.74	0.149	OK	0.31
1.2D+1.6Lr+1.0(CL7)	11.57	1.91	3.59	1.91	3.82	27.22	0.33	OK	0.74	0.156	OK	0.32
1.2D+1.6Lr+1.0(CL8)	11.58	1.91	3.59	1.91	3.82	27.24	0.33	OK	0.74	0.156	OK	0.32
1.2D + 1.6WL1+1.0L+0.5Lr	9.03	1.49	2.80	1.49	2.98	21.24	0.26	OK	0.74	0.122	OK	0.25
1.2D + 1.6WL2+1.0L+0.5Lr	7.14	1.18	2.21	1.18	2.36	16.80	0.2	OK	0.74	0.096	OK	0.20
1.2D + 1.6WL3+1.0L+0.5Lr	14.81	2.45	4.59	2.45	4.89	34.84	0.42	OK	0.74	0.199	OK	0.41
1.2D + 1.6WL4+1.0L+0.5Lr	16.38	2.71	5.08	2.71	5.41	38.53	0.47	OK	0.74	0.221	OK	0.45
1.2D + 1.6WL5+1.0L+0.5Lr	11.69	1.93	3.62	1.93	3.86	27.50	0.33	OK	0.74	0.157	OK	0.32
1.2D + 1.6WL6+1.0L+0.5Lr	13.04	2.16	4.04	2.16	4.30	30.68	0.37	OK	0.74	0.176	OK	0.36
1.2D + 1.6WL7+1.0L+0.5Lr	11.56	1.91	3.58	1.91	3.81	27.19	0.33	OK	0.74	0.156	OK	0.32
1.2D + 1.6WL8+1.0L+0.5Lr	12.77	2.11	3.96	2.11	4.21	30.04	0.36	OK	0.74	0.172	OK	0.35
0.9D+1.6WL1	7.01	1.16	2.17	1.16	2.31	16.49	0.2	OK	0.74	0.094	OK	0.19
0.9D+1.6WL2	5.12	0.85	1.59	0.85	1.69	12.04	0.15	OK	0.74	0.069	OK	0.14
0.9D+1.6WL3	12.59	2.08	3.90	2.08	4.15	29.62	0.36	OK	0.74	0.17	OK	0.35
0.9D+1.6WL4	14.16	2.34	4.39	2.34	4.67	33.31	0.4	OK	0.74	0.191	OK	0.39
0.9D+1.6WL5	9.47	1.57	2.94	1.57	3.13	22.28	0.27	OK	0.74	0.128	OK	0.26
0.9D+1.6WL6	10.82	1.79	3.35	1.79	3.57	25.45	0.31	OK	0.74	0.146	OK	0.30
0.9D+1.6WL7	9.34	1.54	2.90	1.54	3.08	21.97	0.27	OK	0.74	0.126	OK	0.26
0.9D+1.6WL8	10.55	1.74	3.27	1.74	3.48	24.82	0.3	OK	0.74	0.142	OK	0.29

Load Case	kn (Ø=0.75)	p =	A_{st}
Ultimate Conditions			
1.4D	0.43	0.001	265
1.4D+1.7(CL1)+0.5Lr	0.42	0.001	265
1.4D+1.7(CL2)+0.5Lr	0.45	0.001	265
1.4D+1.7(CL3)+0.5Lr	0.45	0.001	265
1.4D+1.7(CL4)+0.5Lr	0.48	0.001	265
1.4D+1.7(CL5)+0.5Lr	0.43	0.001	265
1.4D+1.7(CL6)+0.5Lr	0.43	0.001	265
1.4D+1.7(CL5)+0.5Lr	0.47	0.001	265
1.4D+1.7(CL6)+0.5Lr	0.47	0.001	265
1.2D+1.6Lr+1.0(CL1)	0.4	0.001	265
1.2D+1.6Lr+1.0(CL2)	0.42	0.001	265
1.2D+1.6Lr+1.0(CL3)	0.41	0.001	265
1.2D+1.6Lr+1.0(CL4)	0.44	0.001	265
1.2D+1.6Lr+1.0(CL5)	0.41	0.001	265
1.2D+1.6Lr+1.0(CL6)	0.41	0.001	265
1.2D+1.6Lr+1.0(CL7)	0.42	0.001	265
1.2D+1.6Lr+1.0(CL8)	0.43	0.001	265
1.2D + 1.6WL1+1.0L+0.5Lr	0.33	0.001	265
1.2D + 1.6WL2+1.0L+0.5Lr	0.26	0.001	265
1.2D + 1.6WL3+1.0L+0.5Lr	0.54	0.001	265
1.2D + 1.6WL4+1.0L+0.5Lr	0.6	0.001	265
1.2D + 1.6WL5+1.0L+0.5Lr	0.43	0.001	265
1.2D + 1.6WL6+1.0L+0.5Lr	0.48	0.001	265
1.2D + 1.6WL7+1.0L+0.5Lr	0.42	0.001	265
1.2D + 1.6WL8+1.0L+0.5Lr	0.47	0.001	265
0.9D+1.6WL1	0.26	0.001	265
0.9D+1.6WL2	0.19	0	0
0.9D+1.6WL3	0.46	0.001	265
0.9D+1.6WL4	0.52	0.001	265
0.9D+1.6WL5	0.35	0.001	265
0.9D+1.6WL6	0.4	0.001	265
0.9D+1.6WL7	0.34	0.001	265
0.9D+1.6WL8	0.39	0.001	265

Location	Mux (T-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)
At bottom	2.18	0.36	0.48	0.001	265.00	477.00	12ϕT	237.10	300	200
At top	0.00	0.00	0.00	0.0018	441.00	441.00	12ϕT	256.46	300	200

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :	At Bottom	**12ϕT**	**200**	c/c	At Top	**12ϕT**	**200**	c/c
Along Z Direction :	At Bottom	**12ϕT**	**200**	c/c	At Top	**12ϕT**	**200**	c/c

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED 001	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	C01	2
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED 001	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

3.3 DESIGN OF FOUNDATION MKD F3

	F7			F8		
LOAD ITEM	Fx Ton	Fy Ton	Fz Ton	Fx Ton	Fy Ton	Fz Ton
1 DL	-0.40	20.77	-1.79	-0.50	25.04	1.95
2 LLROOF	0.00	6.00	-0.21	0.00	7.16	0.25
3 CL1(L+XS)	0.19	1.88	-0.18	0.10	1.95	0.18
4 CL2(L-XS)	0.19	1.90	-0.18	0.10	1.95	0.18
5 CL3(R+XS)	0.58	4.69	-0.45	0.36	4.76	0.43
6 CL4(R-XS)	0.35	4.67	-0.44	0.12	4.83	0.44
7 CL5(L+ZT)	0.19	1.41	-0.07	0.10	2.53	0.29
8 CL6(L-ZT)	0.19	2.37	-0.29	0.09	1.37	0.07
9 CL7(R+ZT)	0.46	4.20	-0.34	0.24	5.37	0.54
10 CL8(R-ZT)	0.46	5.15	-0.56	0.24	4.22	0.33
21 WL1(+X)S	0.56	-0.97	0.07	1.17	-1.68	-0.09
22 WL2(+X)P	1.69	-2.03	0.12	1.82	-4.13	-0.21
23 WL3(-X)S	-2.00	-2.45	0.19	-2.08	-3.15	-0.17
24 WL4(-X)P	-0.94	-3.38	0.24	-0.98	-5.25	-0.27
25 WL5(+Z)S	0.88	-20.58	3.13	0.91	16.31	3.15
26 WL6(+Z)P	1.94	-24.19	3.57	2.08	15.88	3.54
27 WL7(-Z)S	0.88	16.35	-2.77	0.91	-22.47	-3.40
28 WL8(-Z)P	1.94	16.91	-3.01	2.02	-27.05	-3.90

FOUNDATION SIZE 3.00m X 4.75m
DEPTH OF THE FOUNDATION 2.25m from FFL
THE WT OF SOIL +SELF WT 3X4.75X2.25X2 = 64.125 T

WORKING STRESS DESIGN

	F7			F8			Load at bottom of foundation				
Load Case	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)	V (T)	Hx (T)	MX (T-m)	Hz (T)	MZ (T-m)
DL+Lr	-0.56	29.08	-2.50	-0.70	35.05	2.72	128.26	-1.26	8.62	0.23	3.15
DL+CL1	-0.24	35.29	-2.91	-0.53	41.95	3.16	141.36	-0.77	9.62	0.25	1.93
DL+CL2	-0.24	35.31	-2.91	-0.53	41.95	3.16	141.38	-0.77	9.57	0.25	1.93
DL+CL3	0.42	40.05	-3.36	-0.09	46.72	3.58	150.89	0.33	9.55	0.22	0.82
DL+CL4	0.03	40.02	-3.36	-0.49	46.84	3.59	150.98	-0.45	9.79	0.23	1.13
DL+CL5	-0.24	34.48	-2.72	-0.52	42.93	3.34	141.54	-0.77	12.94	0.62	1.91
DL+CL6	-0.24	36.11	-3.09	-0.54	40.97	2.97	141.20	-0.78	6.86	-0.12	1.94
DL+CL7	0.22	39.22	-3.18	-0.28	47.76	3.77	151.10	-0.06	13.01	0.60	0.15
DL+CL8	0.22	40.84	-3.55	-0.29	45.80	3.41	150.77	-0.07	7.04	-0.14	0.17
DL+0.75*CL1+0.75*Lr	-0.29	36.41	-2.66	-0.50	43.45	2.91	143.99	-0.79	10.15	0.26	1.98
DL+0.75*CL2+0.75*Lr	-0.29	36.42	-2.66	-0.50	43.45	2.91	144.00	-0.79	10.12	0.26	1.98
DL+0.75*CL3+0.75*Lr	0.09	39.22	-2.93	-0.24	46.25	3.17	149.59	-0.15	10.10	0.24	0.37
DL+0.75*CL4+0.75*Lr	-0.13	39.19	-2.92	-0.47	46.32	3.17	149.64	-0.61	10.25	0.25	1.51
DL+0.75*CL5+0.75*Lr	-0.29	35.94	-2.55	-0.50	44.02	3.02	144.09	-0.79	12.10	0.47	1.97
DL+0.75*CL6+0.75*Lr	-0.29	36.90	-2.77	-0.50	42.87	2.81	143.89	-0.80	8.17	0.04	1.99
DL+0.75*CL7+0.75*Lr	-0.02	38.73	-2.81	-0.35	46.86	3.28	149.72	-0.37	12.14	0.46	0.93
DL+0.75*CL8+0.75*Lr	-0.02	39.68	-3.03	-0.36	45.71	3.06	149.52	-0.38	8.21	0.03	0.95
DL+0.75*Lr+0.75*WL1	0.42	26.38	-2.14	1.27	30.93	2.32	121.43	1.69	6.60	0.18	4.23
DL+0.75*Lr+0.75*WL2	2.22	24.68	-2.05	2.31	27.02	2.12	115.82	4.53	3.34	0.07	11.33
DL+0.75*Lr+0.75*WL3	-3.68	34.00	-1.95	-3.93	28.59	2.18	116.72	-7.61	6.78	0.24	19.02
DL+0.75*Lr+0.75*WL4	-1.98	22.52	-1.87	-2.16	25.22	2.02	111.87	-4.14	4.03	0.15	10.35
DL+0.75*Lr+0.75*WL5	0.9185	-5.0	2.8	0.86	59.71	7.50	118.84	1.78	112.99	10.25	4.44
DL+0.75*Lr+0.75*WL6	2.6185	-10.8	3.5	2.73	59.03	8.12	112.38	5.35	123.20	11.59	13.37
DL+0.75*Lr+0.75*WL7	0.9185	54.1	-6.7	0.86	-2.33	-2.98	115.88	1.78	100.30	-9.65	4.44
DL+0.75*Lr+0.75*WL8	2.6185	55.0	-7.1	2.63	-9.66	-3.78	109.45	5.25	114.40	-10.85	13.11
0.6*DL+WL1	0.538875	17.1	-1.5	1.42	19.84	1.62	101.11	1.96	3.92	0.11	4.91
0.6*DL+WL2	2.338875	15.4	-1.4	2.46	15.93	1.41	95.50	4.80	0.66	0.00	12.01
0.6*DL+WL3	-3.561125	14.8	-1.3	-3.78	17.50	1.47	96.39	-7.34	4.11	0.17	18.34
0.6*DL+WL4	-1.861125	13.3	-1.2	-2.01	14.13	1.31	91.55	-3.87	1.35	0.09	9.67
0.6*DL+WL5	1.038875	-14.2	3.4	1.01	48.62	6.79	98.52	2.05	110.32	10.18	5.12
0.6*DL+WL6	2.738875	-20.0	4.1	2.88	47.94	7.41	92.06	5.62	120.52	11.52	14.05
0.6*DL+WL7	1.038875	44.9	-6.0	1.01	-13.42	-3.69	95.56	2.05	102.98	-9.72	5.12
0.6*DL+WL8	2.738875	45.8	-6.4	2.78	-20.75	-4.49	89.13	5.51	117.07	-10.92	13.79

Check for overturning and sliding:

	Load at Foundation Base					Sliding Coeffecient, μ = 0.4			
Load Case	V (T)	Hx (T)	Mz (Tm)	Hz (T)	Mx (Tm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
DL+Lr	128.26	-1.26	3.15	0.23	8.62	61.13	35.33	40.76	227.26
DL+CL1	141.36	-0.77	1.93	0.25	9.62	109.84	34.89	73.22	224.82
DL+CL2	141.38	-0.77	1.93	0.25	9.57	109.85	35.07	73.23	230.70
DL+CL3	150.89	0.33	0.82	0.22	9.55	277.34	37.54	184.90	274.82
DL+CL4	150.98	-0.45	1.13	0.23	9.79	199.59	36.61	133.06	257.53
DL+CL5	141.54	-0.77	1.91	0.62	12.94	110.93	25.98	73.95	91.76
DL+CL6	141.20	-0.78	1.94	-0.12	6.86	109.09	48.89	72.73	469.20
DL+CL7	151.10	-0.06	0.15	0.60	13.01	1529.13	27.59	1019.42	101.45
DL+CL8	150.77	-0.07	0.17	-0.14	7.04	1327.76	50.86	885.17	425.82
DL+0.75*CL1+0.75*Lr	143.99	-0.79	1.98	0.26	10.15	108.98	33.70	72.65	222.16
DL+0.75*CL2+0.75*Lr	144.00	-0.79	1.98	0.26	10.12	108.99	33.80	72.66	225.43
DL+0.75*CL3+0.75*Lr	149.59	-0.15	0.37	0.24	10.10	612.76	35.17	408.50	248.81
DL+0.75*CL4+0.75*Lr	149.64	-0.61	1.51	0.25	10.25	148.29	34.68	98.86	240.15
DL+0.75*CL5+0.75*Lr	144.09	-0.79	1.97	0.47	12.10	109.60	28.29	73.07	121.53
DL+0.75*CL6+0.75*Lr	143.89	-0.80	1.99	0.04	8.17	108.55	41.84	72.37	1421.15
DL+0.75*CL7+0.75*Lr	149.72	-0.37	0.93	0.46	12.14	240.59	29.30	160.39	129.70
DL+0.75*CL8+0.75*Lr	149.52	-0.38	0.95	0.03	8.21	236.97	43.25	157.98	2136.00
DL+0.75*Lr+0.75*WL1	121.43	1.69	4.23	0.18	6.60	43.03	43.71	28.69	268.54
DL+0.75*Lr+0.75*WL2	115.82	4.53	11.33	0.07	3.34	15.33	82.43	10.22	672.65
DL+0.75*Lr+0.75*WL3	116.72	-7.61	19.02	0.24	6.78	9.21	40.86	6.14	197.09
DL+0.75*Lr+0.75*WL4	111.87	-4.14	10.35	0.15	4.03	16.22	65.99	10.81	292.71
DL+0.75*Lr+0.75*WL5	118.84	1.78	4.44	10.25	112.99	40.13	2.50	26.75	4.64
DL+0.75*Lr+0.75*WL6	112.38	5.35	13.37	11.59	123.20	12.61	2.17	8.40	3.88
DL+0.75*Lr+0.75*WL7	115.88	1.78	4.44	-9.65	100.30	39.13	2.74	26.08	4.80
DL+0.75*Lr+0.75*WL8	109.45	5.25	13.11	-10.85	114.40	12.52	2.27	8.35	4.04
0.6*DL+WL1	101.11	1.96	4.91	0.11	3.92	30.91	61.24	20.61	357.51
0.6*DL+WL2	95.50	4.80	12.01	0.00	0.66	11.93	343.71	7.95	33955.73
0.6*DL+WL3	96.39	-7.34	18.34	0.17	4.11	7.88	55.74	5.26	227.98
0.6*DL+WL4	91.55	-3.87	9.67	0.09	1.35	14.20	161.19	9.46	430.18
0.6*DL+WL5	98.52	2.05	5.12	10.18	110.32	28.88	2.12	19.25	3.87
0.6*DL+WL6	92.06	5.62	14.05	11.52	120.52	9.83	1.81	6.55	3.20
0.6*DL+WL7	95.56	2.05	5.12	-9.72	102.98	28.01	2.20	18.68	3.93
0.6*DL+WL8	89.13	5.51	13.79	-10.92	117.07	9.70	1.81	6.46	3.27

CHECK FOR BEARING PRESSURE

THE NET SAFE BEARING CAPACITY OF SOIL IS 50T/m^2
MAXIMUM GROSS BEARING PRESSURE OBTAINED 16T/m^2 IN LC 23 . HENCE IT IS SAFE.
REFER TO THE STAAD FILE ATTACHED.

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	C01	2
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Foundation has been designed with the Design strength Load Combinations of Staad File (LC 100 - LC 132)

REINFORCEMENT CALCULATION OF FOUNDATION:

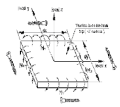
Foundation Thickness =	400	mm
Width =	1000	mm
Clear Cover =	75	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	

STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	10.738	T-M	Corresponding Mxy	-0.10	T-M	*(Plate No. - 139 , L/C No. - 122)*
Min. Mx	-3.487	T-M	Corresponding Mxy	0.27	T-M	*(Plate No. - 138 , L/C No. - 132)*
Max. My	7.117	T-M	Corresponding Mxy	-0.10	T-M	*(Plate No. - 139 , L/C No. - 122)*
Min. My	-4.211	T-M	Corresponding Mxy	0.05	T-M	*(Plate No. - 52 , L/C No. - 104)*

FOR OPPOSITE FACE REINFORCEMENT :

Mx1 =	Mx + abs(Mxy) =	10.738 + 0.101 =	10.839 T-M
My1 =	My + abs(Mxy) =	7.117 + 0.101 =	7.218 T-M
Mx2 =	Mx+abs(Mxy^2/My) =	10.738 + 0.001 =	10.739 T-M
My2 =	My+abs(Mxy^2/Mx) =	7.117 + 0.001 =	7.118 T-M



DESIGN MOMENT CALCULATION FOR OPPOSITE FACE REINFORCEMENT (Bottom) :

Mxd = 10.839 T-m
Myd = 7.218 T-m

ALONG X - DIRECTION :

Mu = 10.839 T-m
Φ*Kn = Mu/bd^2 deff. = 295 mm
Mu/bd^2 = 10.839X10^7/(1000X295^2) = 1.246 N/mm^2 < 6.9 N/mm^2
Pt = 0.0034 Reqd. Ast = 0.0034X1000X295/100 = 10.03 cm^2
Provided **20** mm @ **200** C/C (Provide Ast = 15.71 cm2) at Horizontal

Total Provide Ast = 15.71 cm2 at Horizontal **O.K.**
Total Provide Pt = 0.01

ALONG Y - DIRECTION :

Mu = 7.218 T-m
Φ*Kn = Mu/bd^2 deff. = 315 mm
Mu/bd^2 = 7.218X10^7/(1000X315^2) = 0.727 N/mm^2 < 6.9 N/mm^2
Pt = 0.0020 Reqd. Ast = 0.002X1000X315/100 = 6.3 cm^2
Provided **20** mm @ **200** C/C (Provide Ast = 15.71 cm2) at Horizontal

Total Provide Ast = 15.71 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.01

FOR RETAINED FACE REINFORCEMENT :

Mx1 =	Mx - abs(Mxy) =	-3.487 - 0.265 =	-3.752 T-M
My1 =	My - abs(Mxy) =	-4.211 - 0.048 =	-4.259 T-M
Mx2 =	Mx-abs(Mxy^2/My) =	-3.487 - 0.017 =	-3.504 T-M
My2 =	My-abs(Mxy^2/Mx) =	-4.211 - 0.001 =	-4.21 T-M

DESIGN MOMENT CALCULATION FOR RETAINED FACE REINFORCEMENT (Top) :

Mxd = -3.752 T-M
Myd = -4.259 T-M

ALONG X - DIRECTION :

Mu = 3.752 T-m
Φ*Kn = Mu/bd^2 deff. = 295 mm
Mu/bd^2 = 3.752X10^7/(1000X295^2) = 0.431 N/mm^2 < 6.9 N/mm^2
Pt = 0.0011 Reqd. Ast = 0.0011X1000X295/100 = 3.25 cm^2
Provided **20** mm @ **200** C/C (Provide Ast = 15.71 cm2) at Horizontal

Total Provide Ast = 15.71 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.01

ALONG Y - DIRECTION :

Mu = 4.259 T-m
Φ*Kn = Mu/bd^2 deff. = 315 mm
Mu/bd^2 = 4.259X10^7/(1000X315^2) = 0.429 N/mm^2 < 6.9 N/mm^2
Pt = 0.0011 Reqd. Ast = 0.0011X1000X315/100 = 3.47 cm^2
Provided **20** mm @ **200** C/C (Provide Ast = 15.71 cm2) at Horizontal

Total Provide Ast = 15.71 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.01

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	C01	2
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

SHEAR CHECK

(FROM FACTORED LOAD COMBINATION)

ALONG X - DIRECTION :

SQx = -0.419 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 131 , L/C No. - 122)*

Vu = 12.361 T

Provided Pt = 0.0053

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0053X12.361X10^4X295/3.752/10^7)X1000X295/10^4

= 30.507 T

[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0053)X1000X295/10^4

= 30.582 T

[c] = 0.29*λ*fc^0.5*b*d.eff

= 0.29X1X5.92X1000X295/10^4

= 50.612 T

HENCE, Vc = 30.507 T

Φt*Vc = 0.75X30.507 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 22.88 T **> 12.361T , Hence O.K.**

ALONG Y - DIRECTION :

SQy = -0.384 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 185 , L/C No. - 122)*

Vu = 12.096 T

Pt provided = 0.005

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.005X12.096X10^4X315/4.259/10^7)X1000X315/10^4

= 32.212 T

[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.005)X1000X315/10^4

= 32.495 KN

[c] = 0.29*λ*fc^0.5*b*d.eff

= 0.29X1X5.92X1000X315/10^4

= 54.043 T

HENCE, Vc = 32.212 T

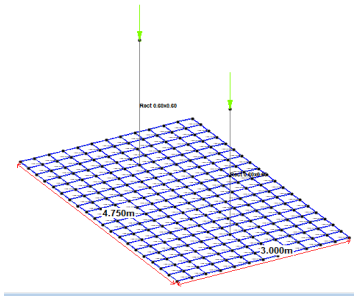
Φt*Vc = 0.75X32.212 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 24.159 T **> 12.096T , Hence O.K.**

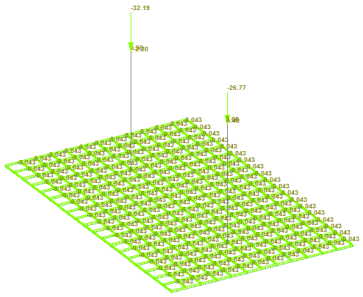
DESIGN OF PEDESTAL: REFER TO STAAD FILE OF FOUNDATION MKD. F3

Size of Column =	600	X	600	mm^2						
Reqired area of Reinforcement =	3600	mm^2								
Reinforcement Provided =	4	no.	25	dia bars	+	4	no.	25	dia bars	
Area of Reinforcement Provided =	3927	mm^2								

STAAD MODEL OF RAFT WITH PLATE ELEMENT



STAAD MODEL OF RAFT WITH DEAD + LIVE+ SURCHARGE LOAD



Surcharge Load on Foundation =		2 T/m^2
Soil Weight over the foundation =	1.35 x 1.8 T/m2 :	2.4 T/m^2
SOIL+SURCHARGE LOAD =		4.4 T/m^2

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	C01	2
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

3.4 Design of Foundation mkd. F4

	A7			A8			B7			B8		
LOAD ITEM	Fx Ton	Fy Ton	Fz Ton	Fx Ton	Fy Ton	Fz Ton	Fx Ton	Fy Ton	Fz Ton	Fx Ton	Fy Ton	Fz Ton
1 DL	-0.1	19.6	-1.5	-0.1	21.8	1.5	0.5	24.2	-2.4	0.5	27.6	2.2
2 LLROOF	0.0	1.4	-0.1	0.0	1.2	0.0	0.0	7.0	-0.3	0.0	7.5	0.3
3 CL1(L+XS)	0.0	-1.6	0.2	0.0	-1.3	-0.2	0.2	6.2	-0.6	0.3	6.1	0.6
4 CL2(L-XS)	0.0	0.0	0.0	0.0	0.4	0.0	-0.5	4.6	-0.5	-0.4	4.4	0.5
5 CL3(R+XS)	0.0	0.2	0.0	0.0	-0.4	0.0	-0.2	1.6	-0.2	0.1	2.4	0.2
6 CL4(R-XS)	0.0	1.8	-0.2	0.0	1.4	0.2	-0.7	0.0	-0.1	-0.5	0.6	0.1
7 CL5(L+ZT)	0.0	-1.0	0.1	0.0	-0.1	-0.1	-0.2	5.0	-0.5	-0.1	5.7	0.6
8 CL6(L-ZT)	0.0	-0.6	0.1	0.0	-0.7	-0.1	-0.1	5.8	-0.6	-0.1	4.8	0.5
9 CL7(R+ZT)	0.0	0.8	-0.1	0.0	0.8	0.1	-0.5	0.4	0.0	-0.2	1.9	0.2
10 CL8(R-ZT)	0.0	1.2	-0.1	0.0	0.2	0.1	-0.5	1.2	-0.2	-0.2	1.1	0.0
21 WL1(+X)S	0.7	-7.4	0.7	0.7	-8.5	-0.7	3.4	4.3	-0.1	3.7	4.7	0.1
22 WL2(+X)P	0.4	-7.6	0.7	0.4	-9.1	-0.6	2.7	2.7	-0.1	3.2	3.3	0.2
23 WL3(-X)S	-0.2	5.7	-0.6	-0.2	7.0	0.6	-2.2	-7.4	0.3	-2.7	-9.7	-0.4
24 WL4(-X)P	-0.6	5.9	-0.6	-0.6	6.7	0.7	-3.1	-9.0	0.3	-3.5	-11.7	-0.3
25 WL5(+Z)S	-0.3	-12.7	2.1	-0.3	12.3	2.3	-0.8	-17.0	3.0	-0.4	11.7	3.1
26 WL6(+Z)P	-0.6	-14.3	2.4	-0.8	13.4	2.6	-1.6	-21.0	3.4	-1.2	10.8	3.6
27 WL7(-Z)S	-0.3	11.8	-2.0	-0.6	-12.8	-2.3	-0.5	11.2	-2.5	-1.6	-17.7	-3.3
28 WL8(-Z)P	-0.7	12.4	-2.1	-0.7	-15.8	-2.6	-1.1	11.2	-2.9	-1.5	-20.9	-3.7

FOUNDATION SIZE (F4) 4.75m X 4.25m
DEPTH OF THE FOUNDATION 2.25m from FFL
THE WT OF SOIL +SELF WT 4.25X4.75X2.25X2 = 90.84375 T

WORKING STRESS DESIGN	A7			A8			B7			B8		
Load Case	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)
DL+Lr	-0.10	20.32	-1.53	-0.12	19.03	1.51	0.75	33.90	-3.32	-0.29	33.60	3.06
DL+CL1	-0.14	17.99	-1.30	-0.15	17.09	1.31	1.16	46.06	-4.42	0.59	45.82	4.18
DL+CL2	-0.14	19.64	-1.47	-0.15	18.75	1.48	-0.12	43.23	-4.31	-0.69	43.01	4.07
DL+CL3	-0.11	19.80	-1.46	-0.11	18.67	1.48	0.39	38.17	-3.73	0.60	38.62	3.50
DL+CL4	-0.11	21.44	-1.64	-0.12	20.58	1.66	-0.48	35.42	-3.59	-0.90	35.29	3.35
DL+CL5	-0.14	18.63	-1.36	-0.15	18.14	1.43	0.50	43.97	-4.22	-0.05	45.02	4.27
DL+CL6	-0.14	19.00	-1.41	-0.15	17.69	1.36	0.54	45.32	-4.52	-0.05	43.81	3.98
DL+CL7	-0.11	20.44	-1.53	-0.12	19.85	1.60	-0.06	36.12	-3.51	-0.16	37.56	3.57
DL+CL8	-0.11	20.80	-1.58	-0.11	19.40	1.54	-0.03	37.47	-3.81	-0.15	36.34	3.27
DL+0.75*CL1+0.75*Lr	-0.14	18.93	-1.39	-0.15	17.62	1.37	0.91	40.28	-3.84	0.31	39.21	3.63
DL+0.75*CL2+0.75*Lr	-0.14	20.17	-1.51	-0.15	18.87	1.50	0.16	38.62	-3.78	-0.44	37.56	3.57
DL+0.75*CL3+0.75*Lr	-0.12	20.29	-1.51	-0.12	18.81	1.50	0.46	35.64	-3.43	0.32	34.98	3.23
DL+0.75*CL4+0.75*Lr	-0.11	21.52	-1.64	-0.13	20.24	1.63	-0.05	34.02	-3.35	-0.56	33.02	3.14
DL+0.75*CL5+0.75*Lr	-0.14	19.42	-1.43	-0.15	18.41	1.46	0.53	39.05	-3.72	-0.07	38.74	3.69
DL+0.75*CL6+0.75*Lr	-0.14	19.69	-1.47	-0.15	18.08	1.41	0.55	39.85	-3.90	-0.06	38.03	3.51
DL+0.75*CL7+0.75*Lr	-0.11	20.77	-1.56	-0.12	19.69	1.59	0.19	34.43	-3.30	-0.13	34.35	3.27
DL+0.75*CL8+0.75*Lr	-0.12	21.04	-1.60	-0.12	19.35	1.54	0.22	35.23	-3.48	-0.12	33.64	3.10
DL+0.75*Lr+0.75*WL1	0.45	14.59	-1.01	0.17	13.97	1.00	6.14	37.55	-3.11	4.86	37.36	2.92
DL+0.75*Lr+0.75*WL2	0.17	14.42	-1.01	0.03	13.70	1.02	4.91	34.93	-3.15	4.48	35.95	3.04
DL+0.75*Lr+0.75*WL3	-0.25	24.42	-1.97	-0.21	23.73	1.98	-2.91	18.79	-2.43	-4.70	17.04	2.16
DL+0.75*Lr+0.75*WL4	-0.54	24.53	-1.98	-0.36	23.34	2.01	-4.24	16.24	-2.47	-4.58	15.65	2.23
DL+0.75*Lr+0.75*WL5	-0.30	10.59	0.07	-0.25	27.99	3.11	-0.62	3.4	1.8	0.09	50.18	7.40
DL+0.75*Lr+0.75*WL6	-0.58	9.43	0.25	-0.39	28.72	3.33	-1.98	-3.0	2.5	0.25	50.60	7.98
DL+0.75*Lr+0.75*WL7	-0.35	28.97	-2.98	-0.43	9.84	-0.15	-0.21	48.5	-7.0	-1.48	4.38	-2.57
DL+0.75*Lr+0.75*WL8	-0.64	29.40	-3.09	-0.33	7.37	-0.41	-1.17	48.6	-7.5	0.24	1.67	-3.19
0.6*DL+WL1	0.68	4.37	-0.21	0.32	4.64	0.22	5.97	28.7	-2.3	4.90	28.92	2.13
0.6*DL+WL2	0.30	4.13	-0.21	0.13	4.29	0.25	4.74	26.1	-2.3	4.52	27.51	2.25
0.6*DL+WL3	-0.25	17.46	-1.50	-0.19	17.65	1.53	-3.08	10.0	-1.6	-4.66	8.61	1.37
0.6*DL+WL4	-0.65	17.61	-1.51	-0.38	17.14	1.57	-4.41	7.4	-1.6	-4.53	7.21	1.44
0.6*DL+WL5	-0.32	-0.97	1.23	-0.24	23.34	3.03	-0.80	-5.4	2.7	0.13	41.75	6.61
0.6*DL+WL6	-0.69	-2.52	1.47	-0.42	24.31	3.33	-2.15	-11.8	3.3	0.29	42.17	7.19
0.6*DL+WL7	-0.39	23.54	-2.84	-0.49	-0.87	-1.31	-0.38	39.6	-6.2	-1.44	-4.05	-3.36
0.6*DL+WL8	-0.77	24.11	-2.99	-0.35	-4.16	-1.66	-1.34	39.8	-6.7	0.28	-6.77	-3.98

SLIDING AND OVERTURNING CHECK

	Load at bottom of foundation					OVERTURNING CHECK		SLIDING CHECK, μ = 0.4	
LOAD CASES	V (T)	Hx (T)	MX (T-m)	Hz (T)	MZ (T-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
DL+Lr	197.70	0.2	2.8	-0.27	45.64	10.29	9.20	329.84	288.88
DL+CL1	217.80	1.5	2.1	-0.23	94.54	5.47	4.90	59.82	378.17
DL+CL2	215.48	-1.1	2.1	-0.23	79.32	6.45	5.77	78.18	374.75
DL+CL3	206.10	0.8	1.5	-0.21	63.23	7.74	6.93	108.10	393.03
DL+CL4	203.57	-1.6	1.9	-0.22	49.92	9.68	8.67	50.73	367.21
DL+CL5	216.61	0.2	1.1	0.13	83.94	6.13	5.48	560.35	688.34
DL+CL6	216.66	0.2	5.3	-0.59	84.41	6.10	5.45	434.96	147.61
DL+CL7	204.81	-0.4	1.5	0.14	54.54	8.92	7.98	184.93	589.91
DL+CL8	204.86	-0.4	4.9	-0.57	54.79	8.88	7.95	205.70	143.04
DL+0.75*CL1+0.75*Lr	206.89	0.9	3.8	-0.23	71.06	6.92	6.19	88.18	366.99
DL+0.75*CL2+0.75*Lr	206.06	-0.6	3.7	-0.22	60.83	8.04	7.20	145.34	366.54
DL+0.75*CL3+0.75*Lr	200.56	0.5	3.4	-0.21	51.79	9.20	8.23	148.03	378.85
DL+0.75*CL4+0.75*Lr	199.65	-0.9	3.6	-0.22	42.58	11.14	9.96	93.65	366.32
DL+0.75*CL5+0.75*Lr	206.47	0.2	1.8	-0.01	64.38	7.62	6.82	478.59	12829.04
DL+0.75*CL6+0.75*Lr	206.48	0.2	5.7	-0.44	64.68	7.58	6.78	415.44	185.92
DL+0.75*CL7+0.75*Lr	200.09	-0.2	1.6	0.00	45.74	10.39	9.30	476.23	30490.12
DL+0.75*CL8+0.75*Lr	200.11	-0.1	5.5	-0.43	45.91	10.35	9.26	564.43	184.33
DL+0.75*Lr+0.75*WL1	194.32	11.6	1.6	-0.20	103.20	4.47	4.00	6.69	394.55
DL+0.75*Lr+0.75*WL2	189.84	9.6	0.7	-0.10	92.38	4.88	4.37	7.92	752.32
DL+0.75*Lr+0.75*WL3	174.82	-8.1	4.0	-0.27	39.89	10.41	9.31	8.66	260.01
DL+0.75*Lr+0.75*WL4	170.61	-9.7	2.9	-0.21	49.88	8.12	7.27	7.02	329.08
DL+0.75*Lr+0.75*WL5	183.00	-1.1	117.7	12.43	26.72	16.27	14.55	67.31	5.89
DL+0.75*Lr+0.75*WL6	176.57	-2.7	133.4	14.01	21.85	19.19	17.17	26.16	5.04
DL+0.75*Lr+0.75*WL7	182.51	-2.5	117.2	-12.74	28.67	15.12	13.53	29.42	5.73
DL+0.75*Lr+0.75*WL8	177.86	-1.9	128.7	-14.24	26.30	16.06	14.37	37.45	5.00
0.6*DL+WL1	157.50	11.9	0.9	-0.12	107.51	3.48	3.11	5.31	522.29
0.6*DL+WL2	152.88	9.7	2.2	-0.02	96.57	3.76	3.36	6.31	3076.87
0.6*DL+WL3	144.53	-8.2	2.0	-0.19	46.92	7.32	6.55	7.07	307.72
0.6*DL+WL4	140.23	-10.0	1.2	-0.12	57.14	5.83	5.22	5.62	470.87
0.6*DL+WL5	149.53	-1.2	130.4	13.56	25.40	13.98	12.51	48.79	4.41
0.6*DL+WL6	142.95	-3.0	147.3	15.28	21.11	16.09	14.39	19.21	3.74
0.6*DL+WL7	149.11	-2.7	126.2	-13.71	27.43	12.91	11.55	22.10	4.35
0.6*DL+WL8	143.79	-2.2	139.3	-15.33	26.29	12.99	11.62	26.42	3.75

CHECK FOR BEARING PRESSURE

THE NET SAFE BEARING CAPACITY OF SOIL IS 50T/m^2
MAXIMUM GROSS BEARING PRESSURE OBTAINED 24T/m^2 IN LC 22. HENCE IT IS SAFE.
REFER TO THE STAAD FILE ATTACHED.

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	C01	2
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Foundation has been designed with the Design strength Load Combinations of Staad File (LC 100 - LC 132)

REINFORCEMENT CALCULATION OF FOUNDATION:

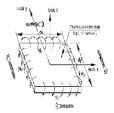
Foundation Thickness =	400	mm
Width =	1000	mm
Clear Cover =	75	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	

STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	9.249	T-M	Corresponding Mxy	-0.43	T-M	*(Plate No. - 27 , L/C No. - 128)*
Min. Mx	-9.063	T-M	Corresponding Mxy	-0.44	T-M	*(Plate No. - 397 , L/C No. - 121)*
Max. My	5.241	T-M	Corresponding Mxy	-1.11	T-M	*(Plate No. - 375 , L/C No. - 101)*
Min. My	-9.181	T-M	Corresponding Mxy	-0.24	T-M	*(Plate No. - 447 , L/C No. - 101)*

FOR OPPOSITE FACE REINFORCEMENT :

Mx1 =	Mx + abs(Mxy) =	9.249 + 0.432 =	9.681 T-M
My1 =	My + abs(Mxy) =	5.241 + 1.113 =	6.354 T-M
Mx2 =	Mx+abs(Mxy^2/My) =	9.249 + 0.036 =	9.285 T-M
My2 =	My+abs(Mxy^2/Mx) =	5.241 + 0.134 =	5.375 T-M



DESIGN MOMENT CALCULATION FOR OPPOSITE FACE REINFORCEMENT (Bottom) :

Mxd = 9.681 T-m
Myd = 6.354 T-m

ALONG X - DIRECTION :

Mu = 9.681 T-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 9.681X10^7/(1000X295^2) = 1.112 N/mm^2 < deff. = 295 mm
Pt = 0.0030 Reqd. Ast = 0.003X1000X295/100 = 8.85 cm^2 < 6.9 N/mm^2
Provided **20** mm @ **200** C/C (Provide Ast = 15.71 cm2) at Horizontal

Total Provide Ast = 15.71 cm2 at Horizontal **O.K.**
Total Provide Pt = 0.01

ALONG Y - DIRECTION :

Mu = 6.354 T-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 6.354X10^7/(1000X315^2) = 0.64 N/mm^2 < deff. = 315 mm
Pt = 0.0017 Reqd. Ast = 0.0017X1000X315/100 = 5.36 cm^2 < 6.9 N/mm^2
Provided **20** mm @ **200** C/C (Provide Ast = 15.71 cm2) at Horizontal

Total Provide Ast = 15.71 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.01

FOR RETAINED FACE REINFORCEMENT :

Mx1 =	Mx - abs(Mxy) =	-9.063 - 0.438 =	-9.501 T-M
My1 =	My - abs(Mxy) =	-9.181 - 0.238 =	-9.419 T-M
Mx2 =	Mx-abs(Mxy^2/My) =	-9.063 - 0.021 =	-9.084 T-M
My2 =	My-abs(Mxy^2/Mx) =	-9.181 - 0.006 =	-9.175 T-M

DESIGN MOMENT CALCULATION FOR RETAINED FACE REINFORCEMENT (Top) :

Mxd = -9.501 T-M
Myd = -9.419 T-M

ALONG X - DIRECTION :

Mu = 9.501 T-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 9.501X10^7/(1000X295^2) = 1.092 N/mm^2 < deff. = 295 mm
Pt = 0.0030 Reqd. Ast = 0.003X1000X295/100 = 8.85 cm^2 < 6.9 N/mm^2
Provided **20** mm @ **200** C/C (Provide Ast = 15.71 cm2) at Horizontal

Total Provide Ast = 15.71 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.01

ALONG Y - DIRECTION :

Mu = 9.419 T-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 9.419X10^7/(1000X315^2) = 0.949 N/mm^2 < deff. = 315 mm
Pt = 0.0026 Reqd. Ast = 0.0026X1000X315/100 = 8.19 cm^2 < 6.9 N/mm^2
Provided **20** mm @ **200** C/C (Provide Ast = 15.71 cm2) at Horizontal

Total Provide Ast = 15.71 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.01

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	C01	2
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

SHEAR CHECK

(FROM FACTORED LOAD COMBINATION)

ALONG X - DIRECTION :

SQx = 0.577 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 391 , L/C No. - 122)*

Vu = 17.022 T

Provided Pt = 0.0053

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0053X17.022X10^4X295/9.501/10^7)X1000X295/10^4

= 29.329 T

[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0053)X1000X295/10^4

= 30.582 T

[c] = 0.29*λ*fc^0.5*b*d.eff

= 0.29X1X5.92X1000X295/10^4

= 50.612 T

HENCE, Vc = 29.329 T

Φt*Vc = 0.75X29.329 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 21.997 T **> 17.022T , Hence O.K.**

ALONG Y - DIRECTION :

SQy = 0.459 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 451 , L/C No. - 117)*

Vu = 14.459 T

Pt provided = 0.005

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.005X14.459X10^4X315/9.419/10^7)X1000X315/10^4

= 31.112 T

[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.005)X1000X315/10^4

= 32.495 KN

[c] = 0.29*λ*fc^0.5*b*d.eff

= 0.29X1X5.92X1000X315/10^4

= 54.043 T

HENCE, Vc = 31.112 T

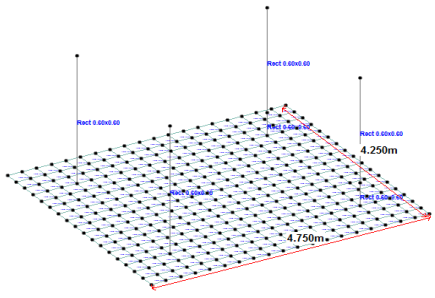
Φt*Vc = 0.75X31.112 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 23.334 T **> 14.459T , Hence O.K.**

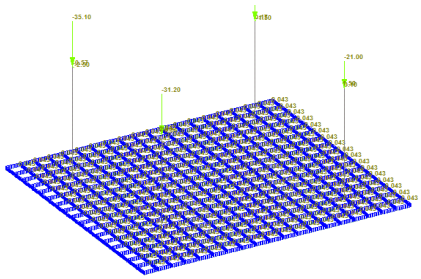
DESIGN OF PEDESTAL: REFER TO STAAD FILE OF FOUNDATION MKD. F4

Size of Column =	600	X	600	mm^2						
Reqired area of Reinforcement =	3600	mm^2								
Reinforcement Provided =	4	no.	25	dia bars	+	4	no.	25	dia bars	
Area of Reinforcement Provided =	3927.0	mm^2								

STAAD MODEL OF RAFT WITH PLATE ELEMENT



STAAD MODEL OF RAFT WITH DEAD + LIVE+ SURCHARGE LOAD



Surcharge Load on Foundation =		2 T/m^2
Soil Weight over the foundation =	1.35 x 1.8 T/m2 =	2.4 T/m^2
SOIL+SURCHARGE LOAD =		4.40 T/m^2

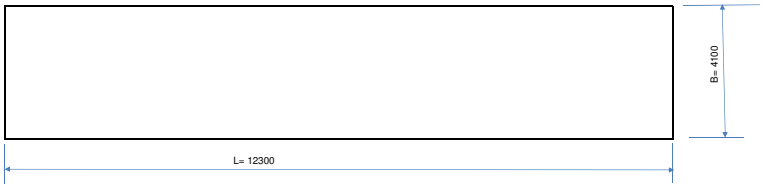
Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01
Date	29-04-2020	01-06-2020	26.10.2023
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Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	C01	2
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED 001	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

4. Design of Engine Foundation

THE ASSIGNMENT LOAD DATA REPRESENT ONLY VERTICAL LOAD. (Refer doc number MD-216-49PR-EG-ME-GAD-V553-1001_RA01_ENGINE_&_GENERATOR)
SINCE THE MACHINE IS SUPPORTED ON SPRING SO NO DYNAMIC CONDITION IS GENERATED.
HENCE FOUNDATION IS DESIGNED FOR VERTICAL LOAD ONLY



TOTAL WEIGHT OF ENGINE = 154 T (Refer the drawing of engine attached)

WEIGHT OF THE FOUNDATION-

L= 12 M
B= 4 M
DEPTH (D)= 1.0 M (From FFL)

WEIGHT OF FOUNDATION (12.3X4.1X1X2.5) 126.075 T

TOTAL WEIGHT OF MACHINE + FOUNDATION= (154+126.1)
280 T

CALCULATION OF BEARING PRESSURE:

BEARING PRESSURE = TOTAL WT / AREA

= 5.55 T/m^2 < 50 T/m^2 (HENCE OK)

REINFORCEMENT :

REINFORCEMENT PROVIDED @ 30Kg/m^3 (**refer ACI 351.3R-04**)

TOTAL QUANTITY OF CONCRETE = 50.43 CUM

REINFORCEMENT REQUIRED= 1.51 MT
(each foundation)

REINFORCEMENT PROVIDED IN EACH FOUNDATION:
16 DIA BAR = 1917kg
12 DIA BAR = 554Kg

TOTAL WEIGHT = 2471Kg > 1510 Kg (HENCE OK)

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	C01	2
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

REFERENCES

[1] Structural Loading and Forces ; SBC Code 301: 2007
[2] Saudi Concrete Structures Code ; SBC 304 : 2018
[3] Civil,Structural and Architectural Design Criteria ; MD-216-1000-EG-ST-CRT-1001
[4] ACI 318 : 2014 Building Code Requirement of Structural Concrete
[5] Geotechnical Report No. MD-216-3000-EG-CV-RPT-1008
[6] ENGINE HALL DESIGN CALCULATION NO. MD-216-49PR-EG-ST-CAL-2410

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	C01	2			
Date	29-04-2020	01-06-2020	26.10.2023				Date	29-04-2020	01-06-2020	26.10.2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT			
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD			
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

ATTACHMENT 1 Foundation Load Data for Engine Hall Building Columns

MMGP:POWER PLANT:ENGINE HALL FOUNDATION LOAD DATA

1 DL	Dead load	
2 LLROOF	Live load on roof	
3 CL1(L+XS)	Crane load	Max Column concentration at B grid,Surge in direction in +X (from B grid to F grid)
4 CL2(L-XS)	Crane load	Max Column concentration at B grid,Surge in direction in -X (from F grid to B grid)
5 CL3(R+XS)	Crane load	Max Column concentration at F grid,Surge in direction in +X (from B grid to F grid)
6 CL4(R-XS)	Crane load	Max Column concentration at F grid,Surge in direction in -X (from F grid to B grid)
7 CL5(L+ZT)	Crane load	Max Column concentration at B grid,tractive in direction in +Z (from line-1 to line-9)
8 CL6(L-ZT)	Crane load	Max Column concentration at B grid,tractive in direction in -Z (from line-9 to line-1)
9 CL5(R+ZT)	Crane load	Max Column concentration at F grid,tractive in direction in +Z (from line-1 to line-9)
10 CL6(R-ZT)	Crane load	Max Column concentration at F grid,tractive in direction in -Z (from line-9 to line-1)
21 WL1(+X)S	Wind load	Wind load in direction of +X Suction(from A grid to F grid) side wind
22 WL2(+X)P	Wind load	Wind load in direction of +X Pressure(from A grid to F grid) side wind
23 WL3(-X)S	Wind load	Wind load in direction of -X Suction(from F grid to A grid) side wind
24 WL4(-X)P	Wind load	Wind load in direction of -X Pressure(from F grid to A grid) side wind
25 WL5(+Z)S	Wind load	Wind load in direction of +Z Suction(from line-1 to line-9) gable wind
26 WL6(+Z)P	Wind load	Wind load in direction of +Z Pressure(from line-1 to line-9) gable wind
27 WL7(-Z)S	Wind load	Wind load in direction of -Z Suction(from line-9 to line-1) gable wind
28 WL8(-Z)P	Wind load	Wind load in direction of -Z Pressure(from line-9 to line-1) gable wind

All crane and wind load cases are exclusive.

Sign convention

FY	+ve sign indicates vertical load downward
	-ve sign indicates vertical load upward
FX	+ve sign indicates horizonal load in across direction from grid-A to F
	-ve sign indicates horizonal load in across direction from grid-F to A
FZ	+ve sign indicates horizonal load in along direction from line 0 to 8
	-ve sign indicates horizonal load in along direction from line 8 to 0

LOAD DATA TABLE

		Load cases	Horizontal Fx Ton	Vertical Fy Ton	Horizontal Fz Ton
Line-0,1,2,3,4,5,6	F grid Line-0,1,2,3,4,5,6	1 DL	-0.4	21.6	0.0
		2 LLROOF	0.0	6.4	0.0
		3 CL1(L+XS)	0.2	1.9	0.0
		4 CL2(L-XS)	0.2	1.9	0.0
		5 CL3(R+XS)	0.6	4.6	0.0
		6 CL4(R-XS)	0.4	4.6	0.0
		7 CL5(L+ZT)	0.2	1.9	0.0
		8 CL6(L-ZT)	0.2	1.9	0.0
		9 CL5(R+ZT)	0.5	4.6	0.0
		10 CL6(R-ZT)	0.5	4.6	0.0
		21 WL1(+X)S	0.6	-1.4	0.0
		22 WL2(+X)P	1.7	-3.2	0.0
		23 WL3(-X)S	-2.0	-2.9	0.0
		24 WL4(-X)P	-0.9	-4.4	0.0
		25 WL5(+Z)S	0.9	-2.6	0.0
		26 WL6(+Z)P	1.9	-4.9	0.0
		27 WL7(-Z)S	0.9	-2.6	0.0
		28 WL8(-Z)P	1.9	-4.6	0.0
	B grid Line-0,1,2,3,4,5,6	1 DL	0.6	24.0	0.0
		2 LLROOF	0.0	7.6	0.0
		3 CL1(L+XS)	0.2	6.2	0.0
		4 CL2(L-XS)	-0.5	4.6	0.0
		5 CL3(R+XS)	-0.2	1.7	0.0
		6 CL4(R-XS)	-0.7	0.0	0.0
		7 CL5(L+ZT)	-0.1	5.4	0.0
		8 CL6(L-ZT)	-0.1	5.4	0.0
		9 CL5(R+ZT)	-0.5	0.9	0.0
		10 CL6(R-ZT)	-0.5	0.9	0.0
		21 WL1(+X)S	3.5	4.1	0.0
		22 WL2(+X)P	2.9	2.8	0.0
		23 WL3(-X)S	-2.4	-8.7	0.0
		24 WL4(-X)P	-3.1	-10.4	0.0
		25 WL5(+Z)S	-0.6	-3.0	0.0
		26 WL6(+Z)P	-1.4	-5.8	0.0
		27 WL7(-Z)S	-0.5	-2.9	0.0
		28 WL8(-Z)P	-1.2	-4.9	0.0
	A grid Line-0,1,2,3,4,5,6	1 DL	-0.1	19.6	0.0
		2 LLROOF	0.0	1.2	0.0
		3 CL1(L+XS)	-0.1	-1.6	0.0
		4 CL2(L-XS)	-0.1	0.0	0.0
		5 CL3(R+XS)	0.0	0.1	0.0
		6 CL4(R-XS)	0.0	1.8	0.0
		7 CL5(L+ZT)	-0.1	-0.8	0.0
		8 CL6(L-ZT)	-0.1	-0.8	0.0
		9 CL5(R+ZT)	0.0	1.0	0.0
		10 CL6(R-ZT)	0.0	1.0	0.0
		21 WL1(+X)S	0.7	-7.6	0.0
		22 WL2(+X)P	0.4	-8.4	0.0
		23 WL3(-X)S	-0.2	6.4	0.0
		24 WL4(-X)P	-0.6	6.3	0.0
		25 WL5(+Z)S	-0.3	-0.5	0.0
		26 WL6(+Z)P	-0.7	-0.7	0.0
		27 WL7(-Z)S	-0.3	-0.7	0.0
		28 WL8(-Z)P	-0.7	-1.3	0.0
Line-7	F grid Line-7	1 DL	-0.4	20.8	-1.8
		2 LLROOF	0.0	6.0	-0.2
		3 CL1(L+XS)	0.2	1.9	-0.2
		4 CL2(L-XS)	0.2	1.9	-0.2
		5 CL3(R+XS)	0.6	4.7	-0.4
		6 CL4(R-XS)	0.4	4.7	-0.4
		7 CL5(L+ZT)	0.2	1.4	-0.1
		8 CL6(L-ZT)	0.2	2.4	-0.3
		9 CL5(R+ZT)	0.5	4.2	-0.3
		10 CL6(R-ZT)	0.5	5.2	-0.6
		21 WL1(+X)S	0.6	-1.0	0.1
		22 WL2(+X)P	1.7	-2.0	0.1
		23 WL3(-X)S	-2.0	-2.5	0.2
		24 WL4(-X)P	-0.9	-3.4	0.2
		25 WL5(+Z)S	0.9	-20.6	3.1
		26 WL6(+Z)P	1.9	-24.2	3.6
		27 WL7(-Z)S	0.9	16.4	-2.8
		28 WL8(-Z)P	1.9	16.9	-3.0

	Load cases	Horizontal Fx ton	Vertical Fy ton	Horizontal Fz ton
All gable post (C0,D0,E0, C8,D8,E8)	1 DL	0	4.5	0
	2 LLROOF	0	0	0
	3 CL1(L+XS)	0	0	0
	4 CL2(L-XS)	0	0	0
	5 CL3(R+XS)	0	0	0
	6 CL4(R-XS)	0	0	0
	7 CL5(L+ZT)	0	0	0
	8 CL6(L-ZT)	0	0	0
	9 CL5(R+ZT)	0	0	0
	10 CL6(R-ZT)	0	0	0
	21 WL1(+X)S	0	0	-0.76
	22 WL2(+X)P	0	0	-1.82
	23 WL3(-X)S	0	0	-0.76
	24 WL4(-X)P	0	0	-1.82
	25 WL5(+Z)S	0	0	1.82
	26 WL6(+Z)P	0	0	0.88
	27 WL7(-Z)S	0	0	-0.59
	28 WL8(-Z)P	0	0	-1.76

	B grid Line-7	1 DL	0.5	24.2	-2.4
		2 LLROOF	0.0	7.0	-0.3
		3 CL1(L+XS)	0.2	6.2	-0.6
		4 CL2(L-XS)	-0.5	4.6	-0.5
		5 CL3(R+XS)	-0.2	1.6	-0.2
		6 CL4(R-XS)	-0.7	0.0	-0.1
		7 CL5(L+ZT)	-0.2	5.0	-0.5
		8 CL6(L-ZT)	-0.1	5.8	-0.6
		9 CL5(R+ZT)	-0.5	0.4	0.0
		10 CL6(R-ZT)	-0.5	1.2	-0.2
		21 WL1(+X)S	3.4	4.3	-0.1
		22 WL2(+X)P	2.7	2.7	-0.1
		23 WL3(-X)S	-2.2	-7.4	0.3
		24 WL4(-X)P	-3.1	-9.0	0.3
		25 WL5(+Z)S	-0.8	-17.0	3.0
		26 WL6(+Z)P	-1.6	-21.0	3.4
		27 WL7(-Z)S	-0.5	11.2	-2.5
		28 WL8(-Z)P	-1.1	11.2	-2.9
	A grid Line-7	1 DL	-0.1	19.6	-1.5
		2 LLROOF	0.0	1.4	-0.1
		3 CL1(L+XS)	0.0	-1.6	0.2
		4 CL2(L-XS)	0.0	0.0	0.0
		5 CL3(R+XS)	0.0	0.2	0.0
		6 CL4(R-XS)	0.0	1.8	-0.2
		7 CL5(L+ZT)	0.0	-1.0	0.1
		8 CL6(L-ZT)	0.0	-0.6	0.1
		9 CL5(R+ZT)	0.0	0.8	-0.1
		10 CL6(R-ZT)	0.0	1.2	-0.1
		21 WL1(+X)S	0.7	-7.4	0.7
		22 WL2(+X)P	0.4	-7.6	0.7
		23 WL3(-X)S	-0.2	5.7	-0.6
		24 WL4(-X)P	-0.6	5.9	-0.6
		25 WL5(+Z)S	-0.3	-12.7	2.1
		26 WL6(+Z)P	-0.6	-14.3	2.4
		27 WL7(-Z)S	-0.3	11.8	-2.0
		28 WL8(-Z)P	-0.7	12.4	-2.1
Line-8	F grid Line-8	1 DL	-0.5	25.0	1.9
		2 LLROOF	0.0	7.2	0.2
		3 CL1(L+XS)	0.1	2.0	0.2
		4 CL2(L-XS)	0.1	2.0	0.2
		5 CL3(R+XS)	0.4	4.8	0.4
		6 CL4(R-XS)	0.1	4.8	0.4
		7 CL5(L+ZT)	0.1	2.5	0.3
		8 CL6(L-ZT)	0.1	1.4	0.1
		9 CL5(R+ZT)	0.2	5.4	0.5
		10 CL6(R-ZT)	0.2	4.2	0.3
		21 WL1(+X)S	1.2	-1.7	-0.1
		22 WL2(+X)P	1.8	-4.1	-0.2
		23 WL3(-X)S	-2.1	-3.1	-0.2
		24 WL4(-X)P	-1.0	-5.3	-0.3
		25 WL5(+Z)S	0.9	16.3	3.1
		26 WL6(+Z)P	2.1	15.9	3.5
		27 WL7(-Z)S	0.9	-22.5	-3.4
		28 WL8(-Z)P	2.0	-27.1	-3.9
	B grid Line-8	1 DL	0.5	27.6	2.2
		2 LLROOF	0.0	7.5	0.3
		3 CL1(L+XS)	0.3	6.1	0.6
		4 CL2(L-XS)	-0.4	4.4	0.5
		5 CL3(R+XS)	0.1	2.4	0.2
		6 CL4(R-XS)	-0.5	0.6	0.1
		7 CL5(L+ZT)	-0.1	5.7	0.6
		8 CL6(L-ZT)	-0.1	4.8	0.5
		9 CL5(R+ZT)	-0.2	1.9	0.2
		10 CL6(R-ZT)	-0.2	1.1	0.0
		21 WL1(+X)S	3.7	4.7	0.1
		22 WL2(+X)P	3.2	3.3	0.2
		23 WL3(-X)S	-2.7	-9.7	-0.4
		24 WL4(-X)P	-3.5	-11.7	-0.3
		25 WL5(+Z)S	-0.4	11.7	3.1
		26 WL6(+Z)P	-1.2	10.8	3.6
		27 WL7(-Z)S	-1.6	-17.7	-3.3
		28 WL8(-Z)P	-1.5	-20.9	-3.7
	A grid Line-8	1 DL	-0.1	21.8	1.5
		2 LLROOF	0.0	1.2	0.0
		3 CL1(L+XS)	0.0	-1.3	-0.2
		4 CL2(L-XS)	0.0	0.4	0.0
		5 CL3(R+XS)	0.0	-0.4	0.0
		6 CL4(R-XS)	0.0	1.4	0.2
		7 CL5(L+ZT)	0.0	-0.1	-0.1
		8 CL6(L-ZT)	0.0	-0.7	-0.1
		9 CL5(R+ZT)	0.0	0.8	0.1
		10 CL6(R-ZT)	0.0	0.2	0.1
		21 WL1(+X)S	0.7	-8.5	-0.7
		22 WL2(+X)P	0.4	-9.1	-0.6
		23 WL3(-X)S	-0.2	7.0	0.6
		24 WL4(-X)P	-0.6	6.7	0.7
		25 WL5(+Z)S	-0.3	12.3	2.3
		26 WL6(+Z)P	-0.8	13.4	2.6
		27 WL7(-Z)S	-0.6	-12.8	-2.3
		28 WL8(-Z)P	-0.7	-15.8	-2.6

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Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD			
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

ATTACHMENT 2 STAAD Input & Output File of Foundation Mkd. F3

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries5           *
*           Version  20.07.10.64                  *
*           Proprietary Program of                *
*           Bentley Systems, Inc.                 *
*           Date=    APR 29, 2020                 *
*           Time=    17: 9:51                     *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F8-F7_FDN_29.04.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 02-APR-19
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER MTON
     7. JOINT COORDINATES
     8. 13 0 -4.2 0; 14 0 -4.2 -1.5; 16 0 -4.2 1; 18 -1 -4.2 1; 20 1 -4.2 1
     9. 21 0.75 -4.2 1; 22 0.75 -4.2 0.75; 23 1 -4.2 0.75; 24 0.5 -4.2 1
    10. 25 0.5 -4.2 0.75; 26 0.25 -4.2 1; 27 0.25 -4.2 0.75; 28 0 -4.2 0.75
    11. 29 0.75 -4.2 0.5; 30 1 -4.2 0.5; 31 0.5 -4.2 0.5; 32 0.25 -4.2 0.5
    12. 33 0 -4.2 0.5; 34 0.75 -4.2 0.25; 35 1 -4.2 0.25; 36 0.5 -4.2 0.25
    13. 37 0.25 -4.2 0.25; 38 0 -4.2 0.25; 39 0.75 -4.2 0; 40 1 -4.2 0; 41 0.5 -4.2 0
    14. 42 0.25 -4.2 0; 43 0.75 -4.2 -0.25; 44 1 -4.2 -0.25; 45 0.5 -4.2 -0.25
    15. 46 0.25 -4.2 -0.25; 47 0 -4.2 -0.25; 48 0.75 -4.2 -0.5; 49 1 -4.2 -0.5
    16. 50 0.5 -4.2 -0.5; 51 0.25 -4.2 -0.5; 52 0 -4.2 -0.5; 53 0.75 -4.2 -0.75
    17. 54 1 -4.2 -0.75; 55 0.5 -4.2 -0.75; 56 0.25 -4.2 -0.75; 57 0 -4.2 -0.75
    18. 58 0.75 -4.2 -1; 59 1 -4.2 -1; 60 0.5 -4.2 -1; 61 0.25 -4.2 -1; 62 0 -4.2 -1.
    19. 63 0.75 -4.2 -1.25; 64 1 -4.2 -1.25; 65 0.5 -4.2 -1.25; 66 0.25 -4.2 -1.25
    20. 67 0 -4.2 -1.25; 68 0.75 -4.2 -1.5; 69 1 -4.2 -1.5; 70 0.5 -4.2 -1.5
    21. 71 0.25 -4.2 -1.5; 72 0.75 -4.2 -1.75; 73 1 -4.2 -1.75; 74 0.5 -4.2 -1.75
    22. 75 0.25 -4.2 -1.75; 76 0 -4.2 -1.75; 77 0.75 -4.2 -2; 78 1 -4.2 -2
    23. 79 0.5 -4.2 -2; 80 0.25 -4.2 -2; 81 0 -4.2 -2; 82 0.75 -4.2 -2.25
    24. 83 1 -4.2 -2.25; 84 0.5 -4.2 -2.25; 85 0.25 -4.2 -2.25; 86 0 -4.2 -2.25
    25. 90 -0.25 -4.2 1; 91 -0.25 -4.2 0.75; 92 -0.5 -4.2 1; 93 -0.5 -4.2 0.75
    26. 94 -0.75 -4.2 1; 95 -0.75 -4.2 0.75; 96 -1 -4.2 0.75; 97 -0.25 -4.2 0.5
    27. 98 -0.5 -4.2 0.5; 99 -0.75 -4.2 0.5; 100 -1 -4.2 0.5; 101 -0.25 -4.2 0.25
    28. 102 -0.5 -4.2 0.25; 103 -0.75 -4.2 0.25; 104 -1 -4.2 0.25; 105 -0.25 -4.2 0
    29. 106 -0.5 -4.2 0; 107 -0.75 -4.2 0; 108 -1 -4.2 0; 109 -0.25 -4.2 -0.25
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    31. 113 -0.25 -4.2 -0.5; 114 -0.5 -4.2 -0.5; 115 -0.75 -4.2 -0.5; 116 -1 -4.2 -0.5
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    33. 120 -1 -4.2 -0.75; 121 -0.25 -4.2 -1; 122 -0.5 -4.2 -1; 123 -0.75 -4.2 -1.
    34. 124 -1 -4.2 -1; 125 -0.25 -4.2 -1.25; 126 -0.5 -4.2 -1.25
    35. 127 -0.75 -4.2 -1.25; 128 -1 -4.2 -1.25; 129 -0.25 -4.2 -1.5
    36. 130 -0.5 -4.2 -1.5; 131 -0.75 -4.2 -1.5; 132 -1 -4.2 -1.5
    37. 133 -0.25 -4.2 -1.75; 134 -0.5 -4.2 -1.75; 135 -0.75 -4.2 -1.75
    38. 136 -1 -4.2 -1.75; 137 -0.25 -4.2 -2; 138 -0.5 -4.2 -2; 139 -0.75 -4.2 -2
```

39. 140 -1 -4.2 -2; 141 -0.25 -4.2 -2.25; 142 -0.5 -4.2 -2.25
40. 143 -0.75 -4.2 -2.25; 144 -1 -4.2 -2.25; 145 0 -1.95 -2; 146 0 -1.95 0.75
41. 147 0.75 -4.2 -2.5; 148 1 -4.2 -2.5; 149 0.5 -4.2 -2.5; 150 0.25 -4.2 -2.5
42. 151 0 -4.2 -2.5; 152 -0.25 -4.2 -2.5; 153 -0.5 -4.2 -2.5; 154 -0.75 -4.2 -2.5
43. 155 -1 -4.2 -2.5; 156 1 -4.2 1.25; 157 0.75 -4.2 1.25; 158 0.5 -4.2 1.25
44. 159 0.25 -4.2 1.25; 160 0 -4.2 1.25; 161 -0.25 -4.2 1.25; 162 -0.5 -4.2 1.25
45. 163 -0.75 -4.2 1.25; 164 -1 -4.2 1.25; 165 -1.25 -4.2 1; 166 -1.25 -4.2 0.75
46. 167 -1.25 -4.2 0.5; 168 -1.25 -4.2 0.25; 169 -1.25 -4.2 0
47. 170 -1.25 -4.2 -0.25; 171 -1.25 -4.2 -0.5; 172 -1.25 -4.2 -0.75
48. 173 -1.25 -4.2 -1; 174 -1.25 -4.2 -1.25; 175 -1.25 -4.2 -1.5
49. 176 -1.25 -4.2 -1.75; 177 -1.25 -4.2 -2; 178 -1.25 -4.2 -2.25
50. 179 -1.25 -4.2 -2.5; 180 -1.25 -4.2 1.25; 181 1.25 -4.2 1; 182 1.25 -4.2 0.75
51. 183 1.25 -4.2 0.5; 184 1.25 -4.2 0.25; 185 1.25 -4.2 0; 186 1.25 -4.2 -0.25
52. 187 1.25 -4.2 -0.5; 188 1.25 -4.2 -0.75; 189 1.25 -4.2 -1; 190 1.25 -4.2 -1.25
53. 191 1.25 -4.2 -1.5; 192 1.25 -4.2 -1.75; 193 1.25 -4.2 -2; 194 1.25 -4.2 -2.25
54. 195 1.25 -4.2 -2.5; 196 1.25 -4.2 1.25; 197 0.75 -4.2 -2.75; 198 1 -4.2 -2.75
55. 199 0.5 -4.2 -2.75; 200 0.25 -4.2 -2.75; 201 0 -4.2 -2.75
56. 202 -0.25 -4.2 -2.75; 203 -0.5 -4.2 -2.75; 204 -0.75 -4.2 -2.75
57. 205 -1 -4.2 -2.75; 206 -1.25 -4.2 -2.75; 207 1.25 -4.2 -2.75; 208 1 -4.2 1.5
58. 209 0.75 -4.2 1.5; 210 0.5 -4.2 1.5; 211 0.25 -4.2 1.5; 212 0 -4.2 1.5
59. 213 -0.25 -4.2 1.5; 214 -0.5 -4.2 1.5; 215 -0.75 -4.2 1.5; 216 -1 -4.2 1.5
60. 217 -1.25 -4.2 1.5; 218 1.25 -4.2 1.5; 219 0.75 -4.2 -3; 220 1 -4.2 -3
61. 221 0.5 -4.2 -3; 222 0.25 -4.2 -3; 223 0 -4.2 -3; 224 -0.25 -4.2 -3
62. 225 -0.5 -4.2 -3; 226 -0.75 -4.2 -3; 227 -1 -4.2 -3; 228 -1.25 -4.2 -3
63. 229 1.25 -4.2 -3; 230 1 -4.2 1.75; 231 0.75 -4.2 1.75; 232 0.5 -4.2 1.75
64. 233 0.25 -4.2 1.75; 234 0 -4.2 1.75; 235 -0.25 -4.2 1.75; 236 -0.5 -4.2 1.75
65. 237 -0.75 -4.2 1.75; 238 -1 -4.2 1.75; 239 -1.25 -4.2 1.75; 240 1.25 -4.2 1.75
66. 241 1.5 -4.2 1; 242 1.5 -4.2 0.75; 243 1.5 -4.2 0.5; 244 1.5 -4.2 0.25
67. 245 1.5 -4.2 0; 246 1.5 -4.2 -0.25; 247 1.5 -4.2 -0.5; 248 1.5 -4.2 -0.75
68. 249 1.5 -4.2 -1; 250 1.5 -4.2 -1.25; 251 1.5 -4.2 -1.5; 252 1.5 -4.2 -1.75
69. 253 1.5 -4.2 -2; 254 1.5 -4.2 -2.25; 255 1.5 -4.2 -2.5; 256 1.5 -4.2 1.25
70. 257 1.5 -4.2 -2.75; 258 1.5 -4.2 1.5; 259 1.5 -4.2 -3; 260 1.5 -4.2 1.75
71. 261 -1.5 -4.2 1; 262 -1.5 -4.2 0.75; 263 -1.5 -4.2 0.5; 264 -1.5 -4.2 0.25
72. 265 -1.5 -4.2 0; 266 -1.5 -4.2 -0.25; 267 -1.5 -4.2 -0.5; 268 -1.5 -4.2 -0.75
73. 269 -1.5 -4.2 -1; 270 -1.5 -4.2 -1.25; 271 -1.5 -4.2 -1.5; 272 -1.5 -4.2 -1.75
74. 273 -1.5 -4.2 -2; 274 -1.5 -4.2 -2.25; 275 -1.5 -4.2 -2.5; 276 -1.5 -4.2 1.25
75. 277 -1.5 -4.2 -2.75; 278 -1.5 -4.2 1.5; 279 -1.5 -4.2 -3; 280 -1.5 -4.2 1.75
76. MEMBER INCIDENCES
77. 133 81 145; 134 28 146
78. ELEMENT INCIDENCES SHELL
79. 25 20 21 22 23; 26 21 24 25 22; 27 24 26 27 25; 28 26 16 28 27; 29 23 22 29 30
80. 30 22 25 31 29; 31 25 27 32 31; 32 27 28 33 32; 33 30 29 34 35; 34 29 31 36 34
81. 35 31 32 37 36; 36 32 33 38 37; 37 35 34 39 40; 38 34 36 41 39; 39 36 37 42 41
82. 40 37 38 13 42; 41 40 39 43 44; 42 39 41 45 43; 43 41 42 46 45; 44 42 13 47 46
83. 45 44 43 48 49; 46 43 45 50 48; 47 45 46 51 50; 48 46 47 52 51; 49 49 48 53 54
84. 50 48 50 55 53; 51 50 51 56 55; 52 51 52 57 56; 53 54 53 58 59; 54 53 55 60 58
85. 55 55 56 61 60; 56 56 57 62 61; 57 59 58 63 64; 58 58 60 65 63; 59 60 61 66 65
86. 60 61 62 67 66; 61 64 63 68 69; 62 63 65 70 68; 63 65 66 71 70; 64 66 67 14 71
87. 65 69 68 72 73; 66 68 70 74 72; 67 70 71 75 74; 68 71 14 76 75; 69 73 72 77 78
88. 70 72 74 79 77; 71 74 75 80 79; 72 75 76 81 80; 73 78 77 82 83; 74 77 79 84 82
89. 75 79 80 85 84; 76 80 81 86 85; 81 16 90 91 28; 82 90 92 93 91; 83 92 94 95 93
90. 84 94 18 96 95; 85 28 91 97 33; 86 91 93 98 97; 87 93 95 99 98
91. 88 95 96 100 99; 89 33 97 101 38; 90 97 98 102 101; 91 98 99 103 102
92. 92 99 100 104 103; 93 38 101 105 13; 94 101 102 106 105; 95 102 103 107 106
93. 96 103 104 108 107; 97 13 105 109 47; 98 105 106 110 109; 99 106 107 111 110
94. 100 107 108 112 111; 101 47 109 113 52; 102 109 110 114 113

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 95. 103 110 111 115 114; 104 111 112 116 115; 105 52 113 117 57
 96. 106 113 114 118 117; 107 114 115 119 118; 108 115 116 120 119
 97. 109 57 117 121 62; 110 117 118 122 121; 111 118 119 123 122
 98. 112 119 120 124 123; 113 62 121 125 67; 114 121 122 126 125
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100. 118 125 126 130 129; 119 126 127 131 130; 120 127 128 132 131
101. 121 14 129 133 76; 122 129 130 134 133; 123 130 131 135 134
102. 124 131 132 136 135; 125 76 133 137 81; 126 133 134 138 137
103. 127 134 135 139 138; 128 135 136 140 139; 129 81 137 141 86
104. 130 137 138 142 141; 131 138 139 143 142; 132 139 140 144 143
105. 135 83 82 147 148; 136 82 84 149 147; 137 84 85 150 149; 138 85 86 151 150
106. 139 86 141 152 151; 140 141 142 153 152; 141 142 143 154 153
107. 142 143 144 155 154; 143 156 157 21 20; 144 157 158 24 21; 145 158 159 26 24
108. 146 159 160 16 26; 147 160 161 90 16; 148 161 162 92 90; 149 162 163 94 92
109. 150 163 164 18 94; 151 18 165 166 96; 152 96 166 167 100; 153 100 167 168 104
110. 154 104 168 169 108; 155 108 169 170 112; 156 112 170 171 116
111. 157 116 171 172 120; 158 120 172 173 124; 159 124 173 174 128
112. 160 128 174 175 132; 161 132 175 176 136; 162 136 176 177 140
113. 163 140 177 178 144; 164 144 178 179 155; 165 164 180 165 18
114. 166 181 20 23 182; 167 182 23 30 183; 168 183 30 35 184; 169 184 35 40 185
115. 170 185 40 44 186; 171 186 44 49 187; 172 187 49 54 188; 173 188 54 59 189
116. 174 189 59 64 190; 175 190 64 69 191; 176 191 69 73 192; 177 192 73 78 193
117. 178 193 78 83 194; 179 194 83 148 195; 180 196 156 20 181; 181 148 147 197 198
118. 182 147 149 199 197; 183 149 150 200 199; 184 150 151 201 200
119. 185 151 152 202 201; 186 152 153 203 202; 187 153 154 204 203
120. 188 154 155 205 204; 189 155 179 206 205; 190 195 148 198 207
121. 191 208 209 157 156; 192 209 210 158 157; 193 210 211 159 158
122. 194 211 212 160 159; 195 212 213 161 160; 196 213 214 162 161
123. 197 214 215 163 162; 198 215 216 164 163; 199 216 217 180 164
124. 200 218 208 156 196; 201 198 197 219 220; 202 197 199 221 219
125. 203 199 200 222 221; 204 200 201 223 222; 205 201 202 224 223
126. 206 202 203 225 224; 207 203 204 226 225; 208 204 205 227 226
127. 209 205 206 228 227; 210 207 198 220 229; 211 230 231 209 208
128. 212 231 232 210 209; 213 232 233 211 210; 214 233 234 212 211
129. 215 234 235 213 212; 216 235 236 214 213; 217 236 237 215 214
130. 218 237 238 216 215; 219 238 239 217 216; 220 240 230 208 218
131. 221 241 181 182 242; 222 242 182 183 243; 223 243 183 184 244
132. 224 244 184 185 245; 225 245 185 186 246; 226 246 186 187 247
133. 227 247 187 188 248; 228 248 188 189 249; 229 249 189 190 250
134. 230 250 190 191 251; 231 251 191 192 252; 232 252 192 193 253
135. 233 253 193 194 254; 234 254 194 195 255; 235 256 196 181 241
136. 236 255 195 207 257; 237 258 218 196 256; 238 257 207 229 259
137. 239 260 240 218 258; 240 165 261 262 166; 241 166 262 263 167
138. 242 167 263 264 168; 243 168 264 265 169; 244 169 265 266 170
139. 245 170 266 267 171; 246 171 267 268 172; 247 172 268 269 173
140. 248 173 269 270 174; 249 174 270 271 175; 250 175 271 272 176
141. 251 176 272 273 177; 252 177 273 274 178; 253 178 274 275 179
142. 254 180 276 261 165; 255 179 275 277 206; 256 217 278 276 180
143. 257 206 277 279 228; 258 239 280 278 217
144. START GROUP DEFINITION
145. ELEMENT
146. _MAX_BM 25 TO 27 29 TO 31 33 TO 71 73 TO 75 82 TO 84 86 TO 124 126 TO 128 -
147. 130 TO 132 135 TO 258
148. _MAX_SHEAR 25 26 29 30 33 34 37 TO 66 69 70 73 74 83 84 87 88 91 TO 120 123 -
149. 124 127 128 131 132 135 136 141 TO 144 149 TO 258
150. END GROUP DEFINITION
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151. ELEMENT PROPERTY
152. 25 TO 76 81 TO 132 135 TO 258 THICKNESS 0.4
153. SUPPORTS
154. 13 14 16 18 20 TO 86 90 TO 144 147 TO 279 -
155. 280 ELASTIC MAT DIRECT Y SUBGRADE 7080 PRINT
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JOINT	INFLUENCE AREA
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14	0.062500
16	0.062500
18	0.062500
20	0.062500
21	0.062500
22	0.062500
23	0.062500
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274	0.031250
275	0.031250
276	0.031250
277	0.031250
278	0.031250
279	0.015625
280	0.015625
TOTAL	14.250000

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156. DEFINE MATERIAL START
157. ISOTROPIC STEEL
158. E 2.09042E+007
159. POISSON 0.3
160. DENSITY 7.83341
161. ALPHA 1.2E-005
162. DAMP 0.03
163. TYPE STEEL
164. STRENGTH FY 25819.2 FU 41584 RY 1.5 RT 1.2
165. ISOTROPIC CONCRETE
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166. E 2.21467E+006
167. POISSON 0.17
168. DENSITY 2.40262
169. ALPHA 1E-005
170. DAMP 0.05
171. TYPE CONCRETE
172. STRENGTH FCU 2812.28
173. END DEFINE MATERIAL
174. MEMBER PROPERTY AMERICAN
175. 133 134 PRIS YD 0.6 ZD 0.6
176. CONSTANTS
177. MATERIAL CONCRETE ALL
178. LOAD 1 DL+LR
179. *SOIL+SURCHARGE=1X1.8+2=4.4T/SQ.M)
180. SELFWEIGHT Y -1
181. ELEMENT LOAD
182. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.4
183. JOINT LOAD
184. 145 FX 0.5 FY -32.19 FZ -2.2
185. 146 FX 0.4 FY -26.77 FZ 2
186. LOAD 2 DL+CL1
187. SELFWEIGHT Y -1
188. ELEMENT LOAD
189. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.4
190. JOINT LOAD
191. 145 FX 0.4 FY -26.99 FZ -2.13
192. 146 FX 0.21 FY -22.66 FZ 1.96
193. LOAD 3 DL+CL2
194. SELFWEIGHT Y -1
195. ELEMENT LOAD
196. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.4
197. JOINT LOAD
198. 145 FX 0.4 FY -26.99 FZ -2.13
199. 146 FX 0.21 FY -22.67 FZ 1.97
200. LOAD 4 DL+CL3
201. SELFWEIGHT Y -1
202. ELEMENT LOAD
203. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.4
204. JOINT LOAD
205. 145 FX 0.14 FY -29.79 FZ -2.38
206. 146 FX -0.17 FY -25.46 FZ 2.23
207. LOAD 5 DL+CL4
208. SELFWEIGHT Y -1
209. ELEMENT LOAD
210. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.4
211. JOINT LOAD
212. 145 FX 0.37 FY -29.86 FZ -2.38
213. 146 FX 0.05 FY -25.44 FZ 2.23
214. LOAD 6 DL+CL5
215. SELFWEIGHT Y -1
216. ELEMENT LOAD
217. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.4
218. JOINT LOAD
219. 145 FX 0.4 FY -27.56 FZ -2.24
220. 146 FX 0.21 FY -22.19 FZ 1.86
221. LOAD 7 DL+CL6
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222. SELFWEIGHT Y -1
223. ELEMENT LOAD
224. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.4
225. JOINT LOAD
226. 145 FX 0.4 FY -26.41 FZ -2.02
227. 146 FX 0.21 FY -23.14 FZ 2.07
228. LOAD 8 DL+CL7
229. SELFWEIGHT Y -1
230. ELEMENT LOAD
231. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.4
232. JOINT LOAD
233. 145 FX 0.25 FY -30.4 FZ -2.49
234. 146 FX -0.06 FY -24.97 FZ 2.12
235. LOAD 9 DL+CL8
236. SELFWEIGHT Y -1
237. ELEMENT LOAD
238. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.4
239. JOINT LOAD
240. 145 FX 0.26 FY -29.25 FZ -2.27
241. 146 FX -0.06 FY -25.93 FZ 2.34
242. LOAD 10 DL+0.75*CL1+0.75*LR
243. SELFWEIGHT Y -1
244. ELEMENT LOAD
245. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.4
246. JOINT LOAD
247. 145 FX 0.42 FY -31.87 FZ -2.27
248. 146 FX 0.26 FY -26.69 FZ 2.08
249. LOAD 11 DL+0.75*CL2+0.75*LR
250. SELFWEIGHT Y -1
251. ELEMENT LOAD
252. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.4
253. JOINT LOAD
254. 145 FX 0.42 FY -31.87 FZ -2.27
255. 146 FX 0.26 FY -26.7 FZ 2.08
256. LOAD 12 DL+0.75*CL3+0.75*LR
257. SELFWEIGHT Y -1
258. ELEMENT LOAD
259. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.4
260. JOINT LOAD
261. 145 FX 0.23 FY -33.97 FZ -2.46
262. 146 FX -0.03 FY -28.79 FZ 2.28
263. LOAD 13 DL+0.75*CL4+0.75*LR
264. SELFWEIGHT Y -1
265. ELEMENT LOAD
266. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.4
267. JOINT LOAD
268. 145 FX 0.41 FY -34.03 FZ -2.46
269. 146 FX 0.14 FY -28.77 FZ 2.28
270. LOAD 14 DL+0.75*CL5+0.75*LR
271. SELFWEIGHT Y -1
272. ELEMENT LOAD
273. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.4
274. JOINT LOAD
275. 145 FX 0.42 FY -32.3 FZ -2.35
276. 146 FX 0.26 FY -26.33 FZ 2
277. LOAD 15 DL+0.75*CL6+0.75*LR
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278. SELFWEIGHT Y -1
279. ELEMENT LOAD
280. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.4
281. JOINT LOAD
282. 145 FX 0.43 FY -31.44 FZ -2.19
283. 146 FX -0.13 FY -22.92 FZ -2.19
284. LOAD 16 DL+0.75*CL7+0.75*LR
285. SELFWEIGHT Y -1
286. ELEMENT LOAD
287. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.4
288. JOINT LOAD
289. 145 FX 0.32 FY -34.43 FZ -2.54
290. 146 FX 0.05 FY -28.42 FZ 2.19
291. LOAD 17 DL+0.75*CL8+0.75*LR
292. SELFWEIGHT Y -1
293. ELEMENT LOAD
294. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.4
295. JOINT LOAD
296. 145 FX 0.32 FY -33.57 FZ -2.38
297. 146 FX 0.05 FY -29.14 FZ 2.36
298. LOAD 18 DL+0.75*LR+0.75*WL1
299. SELFWEIGHT Y -1
300. ELEMENT LOAD
301. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.4
302. JOINT LOAD
303. 145 FX -0.38 FY -29.14 FZ -2.07
304. 146 FX -0.02 FY -24.55 FZ 1.89
305. LOAD 19 DL+0.75*LR+0.75*WL2
306. SELFWEIGHT Y -1
307. ELEMENT LOAD
308. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.4
309. JOINT LOAD
310. 145 FX -0.87 FY -27.31 FZ -1.97
311. 146 FX -0.86 FY -23.75 FZ 1.85
312. LOAD 20 DL+0.75*LR+0.75*WL3
313. SELFWEIGHT Y -1
314. ELEMENT LOAD
315. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.4
316. JOINT LOAD
317. 145 FX 2.06 FY -28.05 FZ -2
318. 146 FX 1.9 FY -23.43 FZ 1.8
319. LOAD 21 DL+0.75*LR+0.75*WL4
320. SELFWEIGHT Y -1
321. ELEMENT LOAD
322. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.4
323. JOINT LOAD
324. 145 FX 1.23 FY -26.47 FZ -1.93
325. 146 FX 1.1 FY -22.74 FZ 1.77
326. LOAD 22 DL+0.75*LR+0.75*WL5
327. SELFWEIGHT Y -1
328. ELEMENT LOAD
329. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.4
330. JOINT LOAD
331. 145 FX -0.18 FY -42.64 FZ -4.49
332. 146 FX -0.26 FY -9.84 FZ -0.4
333. LOAD 23 DL+0.75*LR+0.75*WL6
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334. SELFWEIGHT Y -1
335. ELEMENT LOAD
336. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.4
337. JOINT LOAD
338. 145 FX -1.06 FY -42.31 FZ -4.79
339. 146 FX -1.05 FY -7.13 FZ -0.74
340. LOAD 24 DL+0.75*LR+0.75*WL7
341. SELFWEIGHT Y -1
342. ELEMENT LOAD
343. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.4
344. JOINT LOAD
345. 145 FX -0.18 FY -13.55 FZ 0.42
346. 146 FX -0.26 FY -37.54 FZ 4.02
347. LOAD 25 DL+0.75*LR+0.75*WL8
348. SELFWEIGHT Y -1
349. ELEMENT LOAD
350. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.4
351. JOINT LOAD
352. 145 FX -1.01 FY -10.12 FZ 0.79
353. 146 FX -1.05 FY -37.96 FZ 4.2
354. LOAD 26 0.6*DL+WL1
355. *SOIL+0.6*SURCHARGE=1.35X1.8+.6*2=3.63T/SQ.M)
356. SELFWEIGHT Y -1
357. ELEMENT LOAD
358. 25 TO 76 81 TO 132 135 TO 258 PR GY -3.63
359. JOINT LOAD
360. 145 FX -0.87 FY -13.34 FZ -1.08
361. 146 FX -0.32 FY -11.49 FZ 1.01
362. LOAD 27 0.6*DL+WL2
363. SELFWEIGHT Y -1
364. ELEMENT LOAD
365. 25 TO 76 81 TO 132 135 TO 258 PR GY -3.63
366. JOINT LOAD
367. 145 FX -1.52 FY -10.9 FZ -0.96
368. 146 FX -1.45 FY -10.43 FZ 0.95
369. LOAD 28 0.6*DL+WL3
370. SELFWEIGHT Y -1
371. ELEMENT LOAD
372. 25 TO 76 81 TO 132 135 TO 258 PR GY -3.63
373. JOINT LOAD
374. 145 FX 2.38 FY -11.88 FZ -0.99
375. 146 FX 2.24 FY -10.01 FZ 0.88
376. LOAD 29 0.6*DL+WL4
377. SELFWEIGHT Y -1
378. ELEMENT LOAD
379. 25 TO 76 81 TO 132 135 TO 258 PR GY -3.63
380. JOINT LOAD
381. 145 FX 1.27 FY -9.77 FZ -0.89
382. 146 FX 1.18 FY -9.09 FZ 0.83
383. LOAD 30 0.6*DL+WL5
384. SELFWEIGHT Y -1
385. ELEMENT LOAD
386. 25 TO 76 81 TO 132 135 TO 258 PR GY -3.63
387. JOINT LOAD
388. 145 FX -0.61 FY -31.33 FZ -4.32
389. 146 FX -0.63 FY 8.11 FZ -2.06
```

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390. LOAD 31 0.6*DL+WL6
391. SELFWEIGHT Y -1
392. ELEMENT LOAD
393. 25 TO 76 81 TO 132 135 TO 258 PR GY -3.63
394. JOINT LOAD
395. 145 FX -1.78 FY -30.9 FZ -4.71
396. 146 FX -1.7 FY 11.72 FZ -2.5
397. LOAD 32 0.6*DL+WL7
398. SELFWEIGHT Y -1
399. ELEMENT LOAD
400. 25 TO 76 81 TO 132 135 TO 258 PR GY -3.63
401. JOINT LOAD
402. 145 FX -0.61 FY 7.45 FZ 2.23
403. 146 FX -0.63 FY -28.82 FZ 3.84
404. LOAD 33 0.6*DL+WL8
405. SELFWEIGHT Y -1
406. ELEMENT LOAD
407. 25 TO 76 81 TO 132 135 TO 258 PR GY -3.63
408. JOINT LOAD
409. 145 FX -1.72 FY 12.03 FZ 2.73
410. 146 FX -1.7 FY -29.38 FZ 4.08
411. **********************************************
412. LOAD 100 1.4*D
413. *SOIL+SURCHARGE=1.35X1.8+2*1.4=5.2T/SQ.M)
414. SELFWEIGHT Y -1
415. ELEMENT LOAD
416. 25 TO 76 81 TO 132 135 TO 258 PR GY -5.23
417. JOINT LOAD
418. 145 FX 0.7 FY -35.05 FZ -2.72
419. 146 FX 0.56 FY -29.08 FZ 2.5
420. LOAD 101 1.4D+1.7(CL1)+0.5LR
421. SELFWEIGHT Y -1
422. ELEMENT LOAD
423. 25 TO 76 81 TO 132 135 TO 258 PR GY -5.23
424. JOINT LOAD
425. 145 FX 0.53 FY -41.95 FZ -3.16
426. 146 FX 0.24 FY -35.29 FZ 2.91
427. LOAD 102 1.4D+1.7(CL2)+0.5LR
428. SELFWEIGHT Y -1
429. ELEMENT LOAD
430. 25 TO 76 81 TO 132 135 TO 258 PR GY -5.23
431. JOINT LOAD
432. 145 FX 0.53 FY -41.95 FZ -3.16
433. 146 FX 0.24 FY -35.31 FZ 2.91
434. LOAD 103 1.4D+1.7(CL3)+0.5LR
435. SELFWEIGHT Y -1
436. ELEMENT LOAD
437. 25 TO 76 81 TO 132 135 TO 258 PR GY -5.23
438. JOINT LOAD
439. 145 FX 0.09 FY -46.72 FZ -3.58
440. 146 FX -0.42 FY -40.05 FZ 3.36
441. LOAD 104 1.4D+1.7(CL4)+0.5LR
442. SELFWEIGHT Y -1
443. ELEMENT LOAD
444. 25 TO 76 81 TO 132 135 TO 258 PR GY -5.23
445. JOINT LOAD
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446. 145 FX 0.49 FY -46.84 FZ -3.59
447. 146 FX -0.03 FY -40.02 FZ 3.36
448. LOAD 105 1.4D+1.7(CL5)+0.5LR
449. SELFWEIGHT Y -1
450. ELEMENT LOAD
451. 25 TO 76 81 TO 132 135 TO 258 PR GY -5.23
452. JOINT LOAD
453. 145 FX 0.52 FY -42.93 FZ -3.34
454. 146 FX 0.24 FY -34.48 FZ 2.72
455. LOAD 106 1.4D+1.7(CL6)+0.5LR
456. SELFWEIGHT Y -1
457. ELEMENT LOAD
458. 25 TO 76 81 TO 132 135 TO 258 PR GY -5.23
459. JOINT LOAD
460. 145 FX 0.54 FY -40.97 FZ -2.97
461. 146 FX -0.16 FY -31.78 FZ -2.97
462. LOAD 107 1.4D+1.7(CL7)+0.5LR
463. SELFWEIGHT Y -1
464. ELEMENT LOAD
465. 25 TO 76 81 TO 132 135 TO 258 PR GY -5.23
466. JOINT LOAD
467. 145 FX 0.52 FY -42.93 FZ -3.34
468. 146 FX 0.24 FY -34.48 FZ 2.72
469. LOAD 108 1.4D+1.7(CL8)+0.5LR
470. SELFWEIGHT Y -1
471. ELEMENT LOAD
472. 25 TO 76 81 TO 132 135 TO 258 PR GY -5.23
473. JOINT LOAD
474. 145 FX 0.54 FY -40.97 FZ -2.97
475. 146 FX 0.24 FY -36.11 FZ 3.09
476. ****************************************
477. LOAD 109 1.2D+1.6LR+1.0(CL1)
478. SELFWEIGHT Y -1
479. ELEMENT LOAD
480. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.9
481. JOINT LOAD
482. 145 FX 0.5 FY -43.45 FZ -2.91
483. 146 FX 0.29 FY -36.41 FZ 2.66
484. LOAD 110 1.2D+1.6LR+1.0(CL2)
485. SELFWEIGHT Y -1
486. ELEMENT LOAD
487. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.9
488. JOINT LOAD
489. 145 FX 0.5 FY -43.45 FZ -2.91
490. 146 FX 0.29 FY -36.42 FZ 2.66
491. LOAD 111 1.2D+1.6LR+1.0(CL3)
492. SELFWEIGHT Y -1
493. ELEMENT LOAD
494. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.9
495. JOINT LOAD
496. 145 FX 0.24 FY -46.25 FZ -3.17
497. 146 FX -0.09 FY -39.22 FZ 2.93
498. LOAD 112 1.2D+1.6LR+1.0(CL4)
499. SELFWEIGHT Y -1
500. ELEMENT LOAD
501. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.9
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502. JOINT LOAD
503. 145 FX 0.47 FY -46.32 FZ -3.17
504. 146 FX 0.13 FY -39.19 FZ 2.92
505. LOAD 113 1.2D+1.6LR+1.0(CL5)
506. SELFWEIGHT Y -1
507. ELEMENT LOAD
508. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.9
509. JOINT LOAD
510. 145 FX 0.5 FY -44.02 FZ -3.02
511. 146 FX 0.29 FY -35.94 FZ 2.55
512. LOAD 114 1.2D+1.6LR+1.0(CL6)
513. SELFWEIGHT Y -1
514. ELEMENT LOAD
515. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.9
516. JOINT LOAD
517. 145 FX 0.5 FY -42.87 FZ -2.81
518. 146 FX 0.29 FY -36.9 FZ 2.77
519. LOAD 115 1.2D+1.6LR+1.0(CL7)
520. SELFWEIGHT Y -1
521. ELEMENT LOAD
522. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.9
523. JOINT LOAD
524. 145 FX 0.35 FY -46.86 FZ -3.28
525. 146 FX 0.02 FY -38.73 FZ 2.81
526. LOAD 116 1.2D+1.6LR+1.0(CL8)
527. SELFWEIGHT Y -1
528. ELEMENT LOAD
529. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.9
530. JOINT LOAD
531. 145 FX 0.36 FY -45.71 FZ -3.06
532. 146 FX 0.02 FY -39.68 FZ 3.03
533. ************************************8
534. LOAD 117 1.2D + 1.6WL1+1.0L+0.5LR
535. SELFWEIGHT Y -1
536. ELEMENT LOAD
537. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.9
538. JOINT LOAD
539. 145 FX -1.27 FY -30.93 FZ -2.32
540. 146 FX -0.42 FY -26.38 FZ 2.14
541. LOAD 118 1.2D + 1.6WL2+1.0L+0.5LR
542. SELFWEIGHT Y -1
543. ELEMENT LOAD
544. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.9
545. JOINT LOAD
546. 145 FX -2.31 FY -27.02 FZ -2.12
547. 146 FX -2.22 FY -24.68 FZ 2.05
548. LOAD 119 1.2D + 1.6WL3+1.0L+0.5LR
549. SELFWEIGHT Y -1
550. ELEMENT LOAD
551. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.9
552. JOINT LOAD
553. 145 FX 3.93 FY -28.59 FZ -2.18
554. 146 FX 3.68 FY -24 FZ 1.95
555. LOAD 120 1.2D + 1.6WL4+1.0L+0.5LR
556. SELFWEIGHT Y -1
557. ELEMENT LOAD
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558. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.9
559. JOINT LOAD
560. 145 FX 2.16 FY -25.22 FZ -2.02
561. 146 FX 1.98 FY -22.52 FZ 1.87
562. LOAD 121 1.2D + 1.6WL5+1.0L+0.5LR
563. SELFWEIGHT Y -1
564. ELEMENT LOAD
565. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.9
566. JOINT LOAD
567. 145 FX -0.86 FY -59.71 FZ -7.5
568. 146 FX -0.92 FY 5 FZ -2.76
569. LOAD 122 1.2D + 1.6WL6+1.0L+0.5LR
570. SELFWEIGHT Y -1
571. ELEMENT LOAD
572. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.9
573. JOINT LOAD
574. 145 FX -2.73 FY -59.03 FZ -8.12
575. 146 FX -2.62 FY 10.77 FZ -3.47
576. LOAD 123 1.2D + 1.6WL7+1.0L+0.5LR
577. SELFWEIGHT Y -1
578. ELEMENT LOAD
579. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.9
580. JOINT LOAD
581. 145 FX -0.86 FY 2.33 FZ 2.98
582. 146 FX -0.92 FY -54.09 FZ 6.67
583. LOAD 124 1.2D + 1.6WL8+1.0L+0.5LR
584. SELFWEIGHT Y -1
585. ELEMENT LOAD
586. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.9
587. JOINT LOAD
588. 145 FX -2.63 FY 9.66 FZ 3.78
589. 146 FX -2.62 FY -54.99 FZ 7.07
590. ****************************************
591. LOAD 125 0.9D+1.6WL1
592. *SOIL+09*SURCHARGE=1.35X1.8+0.9*2=3.6T/SQ.M)
593. SELFWEIGHT Y -1
594. ELEMENT LOAD
595. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.23
596. JOINT LOAD
597. 145 FX -1.42 FY -19.84 FZ -1.62
598. 146 FX -0.54 FY -17.14 FZ 1.5
599. LOAD 126 0.9D+1.6WL2
600. SELFWEIGHT Y -1
601. ELEMENT LOAD
602. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.23
603. JOINT LOAD
604. 145 FX -2.46 FY -15.93 FZ -1.41
605. 146 FX -2.34 FY -15.44 FZ 1.41
606. LOAD 127 0.9D+1.6WL3
607. SELFWEIGHT Y -1
608. ELEMENT LOAD
609. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.23
610. JOINT LOAD
611. 146 FX 3.78 FY -17.5 FZ -1.47
612. 145 FX 3.56 FY -14.77 FZ 1.3
613. LOAD 128 0.9D+1.6WL4
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614. SELFWEIGHT Y -1
615. ELEMENT LOAD
616. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.23
617. JOINT LOAD
618. 145 FX 1.86 FY -13.29 FZ 1.23
619. 146 FX 0.11 FY -12.92 FZ -1.31
620. LOAD 129 0.9D+1.6WL5
621. SELFWEIGHT Y -1
622. ELEMENT LOAD
623. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.23
624. JOINT LOAD
625. 145 FX -1.01 FY -48.62 FZ -6.79
626. 146 FX -1.04 FY 14.23 FZ -3.4
627. LOAD 130 0.9D+1.6WL6
628. SELFWEIGHT Y -1
629. ELEMENT LOAD
630. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.23
631. JOINT LOAD
632. 145 FX -2.88 FY -47.94 FZ -7.41
633. 146 FX -2.74 FY 20.01 FZ -4.11
634. LOAD 131 0.9D+1.6WL7
635. SELFWEIGHT Y -1
636. ELEMENT LOAD
637. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.23
638. JOINT LOAD
639. 145 FX -1.01 FY 13.42 FZ 3.69
640. 146 FX -1.04 FY -44.86 FZ 6.03
641. LOAD 132 0.9D+1.6WL8
642. SELFWEIGHT Y -1
643. ELEMENT LOAD
644. 25 TO 76 81 TO 132 135 TO 258 PR GY -4.23
645. JOINT LOAD
646. 145 FX -2.78 FY 20.75 FZ 4.49
647. 146 FX -2.74 FY -45.75 FZ 6.43
648. ************************************************************************
649. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**262**	**NUMBER OF MEMBERS**	**2**
NUMBER OF PLATES	**228**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**260**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 173/ 22/ 72 DOF
TOTAL PRIMARY LOAD CASES = 66, TOTAL DEGREES OF FREEDOM = 792
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 58 DOUBLE KILO-WORDS

STAAD SPACE **-- PAGE NO. 18**

REQRD/AVAIL. DISK SPACE = 14.5/ 115288.8 MB

650. START CONCRETE DESIGN

```
651. CODE ACI 1999
652. TRACK 0 ALL
653. MAXMAIN 25 ALL
654. CLB 0.075 MEMB 133 134
655. CLS 0.075 ALL
656. DESIGN COLUMN 133 134
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COLUMN NO. 133 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==

COLUMN NO. 134 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

**********END OF COLUMN DESIGN RESULTS**********

657. END CONCRETE DESIGN
658. FINISH

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	C01	2			
Date	29-04-2020	01-06-2020	26.10.2023				Date	29-04-2020	01-06-2020	26.10.2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT			
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD			
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

ATTACHMENT 3 STAAD Input & Output File of Foundation Mkd. F4

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries5          *
*           Version  20.07.10.64                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    APR 29, 2020                *
*           Time=    16:38:31                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: B8B7_A8A7_29.04.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 02-APR-19
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER MTON
     7. JOINT COORDINATES
     8. 13 0 -4 0; 14 0 -4 -1.5; 16 0 -4 1; 20 1 -4 1; 21 0.75 -4 1; 22 0.75 -4 0.75
     9. 23 1 -4 0.75; 24 0.5 -4 1; 25 0.5 -4 0.75; 26 0.25 -4 1; 27 0.25 -4 0.75
    10. 28 0 -4 0.75; 29 0.75 -4 0.5; 30 1 -4 0.5; 31 0.5 -4 0.5; 32 0.25 -4 0.5
    11. 33 0 -4 0.5; 34 0.75 -4 0.25; 35 1 -4 0.25; 36 0.5 -4 0.25; 37 0.25 -4 0.25
    12. 38 0 -4 0.25; 39 0.75 -4 0; 40 1 -4 0; 41 0.5 -4 0; 42 0.25 -4 0
    13. 43 0.75 -4 -0.25; 44 1 -4 -0.25; 45 0.5 -4 -0.25; 46 0.25 -4 -0.25
    14. 47 0 -4 -0.25; 48 0.75 -4 -0.5; 49 1 -4 -0.5; 50 0.5 -4 -0.5; 51 0.25 -4 -0.5
    15. 52 0 -4 -0.5; 53 0.75 -4 -0.75; 54 1 -4 -0.75; 55 0.5 -4 -0.75
    16. 56 0.25 -4 -0.75; 57 0 -4 -0.75; 58 0.75 -4 -1; 59 1 -4 -1; 60 0.5 -4 -1.
    17. 61 0.25 -4 -1; 62 0 -4 -1; 63 0.75 -4 -1.25; 64 1 -4 -1.25; 65 0.5 -4 -1.25
    18. 66 0.25 -4 -1.25; 67 0 -4 -1.25; 68 0.75 -4 -1.5; 69 1 -4 -1.5; 70 0.5 -4 -1.5
    19. 71 0.25 -4 -1.5; 72 0.75 -4 -1.75; 73 1 -4 -1.75; 74 0.5 -4 -1.75
    20. 75 0.25 -4 -1.75; 76 0 -4 -1.75; 77 0.75 -4 -2; 78 1 -4 -2; 79 0.5 -4 -2
    21. 80 0.25 -4 -2; 81 0 -4 -2; 82 0.75 -4 -2.25; 83 1 -4 -2.25; 84 0.5 -4 -2.25
    22. 85 0.25 -4 -2.25; 86 0 -4 -2.25; 90 -0.25 -4 1; 91 -0.25 -4 0.75; 92 -0.5 -4 1
    23. 93 -0.5 -4 0.75; 97 -0.25 -4 0.5; 98 -0.5 -4 0.5; 101 -0.25 -4 0.25
    24. 102 -0.5 -4 0.25; 105 -0.25 -4 0; 106 -0.5 -4 0; 109 -0.25 -4 -0.25
    25. 110 -0.5 -4 -0.25; 113 -0.25 -4 -0.5; 114 -0.5 -4 -0.5; 117 -0.25 -4 -0.75
    26. 118 -0.5 -4 -0.75; 121 -0.25 -4 -1; 122 -0.5 -4 -1; 125 -0.25 -4 -1.25
    27. 126 -0.5 -4 -1.25; 129 -0.25 -4 -1.5; 130 -0.5 -4 -1.5; 133 -0.25 -4 -1.75
    28. 134 -0.5 -4 -1.75; 137 -0.25 -4 -2; 138 -0.5 -4 -2; 141 -0.25 -4 -2.25
    29. 142 -0.5 -4 -2.25; 145 0 -1.75 -2; 146 0 -1.75 0.75; 147 0.75 -4 -2.5
    30. 148 1 -4 -2.5; 149 0.5 -4 -2.5; 150 0.25 -4 -2.5; 151 0 -4 -2.5
    31. 152 -0.25 -4 -2.5; 153 -0.5 -4 -2.5; 156 1 -4 1.25; 157 0.75 -4 1.25
    32. 158 0.5 -4 1.25; 159 0.25 -4 1.25; 160 0 -4 1.25; 161 -0.25 -4 1.25
    33. 162 -0.5 -4 1.25; 181 1.25 -4 1; 182 1.25 -4 0.75; 183 1.25 -4 0.5
    34. 184 1.25 -4 0.25; 185 1.25 -4 0; 186 1.25 -4 -0.25; 187 1.25 -4 -0.5
    35. 188 1.25 -4 -0.75; 189 1.25 -4 -1; 190 1.25 -4 -1.25; 191 1.25 -4 -1.5
    36. 192 1.25 -4 -1.75; 193 1.25 -4 -2; 194 1.25 -4 -2.25; 195 1.25 -4 -2.5
    37. 196 1.25 -4 1.25; 219 3.25 -4 -2; 220 3.25 -1.75 -2; 221 3.25 -4 0.75
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95. ELEMENT INCIDENCES SHELL
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191. 455 162 504 489 92; 456 153 503 505 441; 457 459 506 504 162
192. START GROUP DEFINITION
193. ELEMENT
194. _MOMENT 25 TO 27 29 TO 31 33 TO 71 73 TO 75 82 86 89 90 93 94 97 98 101 102 -
195. 105 106 109 110 113 114 117 118 121 122 126 130 135 TO 140 143 TO 148 166 -
196. 167 TO 180 203 TO 247 254 TO 298 306 308 TO 326 328 330 TO 334 337 TO 345 -
197. 348 349 375 TO 457
198. _SHEAR 25 26 29 30 33 34 37 TO 66 69 70 73 74 93 94 97 98 101 102 105 106 -
199. 109 110 113 114 117 118 135 136 143 144 166 TO 180 203 TO 247 254 TO 298 -
200. 311 TO 324 338 TO 344 378 TO 384 390 TO 457
201. END GROUP DEFINITION
202. ELEMENT PROPERTY
203. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
204. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
205. 255 TO 298 305 TO 349 375 TO 457 THICKNESS 0.4
206. SUPPORTS
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207. 13 14 16 20 TO 86 90 TO 93 97 98 101 102 105 106 109 110 113 114 117 118 121 -
208. 122 125 126 129 130 133 134 137 138 141 142 147 TO 153 156 TO 162 -
209. 181 TO 196 219 221 223 TO 270 277 TO 324 331 TO 376 419 TO 505 -
210. 506 ELASTIC MAT DIRECT Y SUBGRADE 7080 PRINT
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```
211. DEFINE MATERIAL START
212. ISOTROPIC STEEL
213. E 2.09042E+007
214. POISSON 0.3
215. DENSITY 7.83341
216. ALPHA 1.2E-005
217. DAMP 0.03
218. TYPE STEEL
219. STRENGTH FY 25819.2 FU 41584 RY 1.5 RT 1.2
220. ISOTROPIC CONCRETE
221. E 2.21467E+006
222. POISSON 0.17
223. DENSITY 2.40262
224. ALPHA 1E-005
225. DAMP 0.05
226. TYPE CONCRETE
227. STRENGTH FCU 2812.28
228. END DEFINE MATERIAL
229. MEMBER PROPERTY AMERICAN
230. 133 134 201 202 356 368 PRIS YD 0.6 ZD 0.6
231. CONSTANTS
232. MATERIAL CONCRETE ALL
233. LOAD 1 DL+LR
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234. SELFWEIGHT Y -1
235. ELEMENT LOAD
236. *SOIL+SURCHARGE=1.35X1.8+2=4.4T/SQ.M)
237. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
238. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
239. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.4
240. JOINT LOAD
241. 145 FX -0.57 FY -35.1 FZ -2.5
242. 220 FX 0.17 FY -23 FZ -1.5
243. 146 FX -0.56 FY -31.2 FZ 2.6
244. 222 FX 0.1 FY -21 FZ 1.5
245. LOAD 2 DL+CL1
246. SELFWEIGHT Y -1
247. ELEMENT LOAD
248. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
249. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
250. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.4
251. JOINT LOAD
252. 145 FX -0.87 FY -33.72 FZ -2.77
253. 220 FX 0.17 FY -20.53 FZ -1.31
254. 146 FX -0.77 FY -30.45 FZ 2.94
255. 222 FX 0.14 FY -17.99 FZ 1.3
256. LOAD 3 DL+CL2
257. SELFWEIGHT Y -1
258. ELEMENT LOAD
259. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
260. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
261. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.4
262. JOINT LOAD
263. 145 FX -0.09 FY -32.01 FZ -2.7
264. 220 FX 0.17 FY -22.24 FZ -1.48
265. 146 FX -0.02 FY -28.79 FZ 2.88
266. 222 FX 0.14 FY -19.64 FZ 1.47
267. LOAD 4 DL+CL3
268. SELFWEIGHT Y -1
269. ELEMENT LOAD
270. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
271. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
272. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.4
273. JOINT LOAD
274. 145 FX -0.59 FY -30.02 FZ -2.37
275. 220 FX 0.15 FY -21.42 FZ -1.48
276. 146 FX -0.32 FY -25.81 FZ 2.53
277. 222 FX 0.11 FY -19.8 FZ 1.46
278. LOAD 5 DL+CL4
279. SELFWEIGHT Y -1
280. ELEMENT LOAD
281. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
282. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
283. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.4
284. JOINT LOAD
285. 145 FX 0 FY -28.2 FZ -2.28
286. 220 FX 0.1 FY -23.2 FZ -1.66
287. 146 FX 0.2 FY -24.2 FZ 2.45
288. 222 FX 0.1 FY -21.4 FZ 1.64
289. LOAD 6 DL+CL5
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290. SELFWEIGHT Y -1
291. ELEMENT LOAD
292. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
293. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
294. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.4
295. JOINT LOAD
296. 145 FX -0.5 FY -33.3 FZ -2.82
297. 220 FX 0.2 FY -21.7 FZ -1.43
298. 146 FX -0.4 FY -29.2 FZ 2.82
299. 222 FX 0.1 FY -18.6 FZ 1.36
300. LOAD 7 DL+CL6
301. SELFWEIGHT Y -1
302. ELEMENT LOAD
303. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
304. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
305. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.4
306. JOINT LOAD
307. 145 FX -0.5 FY -32.4 FZ -2.65
308. 220 FX 0.2 FY -21.1 FZ -1.36
309. 146 FX -0.4 FY -30 FZ 3
310. 222 FX 0.1 FY -19 FZ 1.41
311. LOAD 8 DL+CL7
312. SELFWEIGHT Y -1
313. ELEMENT LOAD
314. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
315. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
316. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.4
317. JOINT LOAD
318. 145 FX -0.3 FY -29.5 FZ -2.41
319. 220 FX 0.1 FY -22.6 FZ -1.6
320. 146 FX -0.1 FY -24.6 FZ 2.4
321. 222 FX 0.1 FY -20.4 FZ 1.53
322. LOAD 9 DL+CL8
323. SELFWEIGHT Y -1
324. ELEMENT LOAD
325. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
326. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
327. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.4
328. JOINT LOAD
329. 145 FX -0.3 FY -28.7 FZ -2.23
330. 220 FX 0.1 FY -22 FZ -1.54
331. 146 FX -0.1 FY -25.4 FZ 2.58
332. 222 FX 0.1 FY -20.8 FZ 1.58
333. ************************************************************888
334. LOAD 10 DL+0.75*CL1+0.75*LR
335. SELFWEIGHT Y -1
336. ELEMENT LOAD
337. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
338. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
339. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.4
340. JOINT LOAD
341. 145 FX -0.2 FY -36.5 FZ -2.77
342. 220 FX 0.2 FY -23 FZ -1.5
343. 146 FX -0.2 FY -32.9 FZ 2.95
344. 222 FX 0.1 FY -20.7 FZ 1.51
345. LOAD 11 DL+0.75*CL2+0.75*LR
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346. SELFWEIGHT Y -1
347. ELEMENT LOAD
348. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
349. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
350. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.4
351. JOINT LOAD
352. 145 FX -0.6 FY -35 FZ -2.52
353. 220 FX 0.2 FY -22.4 FZ -1.5
354. 146 FX -0.4 FY -30.7 FZ 2.69
355. 222 FX 0.1 FY -20.8 FZ 1.51
356. LOAD 12 DL+0.75*CL3+0.75*LR
357. SELFWEIGHT Y -1
358. ELEMENT LOAD
359. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
360. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
361. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.4
362. JOINT LOAD
363. 145 FX -0.39 FY -27.75 FZ 2.69
364. 220 FX 0.12 FY -20.29 FZ 1.51
365. 146 FX -0.2 FY -27.52 FZ -2.52
366. 222 FX 0.12 FY -18.81 FZ -1.5
367. LOAD 13 DL+0.75*CL4+0.75*LR
368. SELFWEIGHT Y -1
369. ELEMENT LOAD
370. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
371. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
372. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.4
373. JOINT LOAD
374. 145 FX -0.2 FY -33.7 FZ -2.45
375. 220 FX 0.2 FY -23.7 FZ -1.63
376. 146 FX 0 FY -29.4 FZ 2.63
377. 222 FX 0.1 FY -22 FZ 1.64
378. LOAD 14 DL+0.75*CL5+0.75*LR
379. ELEMENT LOAD
380. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
381. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
382. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.4
383. JOINT LOAD
384. 145 FX -0.44 FY -30.31 FZ 2.91
385. 220 FX 0.14 FY -19.42 FZ 1.43
386. 146 FX 0.08 FY -30.35 FZ -2.86
387. 222 FX 0.15 FY -18.41 FZ -1.46
388. LOAD 15 DL+0.75*CL6+0.75*LR
389. SELFWEIGHT Y -1
390. ELEMENT LOAD
391. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
392. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
393. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.4
394. JOINT LOAD
395. 145 FX -0.5 FY -36.8 FZ -2.73
396. 220 FX 0.2 FY -22.2 FZ -1.41
397. 146 FX -0.5 FY -33.8 FZ 3.04
398. 222 FX 0.1 FY -20.2 FZ 1.47
399. LOAD 16 DL+0.75*CL7+0.75*LR
400. SELFWEIGHT Y -1
401. ELEMENT LOAD
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402. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
403. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
404. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.4
405. JOINT LOAD
406. 145 FX -0.4 FY -34.7 FZ -2.55
407. 220 FX 0.2 FY -23.3 FZ -1.59
408. 146 FX -0.2 FY -29.8 FZ 2.59
409. 222 FX 0.1 FY -21.3 FZ 1.56
410. LOAD 17 DL+0.75*CL8+0.75*LR
411. SELFWEIGHT Y -1
412. ELEMENT LOAD
413. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
414. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
415. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.4
416. JOINT LOAD
417. 145 FX -0.4 FY -34 FZ -2.42
418. 220 FX 0.2 FY -22.9 FZ -1.54
419. 146 FX -0.2 FY -30.4 FZ 2.73
420. 222 FX 0.1 FY -21.6 FZ 1.6
421. LOAD 18 DL+0.75*LR+0.75*WL1
422. SELFWEIGHT Y -1
423. ELEMENT LOAD
424. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
425. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
426. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.4
427. JOINT LOAD
428. 145 FX -3.3 FY -36.7 FZ -2.46
429. 220 FX -0.4 FY -16.4 FZ -1.
430. 146 FX -3.1 FY -32.7 FZ 2.63
431. 222 FX -0.4 FY -15.1 FZ 1.01
432. LOAD 19 DL+0.75*LR+0.75*WL2
433. SELFWEIGHT Y -1
434. ELEMENT LOAD
435. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
436. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
437. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.4
438. JOINT LOAD
439. 145 FX -2.9 FY -35.7 FZ -2.52
440. 220 FX -0.1 FY -15.9 FZ -1.02
441. 146 FX -2.5 FY -31.5 FZ 2.65
442. 222 FX -0.2 FY -14.9 FZ 1.01
443. LOAD 20 DL+0.75*LR+0.75*WL3
444. SELFWEIGHT Y -1
445. ELEMENT LOAD
446. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
447. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
448. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.4
449. JOINT LOAD
450. 145 FX 1.4 FY -26 FZ -2.11
451. 220 FX 0.3 FY -28 FZ -1.98
452. 146 FX 1.1 FY -23.9 FZ 2.31
453. 222 FX 0.2 FY -24.9 FZ 1.97
454. LOAD 21 DL+0.75*LR+0.75*WL4
455. SELFWEIGHT Y -1
456. ELEMENT LOAD
457. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
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458. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
459. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.4
460. JOINT LOAD
461. 145 FX 2 FY -24.5 FZ -2.14
462. 220 FX 0.6 FY -27.7 FZ -2.01
463. 146 FX 1.7 FY -22.7 FZ 2.33
464. 222 FX 0.5 FY -25 FZ 1.98
465. LOAD 22 DL+0.75*LR+0.75*WL5
466. SELFWEIGHT Y -1
467. ELEMENT LOAD
468. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
469. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
470. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.4
471. JOINT LOAD
472. 145 FX -0.3 FY -42 FZ -4.69
473. 220 FX 0.4 FY -31.9 FZ -3.2
474. 146 FX 0 FY -16.7 FZ 0.31
475. 222 FX 0.3 FY -11.1 FZ -0.07
476. LOAD 23 DL+0.75*LR+0.75*WL6
477. SELFWEIGHT Y -1
478. ELEMENT LOAD
479. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
480. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
481. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.4
482. JOINT LOAD
483. 145 FX 0.3 FY -41.3 FZ -5.05
484. 220 FX 0.7 FY -32.8 FZ -3.43
485. 146 FX 0.7 FY -13.7 FZ 0.02
486. 222 FX 0.6 FY -9.9 FZ -0.25
487. LOAD 24 DL+0.75*LR+0.75*WL7
488. SELFWEIGHT Y -1
489. ELEMENT LOAD
490. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
491. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
492. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.4
493. JOINT LOAD
494. 145 FX 0.6 FY -20 FZ 0.09
495. 220 FX 0.6 FY -13.1 FZ 0.2
496. 146 FX -0.1 FY -37.8 FZ 4.47
497. 222 FX 0.4 FY -29.5 FZ 2.98
498. LOAD 25 DL+0.75*LR+0.75*WL8
499. SELFWEIGHT Y -1
500. ELEMENT LOAD
501. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
502. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
503. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.4
504. JOINT LOAD
505. 145 FX 0.5 FY -17.5 FZ 0.39
506. 220 FX 0.7 FY -10.9 FZ 0.42
507. 146 FX 0.3 FY -37.9 FZ 4.71
508. 222 FX 0.6 FY -29.9 FZ 3.09
509. LOAD 26 0.6*DL+WL1
510. *SOIL+0.6*SURCHARGE=1.35X1.8+.6*2=3.6T/SQ.M)
511. SELFWEIGHT Y -1
512. ELEMENT LOAD
513. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
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514. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
515. 255 TO 298 305 TO 349 375 TO 457 PR GY -3.6
516. JOINT LOAD
517. 145 FX -4.03 FY -21.2 FZ -1.41
518. 220 FX -0.62 FY -4.6 FZ -0.22
519. 146 FX -3.75 FY -18.9 FZ 1.51
520. 222 FX -0.68 FY -4.4 FZ 0.21
521. LOAD 27 0.6*DL+WL2
522. SELFWEIGHT Y -1
523. ELEMENT LOAD
524. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
525. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
526. 255 TO 298 305 TO 349 375 TO 457 PR GY -3.6
527. JOINT LOAD
528. 145 FX -3.49 FY -19.9 FZ -1.49
529. 220 FX -0.28 FY -4 FZ -0.25
530. 146 FX -2.98 FY -17.2 FZ 1.53
531. 222 FX -0.3 FY -4.1 FZ 0.21
532. LOAD 28 0.6*DL+WL3
533. SELFWEIGHT Y -1
534. ELEMENT LOAD
535. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
536. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
537. 255 TO 298 305 TO 349 375 TO 457 PR GY -3.6
538. JOINT LOAD
539. 145 FX 2.35 FY -6.9 FZ -0.94
540. 220 FX 0.26 FY -20.1 FZ -1.53
541. 146 FX 1.91 FY -7.1 FZ 1.08
542. 222 FX 0.25 FY -17.5 FZ 1.5
543. LOAD 29 0.6*DL+WL4
544. SELFWEIGHT Y -1
545. ELEMENT LOAD
546. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
547. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
548. 255 TO 298 305 TO 349 375 TO 457 PR GY -3.6
549. JOINT LOAD
550. 145 FX 3.15 FY -4.9 FZ -0.98
551. 220 FX 0.68 FY -19.8 FZ -1.57
552. 146 FX 2.74 FY -5.5 FZ 1.1
553. 222 FX 0.65 FY -17.6 FZ 1.51
554. LOAD 30 0.6*DL+WL5
555. SELFWEIGHT Y -1
556. ELEMENT LOAD
557. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
558. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
559. 255 TO 298 305 TO 349 375 TO 457 PR GY -3.6
560. JOINT LOAD
561. 145 FX 0.03 FY -28.27 FZ -4.38
562. 220 FX 0.42 FY -25.35 FZ -3.16
563. 146 FX 0.48 FY 2.48 FZ -1.59
564. 222 FX 0.32 FY 0.97 FZ -1.23
565. LOAD 31 0.6*DL+WL6
566. SELFWEIGHT Y -1
567. ELEMENT LOAD
568. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
569. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
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570. 255 TO 298 305 TO 349 375 TO 457 PR GY -3.6
571. JOINT LOAD
572. 145 FX 0.84 FY -27.32 FZ -4.87
573. 220 FX 0.83 FY -26.5 FZ -3.46
574. 146 FX 1.33 FY 6.49 FZ -1.97
575. 222 FX 0.69 FY 2.52 FZ -1.47
576. LOAD 32 0.6*DL+WL7
577. SELFWEIGHT Y -1
578. ELEMENT LOAD
579. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
580. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
581. 255 TO 298 305 TO 349 375 TO 457 PR GY -3.6
582. JOINT LOAD
583. 145 FX 1.26 FY 1.13 FZ 1.99
584. 220 FX 0.69 FY -0.32 FZ 1.38
585. 146 FX 0.22 FY -25.69 FZ 3.96
586. 222 FX 0.39 FY -23.54 FZ 2.84
587. LOAD 33 0.6*DL+WL8
588. SELFWEIGHT Y -1
589. ELEMENT LOAD
590. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
591. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
592. 255 TO 298 305 TO 349 375 TO 457 PR GY -3.6
593. JOINT LOAD
594. 145 FX 1.15 FY 4.35 FZ 2.39
595. 220 FX 0.75 FY 2.68 FZ 1.67
596. 146 FX 0.82 FY -25.76 FZ 4.28
597. 222 FX 0.77 FY -24.11 FZ 2.99
598. ******************************************************************************
599. LOAD 100 1.4D
600. *SOIL+SURCHARGE=1.35X1.8+2*1.4=5.2T/SQ.M)
601. SELFWEIGHT Y -1
602. ELEMENT LOAD
603. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
604. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
605. 255 TO 298 305 TO 349 375 TO 457 PR GY -5.2
606. JOINT LOAD
607. 145 FX -0.8 FY -38.7 FZ -3.1
608. 220 FX 0.2 FY -30.5 FZ -2.1
609. 146 FX -0.8 FY -33.9 FZ 3.3
610. 222 FX 0.1 FY -27.4 FZ 2.1
611. LOAD 101 1.4D+1.7(CL1)+0.5LR
612. SELFWEIGHT Y -1
613. ELEMENT LOAD
614. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
615. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
616. 255 TO 298 305 TO 349 375 TO 457 PR GY -5.2
617. JOINT LOAD
618. 145 FX -1.3 FY -52.8 FZ -4.2
619. 220 FX 0.3 FY -28.9 FZ -1.8
620. 146 FX -1.2 FY -48 FZ 4.4
621. 222 FX -1 FY -51.2 FZ -3.6
622. LOAD 102 1.4D+1.7(CL2)+0.5LR
623. SELFWEIGHT Y -1
624. ELEMENT LOAD
625. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
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626. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
627. 255 TO 298 305 TO 349 375 TO 457 PR GY -5.2
628. JOINT LOAD
629. 145 FX 0 FY -49.9 FZ -4.072
630. 220 FX 0.26611 FY -31.8469 FZ -2.0755
631. 146 FX 0.1245 FY -45.1743 FZ 4.31425
632. 222 FX 0.21575 FY -28.2091 FZ 2.08025
633. LOAD 103 1.4D+1.7(CL3)+0.5LR
634. SELFWEIGHT Y -1
635. ELEMENT LOAD
636. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
637. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
638. 255 TO 298 305 TO 349 375 TO 457 PR GY -5.2
639. JOINT LOAD
640. 145 FX -0.9 FY -46.5 FZ -3.5
641. 220 FX 0.2 FY -30.5 FZ -2.1
642. 146 FX -0.4 FY -40.1 FZ 3.7
643. 222 FX 0.2 FY -28.5 FZ 2.1
644. LOAD 104 1.4D+1.7(CL4)+0.5LR
645. SELFWEIGHT Y -1
646. ELEMENT LOAD
647. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
648. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
649. 255 TO 298 305 TO 349 375 TO 457 PR GY -5.2
650. JOINT LOAD
651. 145 FX 0.1 FY -43.4 FZ -3.3
652. 220 FX 0.2 FY -30.5 FZ -2.1
653. 146 FX 0.5 FY -37.4 FZ 3.6
654. 222 FX 0.1 FY -43.4 FZ -3.3
655. LOAD 105 1.4D+1.7(CL5)+0.5LR
656. SELFWEIGHT Y -1
657. ELEMENT LOAD
658. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
659. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
660. 255 TO 298 305 TO 349 375 TO 457 PR GY -5.2
661. JOINT LOAD
662. 145 FX -0.7 FY -52.1 FZ -4.3
663. 220 FX 0.2 FY -33.5 FZ -2.4
664. 146 FX -0.5 FY -45.9 FZ 4.2
665. 222 FX 0.2 FY -31.3 FZ 2.4
666. LOAD 106 1.4D+1.7(CL6)+0.5LR
667. SELFWEIGHT Y -1
668. ELEMENT LOAD
669. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
670. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
671. 255 TO 298 305 TO 349 375 TO 457 PR GY -5.2
672. JOINT LOAD
673. 145 FX -0.7 FY -50.6 FZ -4
674. 220 FX 0.3 FY -30.9 FZ -2
675. 146 FX -0.5 FY -47.3 FZ 4.5
676. 222 FX 0.2 FY -26.5 FZ 1.9
677. LOAD 107 1.4D+1.7(CL7)+0.5LR
678. SELFWEIGHT Y -1
679. ELEMENT LOAD
680. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
681. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
```

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682. 255 TO 298 305 TO 349 375 TO 457 PR GY -5.2
683. JOINT LOAD
684. 145 FX -0.4 FY -45.7 FZ -3.6
685. 220 FX 0.3 FY -29.9 FZ -1.9
686. 146 FX 0.1 FY -38.1 FZ 3.5
687. 222 FX 0.2 FY -29.6 FZ 2.2
688. LOAD 108 1.4D+1.7(CL8)+0.5LR
689. SELFWEIGHT Y -1
690. ELEMENT LOAD
691. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
692. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
693. 255 TO 298 305 TO 349 375 TO 457 PR GY -5.2
694. JOINT LOAD
695. 145 FX -0.4 FY -44.2 FZ -3.3
696. 220 FX 0.2 FY -31.5 FZ -2.2
697. 146 FX 0 FY -39.4 FZ 3.8
698. 222 FX 0.2 FY -30.2 FZ 2.3
699. LOAD 109 1.2D+1.6LR+1.0(CL1)
700. SELFWEIGHT Y -1
701. ELEMENT LOAD
702. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
703. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
704. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.8
705. JOINT LOAD
706. 145 FX -1 FY -51.2 FZ -3.6
707. 220 FX 0.3 FY -26.8 FZ -1.6
708. 146 FX -0.9 FY -46.5 FZ 3.8
709. 222 FX 0.2 FY -24.1 FZ 1.7
710. LOAD 110 1.2D+1.6LR+1.0(CL2)
711. SELFWEIGHT Y -1
712. ELEMENT LOAD
713. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
714. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
715. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.8
716. JOINT LOAD
717. 145 FX -0.2 FY -50 FZ -3.57
718. 220 FX 0.3 FY -28 FZ -1.81
719. 146 FX -0.2 FY -45 FZ 3.78
720. 222 FX 0.2 FY -26 FZ 1.85
721. LOAD 111 1.2D+1.6LR+1.0(CL3)
722. SELFWEIGHT Y -1
723. ELEMENT LOAD
724. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
725. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
726. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.8
727. JOINT LOAD
728. 145 FX -0.7 FY -48 FZ -3.23
729. 220 FX 0.2 FY -28 FZ -1.81
730. 146 FX -0.5 FY -42 FZ 3.43
731. 222 FX 0.2 FY -26 FZ 1.85
732. LOAD 112 1.2D+1.6LR+1.0(CL4)
733. SELFWEIGHT Y -1
734. ELEMENT LOAD
735. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
736. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
737. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.8
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738. JOINT LOAD
739. 145 FX -0.19 FY -46 FZ -3.141
740. 220 FX 0.2275 FY -29 FZ -1.992
741. 146 FX 0.04775 FY -40 FZ 3.35225
742. 222 FX 0.14875 FY -28 FZ 2.027
743. LOAD 113 1.2D+1.6LR+1.0(CL5)
744. SELFWEIGHT Y -1
745. ELEMENT LOAD
746. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
747. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
748. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.8
749. JOINT LOAD
750. 145 FX -0.64 FY -51 FZ -3.687
751. 220 FX 0.2561 FY -28 FZ -1.7545
752. 146 FX -0.526 FY -45 FZ 3.71975
753. 222 FX 0.1775 FY -25 FZ 1.74575
754. LOAD 114 1.2D+1.6LR+1.0(CL6)
755. SELFWEIGHT Y -1
756. ELEMENT LOAD
757. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
758. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
759. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.8
760. JOINT LOAD
761. 145 FX -0.6 FY -50 FZ -3.51
762. 220 FX 0.3 FY -27 FZ -1.69
763. 146 FX -0.5 FY -46 FZ 3.9
764. 222 FX 0.2 FY -25 FZ 1.8
765. LOAD 115 1.2D+1.6LR+1.0(CL7)
766. SELFWEIGHT Y -1
767. ELEMENT LOAD
768. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
769. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
770. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.8
771. JOINT LOAD
772. 145 FX -0.5 FY -47 FZ -3.27
773. 220 FX 0.2 FY -29 FZ -1.93
774. 146 FX -0.2 FY -41 FZ 3.3
775. 222 FX 0.1 FY -27 FZ 1.91
776. LOAD 116 1.2D+1.6LR+1.0(CL8)
777. SELFWEIGHT Y -1
778. ELEMENT LOAD
779. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
780. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
781. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.8
782. JOINT LOAD
783. 145 FX -0.47 FY -46.2 FZ -3.1
784. 220 FX 0.23 FY -28.3 FZ -1.87
785. 146 FX -0.22 FY -41.4 FZ 3.48
786. 222 FX 0.15 FY -27 FZ 1.97
787. LOAD 117 1.2D + 1.6WL1+1.0L+0.5LR
788. SELFWEIGHT Y -1
789. ELEMENT LOAD
790. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
791. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
792. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.8
793. JOINT LOAD
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794. 145 FX -6.6 FY -44.4 FZ -2.92
795. 220 FX -0.94 FY -13.2 FZ -0.73
796. 146 FX -6.14 FY -39.5 FZ 3.11
797. 222 FX -1.05 FY -12.4 FZ 0.72
798. LOAD 118 1.2D + 1.6WL2+1.0L+0.5LR
799. SELFWEIGHT Y -1
800. ELEMENT LOAD
801. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
802. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
803. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.8
804. JOINT LOAD
805. 145 FX -5.73 FY -42.2 FZ -3.039
806. 220 FX -0.39 FY -12.2 FZ -0.761
807. 146 FX -4.91 FY -36.9 FZ 3.146
808. 222 FX -0.45 FY -12 FZ 0.723
809. LOAD 119 1.2D + 1.6WL3+1.0L+0.5LR
810. SELFWEIGHT Y -1
811. ELEMENT LOAD
812. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
813. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
814. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.8
815. JOINT LOAD
816. 145 FX 3.62 FY -21.4 FZ -2.16
817. 220 FX 0.47 FY -38 FZ -2.81
818. 146 FX 2.91 FY -20.7 FZ 2.43
819. 222 FX 0.43 FY -33.3 FZ 2.78
820. LOAD 120 1.2D + 1.6WL4+1.0L+0.5LR
821. SELFWEIGHT Y -1
822. ELEMENT LOAD
823. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
824. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
825. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.8
826. JOINT LOAD
827. 145 FX -6.4 FY -32 FZ -2.13
828. 220 FX -5.6 FY -30 FZ -2.25
829. 146 FX 3.8 FY -9 FZ -1.37
830. 222 FX 5.1 FY -6 FZ -1.44
831. LOAD 121 1.2D + 1.6WL5+1.0L+0.5LR
832. SELFWEIGHT Y -1
833. ELEMENT LOAD
834. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
835. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
836. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.8
837. JOINT LOAD
838. 145 FX -0.09 FY -55.6 FZ -7.665
839. 220 FX 0.72 FY -46.4 FZ -5.42
840. 146 FX 0.62 FY -5.3 FZ -1.842
841. 222 FX 0.55 FY -3.8 FZ -1.587
842. LOAD 122 1.2D + 1.6WL6+1.0L+0.5LR
843. SELFWEIGHT Y -1
844. ELEMENT LOAD
845. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
846. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
847. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.8
848. JOINT LOAD
849. 145 FX 1.2 FY -54.09 FZ -8.44
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850. 220 FX 1.38 FY -48.22 FZ -5.9
851. 146 FX 1.98 FY 1.08 FZ -2.45
852. 222 FX 1.14 FY -1.37 FZ -1.969
853. LOAD 123 1.2D + 1.6WL7+1.0L+0.5LR
854. SELFWEIGHT Y -1
855. ELEMENT LOAD
856. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
857. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
858. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.8
859. JOINT LOAD
860. 145 FX 1.87 FY -8.6 FZ 2.52
861. 220 FX 1.16 FY -6.3 FZ 1.84
862. 146 FX 0.21 FY -50.4 FZ 7.04
863. 222 FX 0.65 FY -43.1 FZ 4.92
864. LOAD 124 1.2D + 1.6WL8+1.0L+0.5LR
865. SELFWEIGHT Y -1
866. ELEMENT LOAD
867. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
868. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
869. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.8
870. JOINT LOAD
871. 145 FX 1.7 FY -3.4 FZ 3.16
872. 220 FX 1.2 FY -1.5 FZ 2.31
873. 146 FX 1.2 FY -50.5 FZ 7.55
874. 222 FX 1.3 FY -44 FZ 5.16
875. LOAD 125 0.9D+1.6WL1
876. SELFWEIGHT Y -1
877. ELEMENT LOAD
878. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
879. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
880. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.3
881. JOINT LOAD
882. 145 FX -6.4 FY -32.3 FZ -2.13
883. 220 FX -1 FY -6.1 FZ -0.27
884. 146 FX -6 FY -28.7 FZ 2.27
885. 222 FX -1.1 FY -5.8 FZ 0.25
886. LOAD 126 0.9D+1.6WL2
887. SELFWEIGHT Y -1
888. ELEMENT LOAD
889. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
890. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
891. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.3
892. JOINT LOAD
893. 145 FX -5.6 FY -30.2 FZ -2.25
894. 220 FX -0.4 FY -5.1 FZ -0.3
895. 146 FX -4.7 FY -26.1 FZ 2.3
896. 222 FX -0.5 FY -5.4 FZ 0.25
897. LOAD 127 0.9D+1.6WL3
898. SELFWEIGHT Y -1
899. ELEMENT LOAD
900. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
901. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
902. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.3
903. JOINT LOAD
904. 145 FX 3.8 FY -9.35 FZ -1.37
905. 220 FX 0.4 FY -30.91 FZ -2.36
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906. 146 FX 3.1 FY -9.97 FZ 1.59
907. 222 FX 0.4 FY -26.77 FZ 2.3
908. LOAD 128 0.9D+1.6WL4
909. SELFWEIGHT Y -1
910. ELEMENT LOAD
911. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
912. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
913. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.3
914. JOINT LOAD
915. 146 FX 5.1 FY -6.2 FZ -1.4
916. 220 FX 1.1 FY -30.4 FZ -2.4
917. 146 FX 4.4 FY -7.4 FZ 1.6
918. 222 FX 1 FY -27 FZ 2.3
919. LOAD 129 0.9D+1.6WL5
920. SELFWEIGHT Y -1
921. ELEMENT LOAD
922. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
923. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
924. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.3
925. JOINT LOAD
926. 145 FX 0.09 FY -43.6 FZ -6.9
927. 220 FX 0.66 FY -39.3 FZ -5
928. 146 FX 0.8 FY 5.4 FZ -2.7
929. 222 FX 0.51 FY 2.7 FZ -2.1
930. LOAD 130 0.9D+1.6WL6
931. *SOIL+09*SURCHARGE=1.35X1.8+0.9*2=4.3T/SQ.M)
932. SELFWEIGHT Y -1
933. ELEMENT LOAD
934. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
935. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
936. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.3
937. JOINT LOAD
938. 145 FX 1.4 FY -42.1 FZ -7.7
939. 220 FX 1.3 FY -41.1 FZ -5.4
940. 146 FX 2.2 FY 11.8 FZ -3.3
941. 222 FX 1.1 FY 5.2 FZ -2.4
942. LOAD 131 0.9D+1.6WL7
943. SELFWEIGHT Y -1
944. ELEMENT LOAD
945. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
946. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
947. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.3
948. JOINT LOAD
949. 145 FX 2.1 FY 3.5 FZ 3.31
950. 220 FX 1.1 FY 0.8 FZ 2.3
951. 146 FX 0.4 FY -39.6 FZ 6.2
952. 222 FX 0.6 FY -36.5 FZ 4.45
953. LOAD 132 0.9D+1.6WL8
954. SELFWEIGHT Y -1
955. ELEMENT LOAD
956. 25 TO 76 81 82 85 86 89 90 93 94 97 98 101 102 105 106 109 110 113 114 117 -
957. 118 121 122 125 126 129 130 135 TO 140 143 TO 148 166 TO 180 203 TO 247 254 -
958. 255 TO 298 305 TO 349 375 TO 457 PR GY -4.3
959. JOINT LOAD
960. 145 FX 1.87 FY 8.6 FZ 3.951
961. 220 FX 1.19 FY 5.6 FZ 2.767
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962. 146 FX 1.34 FY -39.8 FZ 6.706
963. 222 FX 1.23 FY -37.4 FZ 4.689
964. PERFORM ANALYSIS

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**366**	**NUMBER OF MEMBERS**	**6**
NUMBER OF PLATES	**323**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**360**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 291/ 30/ 96 DOF
TOTAL PRIMARY LOAD CASES = 66, TOTAL DEGREES OF FREEDOM = 1116
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 108 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 15.8/ 115315.8 MB

965. START CONCRETE DESIGN

```
966. CODE ACI 1999
967. TRACK 0 MEMB 133 134 356 368
968. MAXMAIN 25 MEMB 133 134 356 368
969. CLB 0.075 MEMB 133 134 201 202 356 368
970. CLS 0.075 ALL
971. DESIGN COLUMN 133 134 201 202 356 368
```

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COLUMN NO. 133 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

===

COLUMN NO. 134 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

===

COLUMN NO. 201 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 202 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 356 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 368 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

**********END OF COLUMN DESIGN RESULTS**********

972. END CONCRETE DESIGN
973. FINISH

********** END OF THE STAAD.Pro RUN **********

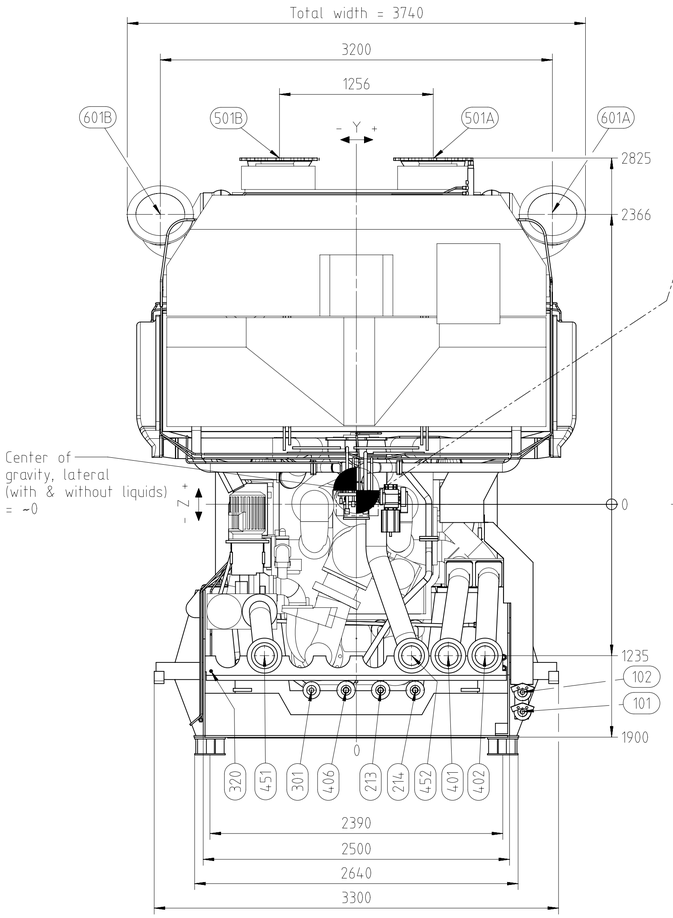
**** DATE= APR 29,2020 TIME= 16:40:30 ****

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*   For technical assistance on STAAD.Pro, please visit     *
*   http://selectservices.bentley.com/en-US/                *
*                                                           *
*   Details about additional assistance from                *
*   Bentley and Partners can be found at program menu       *
*   Help->Technical Support                                 *
*                                                           *
*      Copyright (c) 1997-2014 Bentley Systems, Inc.
*                http://www.bentley.com                     *
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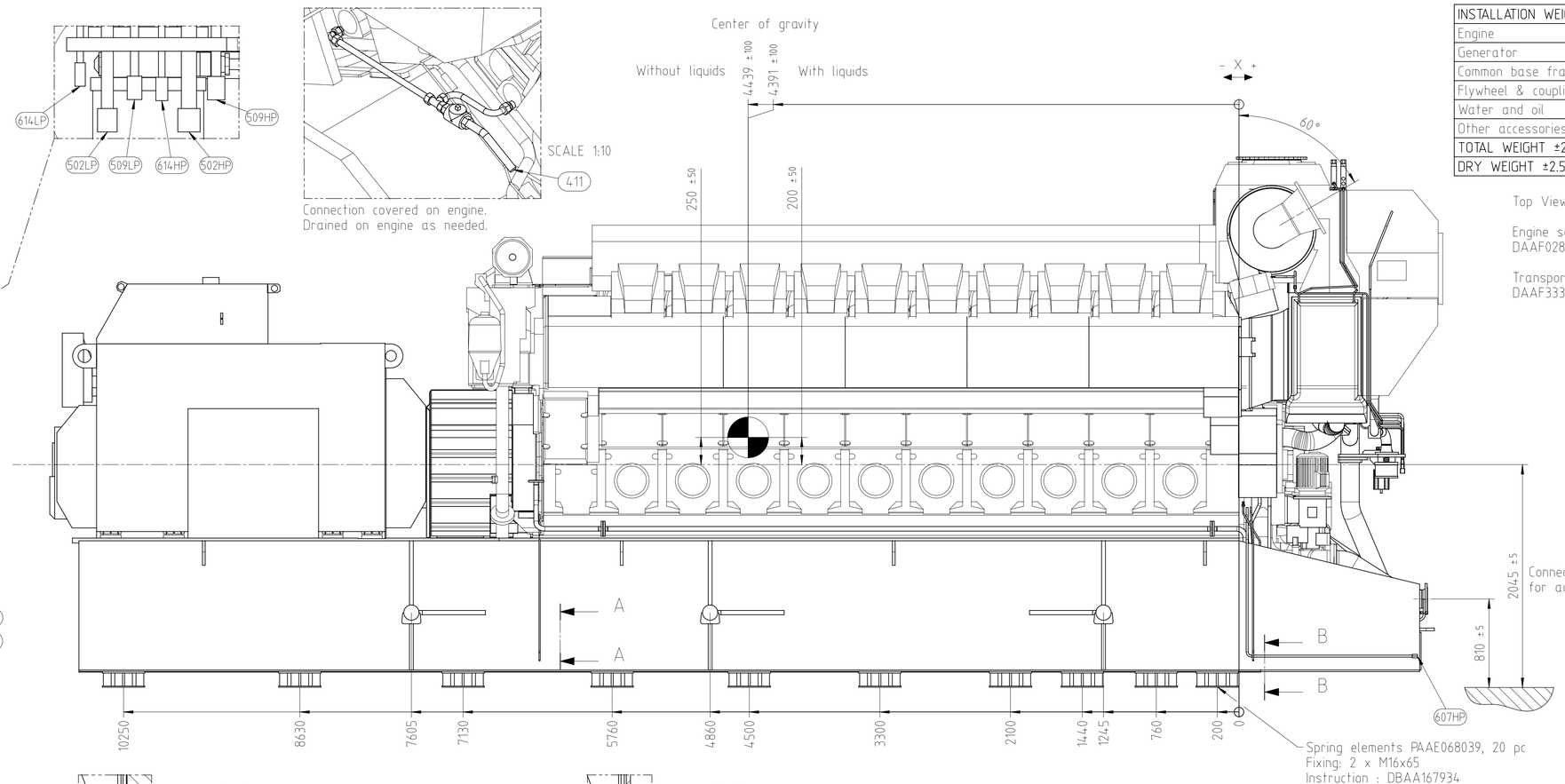
Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	C01	2			
Date	29-04-2020	01-06-2020	26.10.2023				Date	29-04-2020	01-06-2020	26.10.2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT			
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD			
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

ATTACHMENT 4 Load Data Drawing for Engine



Connection covered on engine.
Drained on engine as needed.



Connection dimensions for aux module

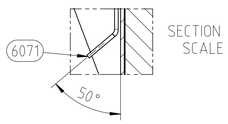
Spring elements PAAE068039, 20 pc
Fixing: 2 x M16x65
Instruction : DBAA167934

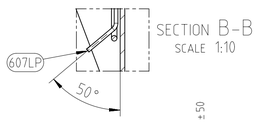
INSTALLATION WEIGHTS	KG
Engine	88800
Generator	33900
Common base frame	21114
Flywheel & coupling	2624
Water and oil	6000
Other accessories	1335
TOTAL WEIGHT ±2.5%	153773
DRY WEIGHT ±2.5%	147773

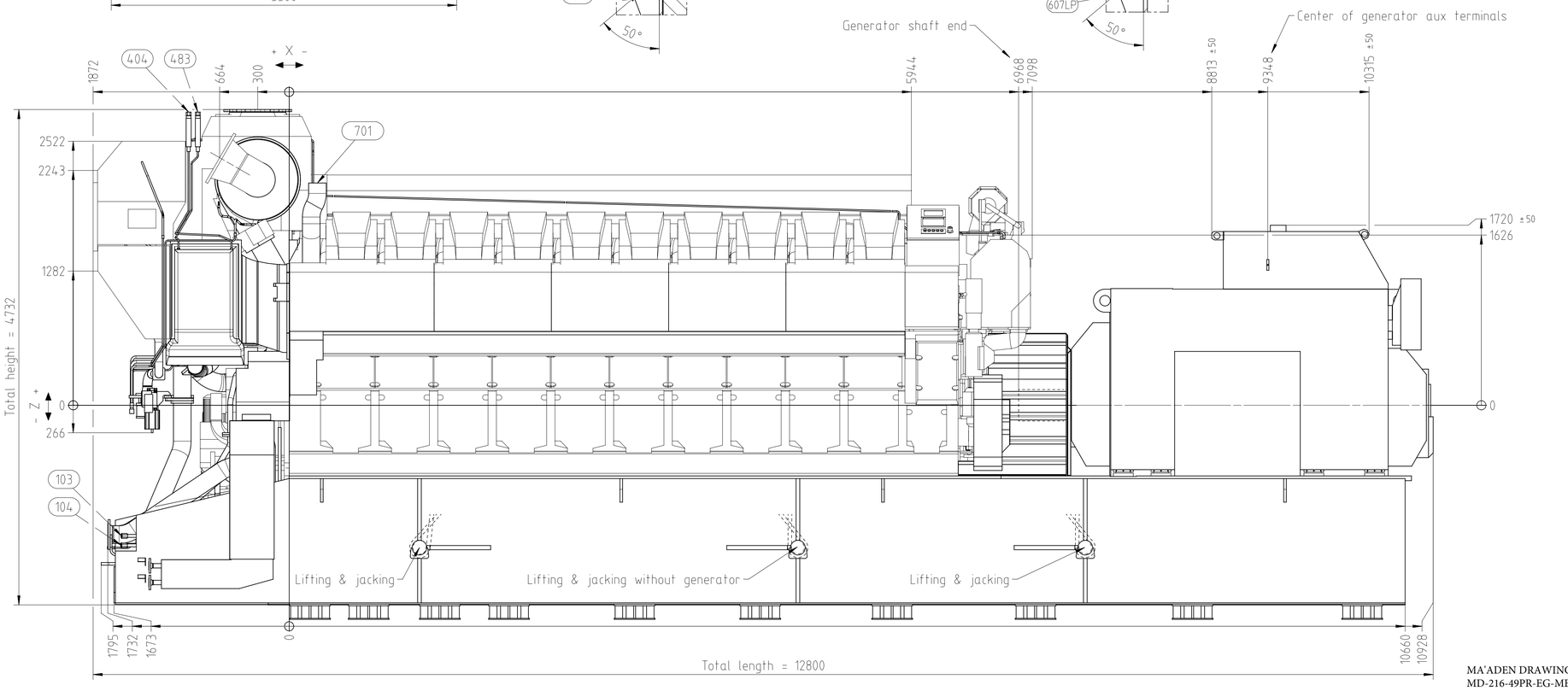
Top View on page 2

Engine service spaces: DAAF028850

Transport Dimensions: DAAF333569







CODE	EXPLANATION	CONNECTION TYPE	SIZE	PN	X	Y	Z
701	CRANKCASE AIR VENT	PIPE EN ISO 6708	DN150	-	-271	+1182	+2123
614LP	SCAVENGING AIR OUTLET TO TC CLEANING VALVE UNIT	FITTING DIN 2353	∅ 10	-	+1502	-125	+15
614HP	SCAVENGING AIR OUTLET TO TC CLEANING VALVE UNIT	FITTING DIN 2353	∅ 12	-	+1502	+25	-9
607LP	CONDENSATE WATER FROM AIR COOLER	PIPE EN ISO 6708	∅ 12	-	+130	-1356	-1800
607HP	CONDENSATE WATER FROM AIR COOLER	FITTING DIN 2353	∅ 18	-	+1608	-1270	-1760
6071	CONDENSATE WATER FROM AIR RECEIVER	PIPE EN ISO 6708	∅ 12	-	-6427	-1356	-1800
601B	AIR INLET TO TURBOCHARGER	FLANGE EN 1092-1	DN400	2.5	+664	-1600	+2366
601A	AIR INLET TO TURBOCHARGER	FLANGE EN 1092-1	DN400	2.5	+664	+1600	+2366
509LP	CLEANING WATER TO COMPRESSOR	FITTING DIN 2353	∅ 15	-	+1502	-25	-9
509HP	CLEANING WATER TO COMPRESSOR	FITTING DIN 2353	∅ 18	-	+1502	+125	-9
502LP	CLEANING WATER TO TURBINE	FITTING DIN 2353	∅ 22	-	+1502	-75	-69
502HP	CLEANING WATER TO TURBINE	FITTING DIN 2353	∅ 28	-	+1502	+75	-69
501B	EXHAUST GAS OUTLET	FLANGE EN 1092-1	DN500	2.5	+300	-628	+2825
501A	EXHAUST GAS OUTLET	FLANGE EN 1092-1	DN500	2.5	+300	+628	+2825
483	LT-WATER AIR VENT	FITTING DIN 2353	∅ 28	-	+872	+934	+2784
452	LT-WATER OUTLET	FLANGE ISO 7005-1	DN150	16	+1732	+450	-1235
451	LT-WATER INLET	FLANGE ISO 7005-1	DN150	16	+1732	-750	-1235
411	HT-WATER DRAIN	PIPE EN ISO 6708	∅ 28	-	-5904	-577	+1270
406	WATER FROM PREHEATER TO HT-CIRCUIT	FLANGE ISO 7005-1	DN40	40	+1338	-84	-1517
404	HT-WATER AIR VENT	FITTING DIN 2353	∅ 28	16	+960	+934	+2784
402	HT-WATER OUTLET	FLANGE ISO 7005-1	DN150	16	+1732	+1050	-1235
401	HT-WATER INLET	FLANGE ISO 7005-1	DN150	16	+1732	+750	-1235
320	INSTRUMENTATION AIR INLET	FITTING DIN 2353	∅ 12	-	+1720	-1186	-1361
301	STARTING AIR INLET	FLANGE ISO 7005-1	DN32	40	+1333	-366	-1517
214	LUBE OIL TO SEPARATOR AND DRAIN	FLANGE ISO 7005-1	DN40	40	+1338	+480	-1517
213	LUBE OIL FROM SEPARATOR AND FILLING	FLANGE ISO 7005-1	DN40	40	+1338	+198	-1517
104	LEAK FUEL DRAIN, DIRTY FUEL	FITTING DIN 2353	∅ 28	-	+1587	+1201	-1338
103	LEAK FUEL DRAIN, CLEAN FUEL	FITTING DIN 2353	∅ 28	-	+1584	+1200	-1248
102	FUEL OIL OUTLET	FLANGE ISO 7005-1	DN 32	40	+1340	+1356	-1533
101	FUEL OIL INLET	FLANGE ISO 7005-1	DN 32	40	+1340	+1356	-1693

MA'ADEN DRAWING NUMBER :
MD-216-49PR-EG-ME-GAD-V553-1001

WÄRTSILÄ

Product: W20V32

GENERATING SET
ABB POWER2 550-M, ABB AMG 1120 L

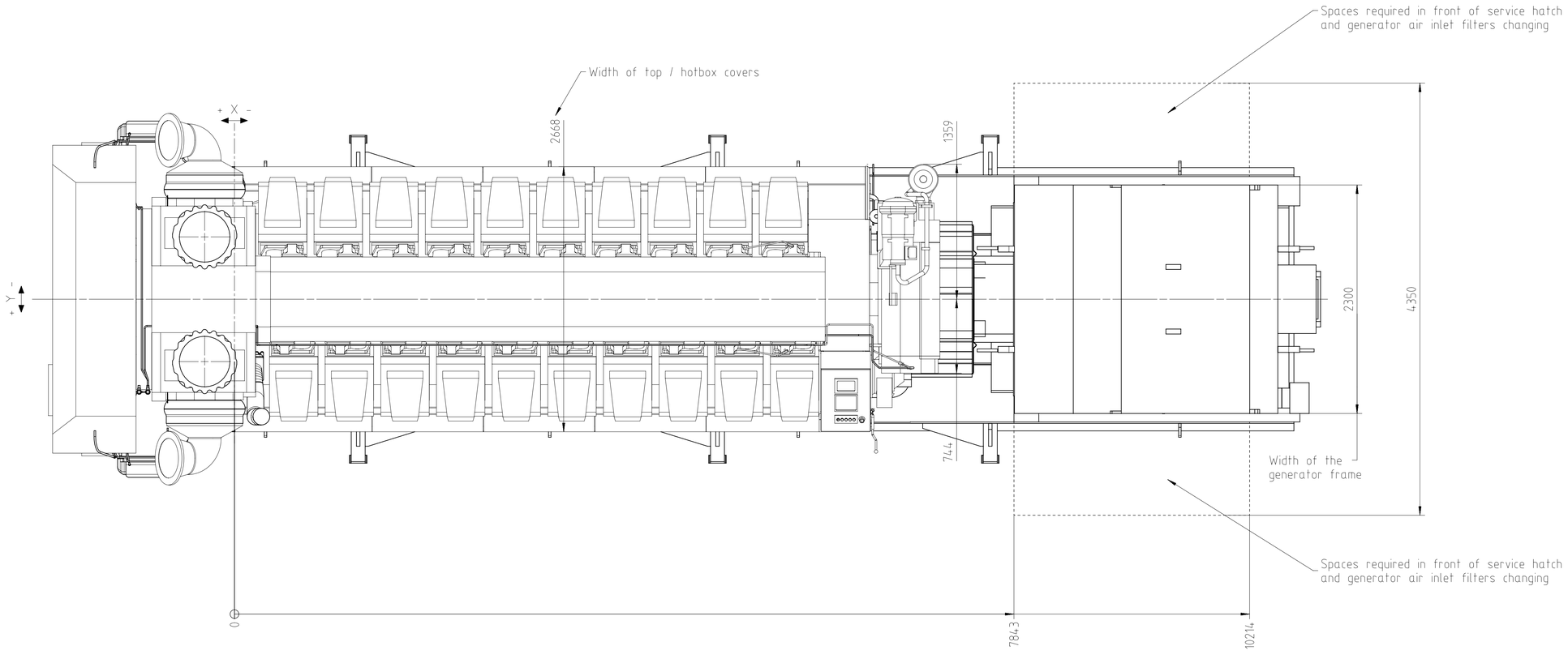
1-C EAM

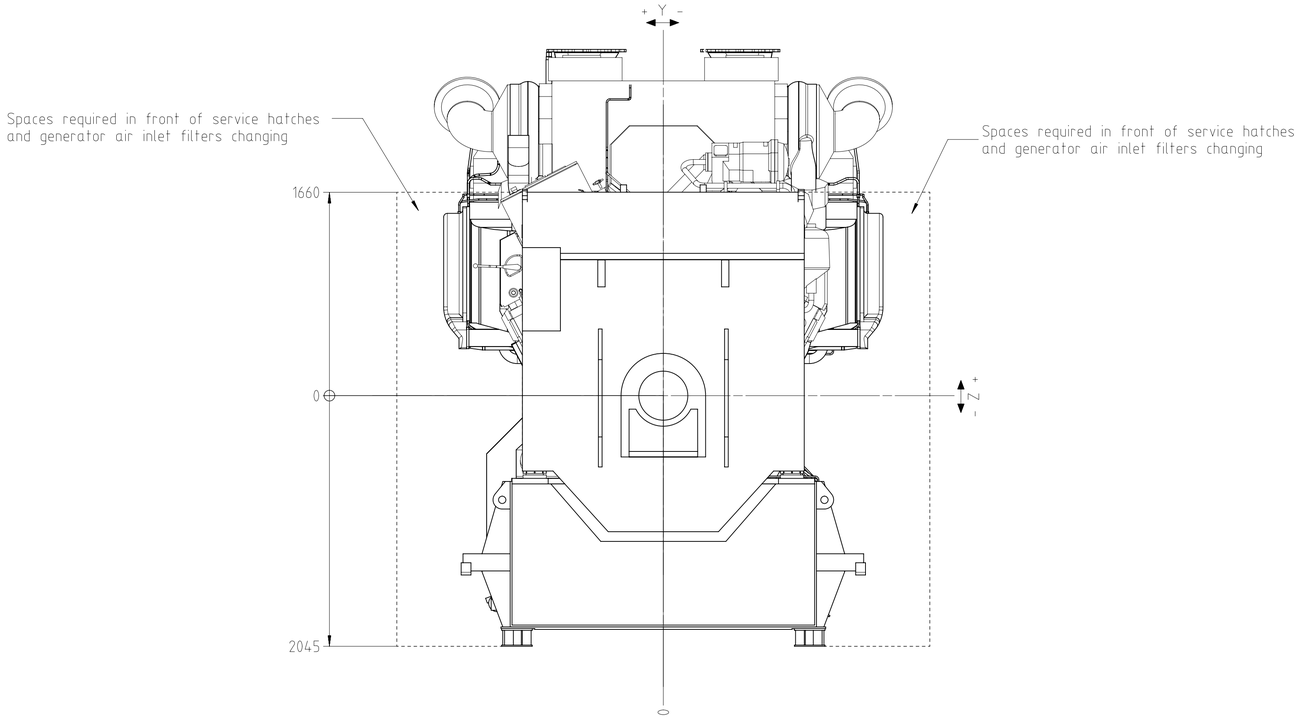
Units	mm kg	NX	Basic Material			Net Weight
Made	27.6.2017	jvix02 Joshi	Scale 1:25	Size A1	Page 1/2	Material ID
Chkd	29.06.2017	tsox03 Somppi	Design Group 58A			
Appd	29.06.2017	kedx01 Edsvik	Drawing ID DAAF383194			Drawing Rev –

Approved

LAD - LAYOUT DRAWING - Internal

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Engine service spaces: DAAF028850

WÄRTSILÄ	Product W20V32	GENERATING SET ABB POWER2 550-M, ABB AMG 1120 L 1-C EAM			
Units mm kg	NX	Basic Material			Net Weight
Made 27.6.2017	jvix02 Joshi	Scale 1:25	Size A1	Page 2/2	Material ID
Chkd 29.06.2017	tsox03 Somppi	Design Group 58A	Drawing ID DAAF383194		Drawing Rev –
Appd 29.06.2017	kedx01 Edsvik				

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CAD - LAYOUT DRAWING - Internal
Approved

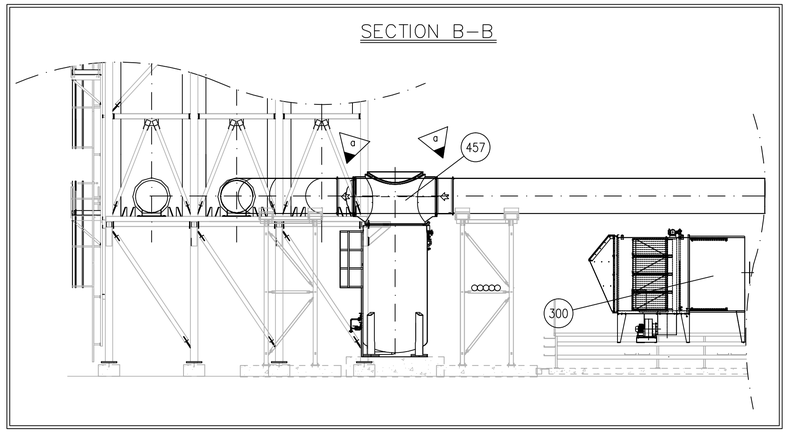
Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	C01	2			
Date	29-04-2020	01-06-2020	26.10.2023				Date	29-04-2020	01-06-2020	26.10.2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT			
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD			
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

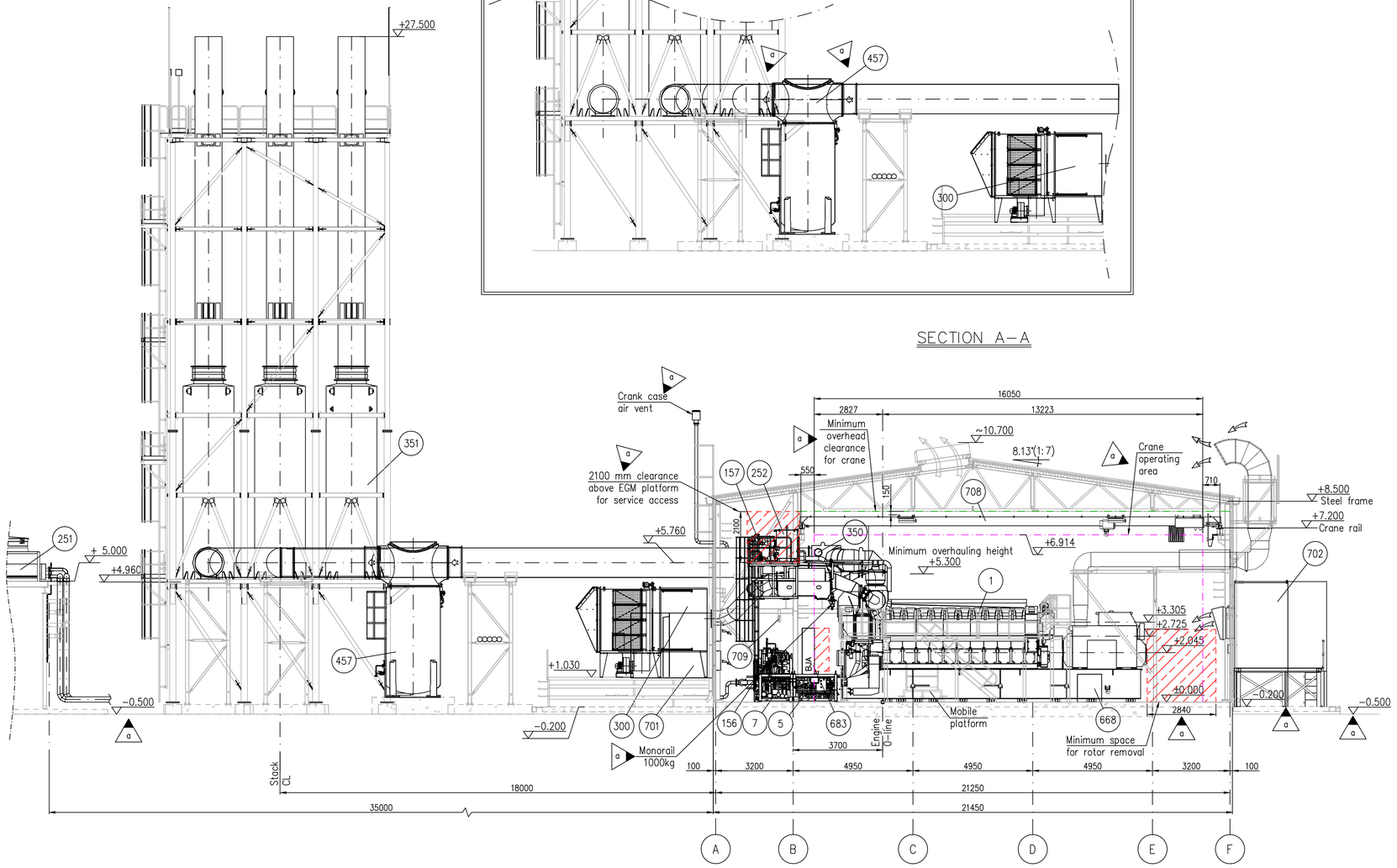
ATTACHMENT 5 Mechanical Input drawing for Engine Hall Building

DRAWING NUMBER: MD-216-49PR-EG-ME-LAD-V553-1002 | REV: A01

SECTION B–B



SECTION A–A



SCALE 1:100

0 1m 2.5m 5m 10m 15m

ENGINE CONFIGURATION: 5+1(+1) x W20V32TS

ENGINE HALL (This list applies to Engine hall section layout)

Item No.	Pcs.	Code	Description	Volume [m³]	Weight [kg] Operative	Mounting level (Floor level)
1	6	SQA	Engine generator set W20V32TS		153773	±0.000
5	6	MOD	Engine Auxiliary Module 3.0 (EAM)		3733	±0.000
7	6	PCC	Engine Fuel Booster 3.0 (EFB)		4230	±0.000
156	6	QBB	Lube oil separator unit		1104	+1.150
157	6	QBF	Oil mist separator		354	+5.770
252	6	VEA	Expansion vessel	0.6	895	+5.770
300	6	NGA	Intake air filter (jet pulse)		–	+1.030
350	6	NHA	Exhaust gas module (EGM)		6150	+4.266
668	6	CFE	Local control panel		–	+1.000
683	6	BJA	Engine auxiliary module control panel		–	+1.150
701	6	EAA	Ventilation unit, aux. area (min. 12m3/s)		~2500	+1.030
702	6	EAA	Ventilation unit, engine hall (min. 18m3/s)		~3000	−0.200
708	1	–	Overhead crane (2 ton capacity)			+7.200
709	1	–	Monorail (250 kg capacity)			

SITE AREA (This list applies to Engine hall section layout)

Item No.	Pcs.	Code	Description	Volume [m³]	Weight [kg] Operative	Mounting level (Floor level)
251	18	VCA	Radiator (NL2)		6130	+5.000
351	6	NHA	Exhaust gas silencer		4790	+4.960
457	3	RCC	Exhaust gas boiler		17800	+0.221

LEGEND (This list applies to Engine Hall section layout)

Colour:
- Items in L&T scope
- Items in Wärtsilä scope

Hatch:
- Space needed for maintenance
- Foundation

Linetype:
- Item above ground
- Cut–line
- Crane operating area
- Center line/grid line
- Minimum overhead clearance for crane

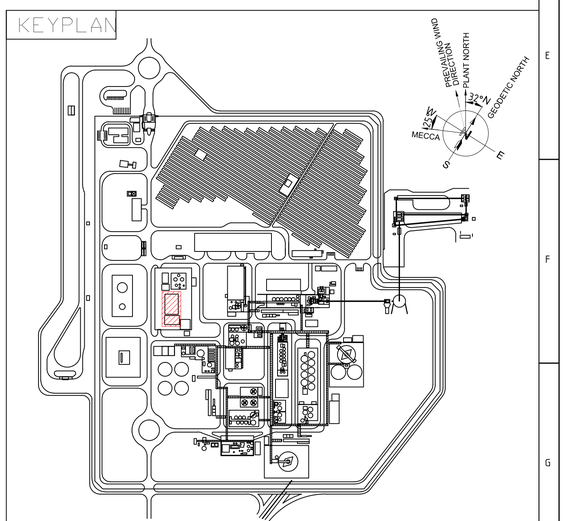
SITE INFO (This list applies to Engine Hall plan, section and Power lant site layouts)

FFL ±0.000 m (+913.6 m from sea level)
FGL −0.500 m (+913.1 m from sea level)
RL −0.300 m (+913.3 m from sea level)

NOTES

– All dimensions are in mm

KEYPLAN



REV	DATE	DESCRIPTION	DRN	CHK	ENG	APP	EMP
A01	25/03/2020	ISSUED FOR APPROVAL	--	SJK	--	NBR	

DOCUMENT NUMBER | REFERENCE DOCUMENTS

WÄRTSILÄ

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DRAWING NUMBER	DBAF489642			REVISION	a
DATE DRAWN	25/03/2020 tosx01	DATE CHECKED	25/03/2020 mspx03	DATE APPROVED	25/03/2020 mnyx02

LARSEN & TOUBRO DRAWING NUMBER	019045_008_M_V_12022_001		REVISION 0
OUTOTEC DRAWING NUMBER	900707 __GAU01 __WG01__MTL01 __49032	DOCUMENT ID OU100674634	REVISION 0

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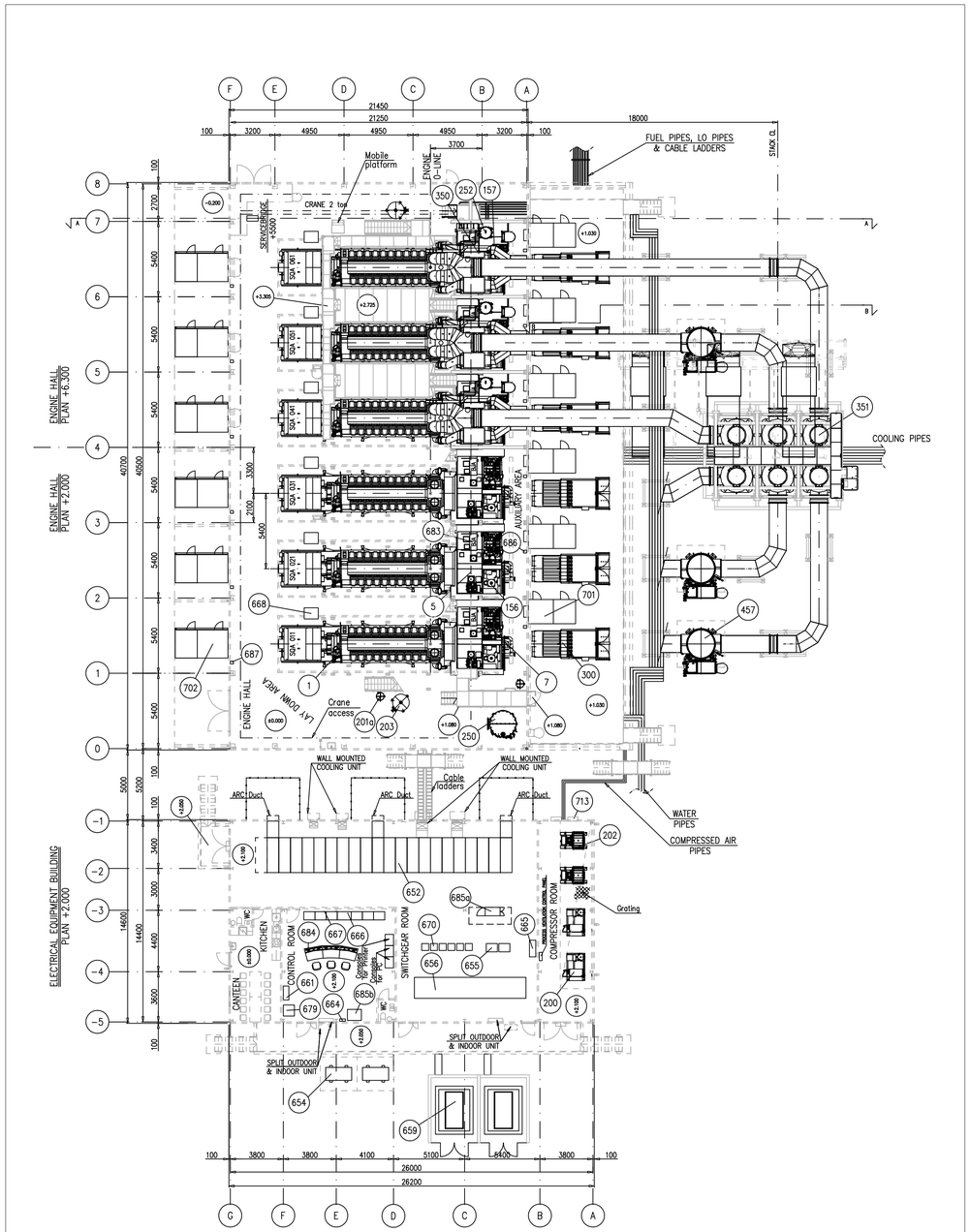
MA'ADEN معادن
Saudi Arabian Mining Company شركة معادن العربية السعودية

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DRAWN	--	--
CHECKED	SJK	17/03/2020
ENGINEER	--	--
APPROVED	NBR	25/03/2020
EMPLOYER		
SCALE	1:100	A1

MANSOURAH - MASSARAH GOLD PROJECT
LAYOUT
POWER GENERATION
CROSS SECTION

DRAWING NUMBER: MD-216-49PR-EG-ME-LAD-V553-1002 | REV: A01



ENGINE HALL (This list applies to Engine Hall plan, Engine hall section and Site layouts.)						
Item No.	Pcs.	Code	Description	Volume [m³]	Weight [kg] Operative	Mounting level (Floor level)
1	6	SQA	Engine generator set W20V32TS		153773	±0.000
5	6	MOD	Engine Auxiliary Module 3.0 (EAM)		3733	±0.000
7	6	PCC	Engine Fuel Booster 3.0 (EFB)		4230	±0.000
156	6	QBB	Lube oil separator unit		1104	+1.150
157	6	QBF	Oil mist separator		354	+5.770
201a	2	TCC	Instrument air bottle	0.2	–	±0.000
203	2	TSB	Starting air bottle	4.8	2023	±0.000
250	1	VBA	Maintenance water tank	6.0	7400	±0.000
252	6	VEA	Expansion vessel	0.6	895	+5.770
300	6	NGA	Intake air filter (jet pulse)		–	+1.030
350	6	NHA	Exhaust gas module (EGM)		6150	+4.266
668	6	CFE	Local control panel		–	+1.000
683	6	BJA	Engine auxiliary module control panel		–	+1.150
686	6	BLQ	Frequency converter (Engine hall inlet ventilation auxiliary side)		–	–
687	6	BLQ	Frequency converter (Engine hall inlet ventilation generator side)		–	–
701	6	EAA	Ventilation unit, aux. area (min. 12m3/s)		~2500	+1.030
702	6	EAA	Ventilation unit, engine hall (min. 18m3/s)		~3000	−0.200

ELECTRICAL EQUIPMENT BUILDING (This list applies to Engine Hall plan, Engine hall section and Site layouts.)						
Item No.	Pcs.	Code	Description	Volume [m³]	Weight [kg] Incl. liquids	Mounting level (Floor level)
200	2	TCA	Instrument and working air unit		570	+2.100
202	2	TSA	Starting air unit		1200	+2.100
652	1	BAC	MV Switchgear			+2.100
655	2	BEY	DC–panel			+2.100
656	1	BFA	LV Switchgear			+2.100
661	1	BLI	Lighting panel *			+2.100
664	1	BLN	Fire detection panel *			+2.100
665	1	BLQ	Ventilation panel			+2.100
666	1	CFA	Control panel, common			+2.100
667	6	CFC	Control panel, engine wise			+2.100
670	6	BLP	Frequency converter, (Radiator)			+2.100
679	1		Office IT equipment *			+2.100
684	1		CCTV *			+2.100
685a	1		Server rack		135	+2.100
685b	1		Server rack (customer) *			+2.100
713	3		Ventilation outlet fan *			–

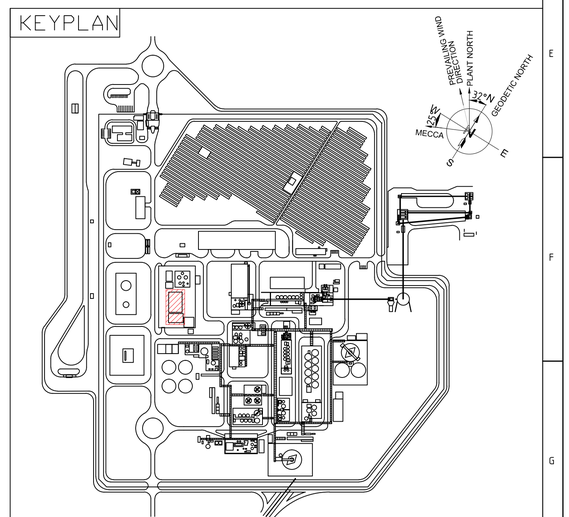
SITE AREA (This list applies to Engine Hall plan, Engine hall section and Site layouts.)						
Item No.	Pcs.	Code	Description	Volume [m³]	Weight [kg] Operative	Mounting level (Floor level)
251	18	VCA	Radiator (NL2)		6130	+5.000
351	6	NHA	Exhaust gas silencer		4790	+4.960
450	1	RHC	Heat recovery container		~19000	–
457	3	RCC	Exhaust gas boiler		17800	+0.221
464	1	RDB	Condensate tank		1350	–
468	1	DAD	Boiler washing water tank	5	~5620	–
500	5	DAA	Oily water collecting pit *	2.5	–	–
512	1	–	Control pit		–	–
551	1	VBB	Water treatment container		~4600	–
552	1	VBC	Treated water tank	10	11450	–
556	1	VBE	Raw water tank	20	22590	–
600	1	VFB	Fire water tank *		–	–
602	1	VFC	Fire pump station *		–	–
654	2	BAZ	Earthing transformer		–	–
659	2	BFB	Station transformer		–	–
663	1	BLM	Black starting unit		~5500	–

(* Item in L&T scope)

NOTES:

1. ALL DIMENSIONS ARE IN mm.

KEYPLAN



QRS		A01	25/03/2020	ISSUED FOR APPROVAL	--	SJK	--	NBR	
DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

WÄRTSILÄ

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Energy Business

DRAWING NUMBER					REVISION
DBAF489644					0
DATE DRAWN	17/01/2020 Jenny Pusula	DATE CHECKED	28/01/2020 Joakim Soderling	DATE APPROVED	30/01/2020 Giridhar Bh

LARSEN & TOUBRO DRAWING NUMBER: O19045_008_M_V_12027_001 — REVISION: 0

OUTOTEC DRAWING NUMBER: 900707__GAU01__WG01__ELD06__49001 — DOCUMENT ID: OU100674709 — REVISION: 0

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DRAWN	--	--
CHECKED	SJK	17/03/2020
ENGINEER	--	--
APPROVED	NBR	25/03/2020
EMPLOYER		
SCALE	1:200	A1

MANSOURAH - MASSARAH GOLD PROJECT
LAYOUT
POWER GENERATION
ENGINE HALL PLAN

DRAWING NUMBER: MD-216-49PR-EG-ME-LAD-V553-1003 — REV: A01

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	C01	2
Date	29-04-2020	01-06-2020	26.10.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYERS COMMENT ON 27-05-2020 & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

AS BUILT

Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

PROJECT:	**MANSOURAH MASSARAH GOLD PROJECT**	**PROJECT NO.: 216**	
STAGE:	**4 - PROJECT EXECUTION.**	**REVIEW LOG DOC. NO. & REV.:**	
ACONEX REF. NO. & DATE:	BECHTEL-WTRAN-002347, 27 May 2020		

STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

DOCUMENT NUMBER	Area / Title	Comment No.	PMC/PMT COMMENT DATE: 27 May 2020	Comment by	EPC Response DATE: 2 June 2020	Status	Remarks
			General Comments (cc)				
MD-216-49PR-EG-CV-CAL-1600-A01	49PR	1	It is recommended to have signature of Wartsila on all Power Plant Structural Design & Calculations for proper coordination. (Pg no. 1)	Mr. Doron Golan	The design and drawings are finalised in due consultation with Wartsila in every stage. The documents are released in complete coordination with Wartsila.	Closed	